Physics Calc

Mathematical Foundations, Classical Theory, and Modern Physics

Physics and Programming

September 2026

Contents

How to use this book **xii**

I Mathematical Toolkit **1**

1 Functions, trigonometry, and calculus **2**

1.1 Functions, graphs, and scale 2

1.2 Angles and oscillations 2

1.3 Derivatives, integrals, and approximation 3

2 Complex numbers, vectors, and linear algebra **4**

2.1 Complex numbers as rotations 4

2.2 Vectors, matrices, and bases 4

3 Fields, partial derivatives, and vector calculus **6**

II First Steps and Newtonian Mechanics **8**

4 What a physical model is **9**

4.1 Quantities, numbers, and units 9

4.2 Graphs and simple relationships 10

4.3 Choose a model and find its limits 10

5 Change and accumulation **11**

5.1 Position, displacement, and average velocity 11

5.2 Velocity at an instant 12

5.3 Acceleration: a change in velocity . . . 12
5.4 Adding velocity to find displacement . . . 13

6 Vectors and first motion models 14

6.1 Arrows and components . . . 14
6.2 Velocity and acceleration in two directions . . . 15
6.3 A ball launched horizontally . . . 15
6.4 Direction and a first vector diagram . . . 16

7 First forces and oscillations 17

7.1 A force changes motion . . . 17
7.2 Gravity near Earth . . . 18
7.3 What changes when the ball rests on a table? . . . 18
7.4 A different force: the spring . . . 19
7.5 Energy as a preview . . . 19
7.6 Checkpoint before the deeper pass . . . 19

8 Newtonian motion and conservation laws 21

8.1 What a force diagram tells us . . . 21
8.2 One force at a time: falling near Earth . . . 22
8.3 Adding a surface: normal force and friction . . . 23
8.4 When the force changes: a spring . . . 23
8.5 Work: force acting through a distance . . . 24
8.6 Potential energy and conservation . . . 24
8.7 Momentum and impulse . . . 25
8.8 A collision, step by step . . . 26
8.9 Rotation: angular momentum and torque . . . 26
8.10 A changing force from motion: drag . . . 27
8.11 An optional extension: a rocket changes mass . . . 27

III Classical Mechanics **29**

9 Gravitation, orbits, and rigid bodies **30**

9.1 Distance from a center . . . 30
9.2 Newtonian gravity and potential energy . . . 31
9.3 The two-body problem and effective potential . . . 32
9.4 Why an axis changes the moment of inertia . . . 33
9.5 Rigid-body rotation . . . 33
9.6 Accelerated frames . . . 34

10 Oscillations, coupled systems, and chaos **35**

10.1 Can units predict a pendulum period? . . . 35
10.2 Simple and damped oscillators . . . 36
10.3 Coupled modes and beats . . . 37
10.4 Linear algebra and normal modes . . . 37
10.5 When the small-angle approximation fails . . . 38
10.6 Nonlinearity and deterministic chaos . . . 38

11 Lagrangian and Hamiltonian mechanics **40**

11.1 From a pendulum to a new coordinate . . . 40
11.2 What stationary action means . . . 41
11.3 Constraints and multipliers . . . 42
11.4 Symmetry and Noether's theorem . . . 42
11.5 Hamiltonian dynamics and phase space . . . 43
11.6 Fields and continuum action . . . 43

IV Continuum Physics, Waves, and Optics **45**

12 Elasticity, fluids, and transport **46**

12.1 A stretched rod: stress and strain . . . 46
12.2 From one rod to a displacement field . . . 47
12.3 Hydrostatics and buoyancy . . . 47

12.4 Counting mass in a small region . . . 48
12.5 Following a fluid parcel . . . 49
12.6 Viscosity and dimensionless parameters . . . 49
12.7 Diffusion, heat conduction, and random walks . . . 50
12.8 A diffusion problem with fixed boundaries . . . 51

13 Waves, sound, and interference **52**

13.1 From a string to a wave equation . . . 52
13.2 Energy, impedance, and boundaries . . . 53
13.3 Complex numbers and Fourier analysis . . . 53
13.4 Dispersion and wave packets . . . 54
13.5 Sound and the Doppler effect . . . 54
13.6 Superposition and coherence . . . 55

14 Geometrical and physical optics **57**

14.1 Rays, refraction, and imaging . . . 57
14.2 Diffraction as Fourier analysis . . . 58
14.3 Polarization and interfaces . . . 59
14.4 Lasers and optical amplification . . . 59

V Thermal and Statistical Physics **61**

15 Thermodynamics **62**

15.1 A system, its boundary, and temperature . . . 62
15.2 The first law: keeping track of energy . . . 62
15.3 Why piston work equals pressure times volume change . . . 63
15.4 Entropy and the second law . . . 63
15.5 Heat capacities and ideal-gas processes . . . 64
15.6 Engines and refrigerators . . . 65
15.7 Thermodynamic potentials and Maxwell relations . . . 65
15.8 Phase equilibrium and stability . . . 66

16 Statistical mechanics **68**

16.1 Probabilities for microscopic states . . . 68
16.2 Microstates, macrostates, and ensembles . . . 69
16.3 The canonical partition function . . . 69
16.4 A second route to the same probabilities . . . 70
16.5 Classical ideal gas and equipartition . . . 71
16.6 Quantum statistics . . . 71
16.7 Blackbody radiation . . . 72
16.8 Interactions, order, and critical behavior . . . 73

VI Electricity and Magnetism **74**

17 Electrostatics, potentials, and matter **75**

17.1 Flux through a closed surface . . . 75
17.2 A field from one charge . . . 76
17.3 Gauss's law from a centered sphere . . . 76
17.4 From a whole surface to a local rule . . . 77
17.5 Potential as energy per charge . . . 77
17.6 An advanced boundary tool: Green functions . . . 77
17.7 Conductors and boundary conditions . . . 78
17.8 Multipoles and dipoles . . . 78
17.9 Capacitance and field energy . . . 79
17.10 Polarization and dielectric response . . . 80

18 Currents, magnetism, and circuits **81**

18.1 Charge transport and resistance . . . 81
18.2 Magnetic forces and steady fields . . . 82
18.3 Magnetized matter . . . 82
18.4 Induction and inductance . . . 83
18.5 Lumped circuits and transients . . . 83

19 Maxwell's equations and electromagnetic waves **85**

19.1 The unified field equations . . . 85

19.2 Potentials and gauge freedom . . . 86

19.3 Wave derivation and polarization . . . 86

19.4 Energy, momentum, and radiation . . . 87

19.5 Plasmas and electromagnetic response . . . 88

VII Relativity and Gravitation **89**

20 Special relativity **90**

20.1 Events, intervals, and Lorentz transformations . . . 90

20.2 Velocity addition, light cones, and causality . . . 91

20.3 Four-momentum and relativistic energy . . . 91

20.4 Collisions and thresholds . . . 92

20.5 Electromagnetic covariance . . . 93

VIII Geometry, Topology, and Continuous Symmetry **94**

21 Topology and global structure **95**

21.1 Open sets, continuity, and homeomorphisms . . . 95

21.2 Connectedness and compactness . . . 95

21.3 Loops, winding, and the fundamental group . . . 96

21.4 Surfaces and the Euler characteristic . . . 96

22 Differential geometry **97**

22.1 Charts, tangent vectors, and covectors . . . 97

22.2 Metrics, lengths, and volume . . . 97

22.3 Connections and covariant differentiation . . . 98

22.4 Geodesics and curvature . . . 98

22.5 Differential forms . . . 99

23 Lie groups and Lie algebras **100**

23.1 Groups and representations . . . 100

23.2 Generators and the exponential map . . . 100

23.3 Commutators and structure constants . . . 101

23.4 $SO(3)$, $SU(2)$, and spin . . . 101

23.5 Symmetry, conservation, and gauge groups . . . 101

IX Relativistic Gravitation **103**

24 General relativity **104**

24.1 Equivalence principle and geometry . . . 104

24.2 Manifolds, tangent spaces, and coordinate changes . . . 105

24.3 Connections, parallel transport, and curvature . . . 105

24.4 Covariant derivatives and geodesics . . . 106

24.5 Einstein's field equations . . . 107

24.6 Schwarzschild geometry and tests . . . 107

24.7 Gravitational waves and black-hole thermodynamics . . . 108

X Quantum Physics **110**

25 Quantum principles and mathematical structure **111**

25.1 A frequency threshold for ejecting an electron . . . 111

25.2 Why matter needs wave amplitudes . . . 112

25.3 From two alternatives to a state . . . 112

25.4 Measurements and observables . . . 113

25.5 Operators and uncertainty . . . 113

25.6 Schrodinger evolution and continuity . . . 114

25.7 Ehrenfest's theorem and pictures of dynamics . . . 115

26 Exactly solvable quantum systems **116**

26.1 Free particles and wave packets . . . 116

26.2 Infinite and finite wells . . . 117
26.3 Tunneling through a barrier . . . 117
26.4 The harmonic oscillator . . . 118
26.5 Angular momentum and spin . . . 119
26.6 Hydrogenic atoms . . . 119

27 Quantum approximation methods and scattering **121**
27.1 Time-independent perturbation theory . . . 121
27.2 Variational principle and WKB . . . 122
27.3 Time-dependent transitions . . . 123
27.4 Adiabatic evolution and geometric phase . . . 124
27.5 Scattering and cross sections . . . 124

28 Identical particles, entanglement, and quantum information **126**
28.1 Exchange symmetry . . . 126
28.2 Density operators and subsystems . . . 127
28.3 Entanglement and correlations . . . 127
28.4 Qubits, interference, and decoherence . . . 128

XI Quantum Computing **130**

29 Quantum computing from bits to algorithms **131**
29.1 Bits, states, and reversible operations . . . 131
29.2 One qubit, one measurement at a time . . . 131
29.3 The Bloch sphere as a picture of one qubit . . . 132
29.4 Gates as rotations and interference . . . 132
29.5 Registers, tensor products, and circuits . . . 133
29.6 A first algorithm: amplitude amplification . . . 133
29.7 Noise, error correction, and what the picture must show . . . 134

XII Atoms, Materials, Nuclei, and Particles **135**

30 Atomic and molecular physics **136**

30.1 Atomic structure beyond hydrogen . . . 136

30.2 Light-matter interaction and selection rules . . . 137

30.3 Magnetic fields and resonance . . . 137

30.4 Molecules and the Born–Oppenheimer approximation . . . 138

31 Condensed matter and collective phenomena **140**

31.1 Crystals, reciprocal space, and phonons . . . 140

31.2 Bloch's theorem and electronic bands . . . 141

31.3 Fermi gases and electron transport . . . 142

31.4 Semiconductors and junctions . . . 142

31.5 Magnetism, superconductivity, and superfluidity . . . 143

31.6 Topology when collective states need it . . . 144

31.7 Topology and emergence . . . 144

32 Nuclear physics **145**

32.1 Nuclear scales and binding . . . 145

32.2 Radioactive decay . . . 146

32.3 Reaction energetics and cross sections . . . 147

32.4 Nuclear models and dense matter . . . 147

33 Relativistic quantum theory and particle physics **149**

33.1 Why quantum fields are needed . . . 149

33.2 Spinors, antiparticles, and quantization . . . 150

33.3 Interactions, diagrams, and effective theories . . . 151

33.4 The Standard Model's structure . . . 151

33.5 Strong and weak interactions . . . 152

33.6 Symmetries, conservation, and open questions . . . 153

XIII Astrophysics, Cosmology, and Synthesis **154**

34 Stars, compact objects, and the expanding universe **155**

34.1 Stellar structure and energy transport . . . 155

34.2 Degeneracy pressure and compact remnants . . . 156

34.3 Homogeneity, isotropy, and expansion . . . 157

34.4 Friedmann equations and cosmic contents . . . 157

34.5 Thermal history and structure formation . . . 158

34.6 Dark sectors and inflation . . . 159

35 Computational physics and model validation **160**

35.1 Discretization and numerical error . . . 160

35.2 Time integration and Hamiltonian systems . . . 161

35.3 PDE stability and eigenvalue problems . . . 162

35.4 Monte Carlo and statistical sampling . . . 162

35.5 Uncertainty and data . . . 163

35.6 A practical verification checklist . . . 164

36 Connections across the subject **165**

36.1 Conservation from local balance . . . 165

36.2 The oscillator as a recurring model . . . 166

36.3 Symmetry, gauge freedom, and observables . . . 166

36.4 Scale and effective descriptions . . . 167

A Answers to chapter exercises **168**

B Constants, units, and useful identities **172**

B.1 Common constants . . . 172

B.2 Integrals and expansions . . . 173

B.3 Natural units and restoring dimensions . . . 173

C Guided diagrams and thought exercises **174**

C.1 Graphs, forces, and conservation 174

C.2 Fields, waves, and thermal processes 174

C.3 Geometry, relativity, and quantum theory 175

D Review problems **176**

E Solutions and reasoning **178**

F Further study and a route through the material **180**

Building this document **182**

How to use this book

This book can be read at more than one depth. A **foundation** chapter develops ideas needed repeatedly later. A **core** chapter forms part of a broad first path through physics. An **advanced** chapter gives a careful entrance to material normally studied in a later course. These labels describe prerequisites, not importance.

The recommended first route is mathematical tools; motion, forces, energy, and momentum; oscillations and waves; thermodynamics; electricity and magnetism; special relativity; and quantum principles. After that route, choose branches according to interest. Differential geometry and Lie theory prepare the mathematical language of general relativity, gauge theory, and modern particle physics. Continuum mechanics, statistical mechanics, condensed matter, nuclear physics, and cosmology can be read as further branches.

Each chapter contains short exercises that require only pencil, paper, and the information in the text. Answers are collected in the appendix. No exercise requires laboratory equipment, purchased materials, or access to a school facility. Sketches are mathematical diagrams: their purpose is to expose directions, slopes, boundaries, or transformations clearly enough to reason from them.

When a chapter feels too abrupt, return to the mathematical toolkit and follow the cross-reference named at the start of the section. A productive pass through a chapter has four goals: explain the physical picture, reproduce one central derivation, solve a representative exercise, and state the assumptions that limit the result.

Part I

Mathematical Toolkit

Chapter 1

Functions, trigonometry, and calculus

Foundation. No prior calculus is assumed. This chapter supplies the notation used throughout the book.

A physical model often begins with a function: one quantity is assigned to each allowed value of another. Writing $x(t)$ means that time is the input and position is the output. A graph of the function records the same relation geometrically. Its slope measures change, while area accumulated under a rate records a total change.

1.1 Functions, graphs, and scale

For a linear function $y = mx + b$, the number m is the change in y per unit change in x, and b is the value at $x = 0$. Powers describe other common scaling laws. If $y = Cx^2$, doubling x multiplies y by four. If $y = C/x^2$, doubling x divides y by four. On logarithmic axes, a power law $y = Cx^n$ becomes $\ln y = \ln C + n \ln x$, so its exponent is a slope.

An equation may define a family rather than one answer. The differential equation $\dot{x} = v$ has solutions $x = vt + C$; an initial condition such as $x(0) = x_0$ fixes $C = x_0$. This distinction between a law and the data selecting one solution will recur in mechanics, waves, diffusion, and quantum theory.

1.2 Angles and oscillations

Radians measure angle by arc length divided by radius. This makes an angle dimensionless and gives the clean derivative rules

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x, \qquad \frac{\mathrm{d}}{\mathrm{d}x}\cos x = -\sin x. \tag{1.1}$$

For a right triangle, $\sin\theta$ is opposite over hypotenuse, $\cos\theta$ is adjacent over hypotenuse, and $\tan\theta = \sin\theta/\cos\theta$. The identity $\sin^2\theta + \cos^2\theta = 1$ is the Pythagorean theorem written in normalized coordinates. A phase $\omega t + \phi$ must be dimensionless, so angular frequency ω has units of inverse time.

1.3 Derivatives, integrals, and approximation

The derivative is a limiting slope,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}. \tag{1.2}$$

Useful rules include $(fg)' = f'g + fg'$, $[f(g(x))]' = f'(g(x))g'(x)$, and $\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{ax} = a\mathrm{e}^{ax}$. An integral adds narrow contributions. The fundamental theorem connects the two operations:

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_a^x f(s)\,\mathrm{d}s = f(x), \qquad \int_a^b f'(x)\,\mathrm{d}x = f(b) - f(a). \tag{1.3}$$

Integration by parts follows from the product rule:

$$\int_a^b u\,\mathrm{d}v = [uv]_a^b - \int_a^b v\,\mathrm{d}u. \tag{1.4}$$

It will be used when deriving equations from an action and when transforming derivatives in Fourier analysis.

A Taylor expansion makes the size of an approximation visible:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \cdots. \tag{1.5}$$

Keeping only the first terms is justified when the omitted contribution is small compared with the accuracy required.

Worked example 1.1 (From a rate to a total)**.** If $v(t) = v_0 + at$, then $x(t) - x(0) = \int_0^t (v_0 + as)\mathrm{d}s = v_0 t + at^2/2$. Differentiating the result returns the velocity, which checks the calculation.

Exercises

Differentiate $x(t) = 3t^2 - 2t + 1$. Integrate the resulting velocity from 0 to T and verify that it equals $x(T) - x(0)$. Explain why the argument of $\sin(\omega t)$ is dimensionless.

Answers are in Appendix A.

Chapter 2

Complex numbers, vectors, and linear algebra

Foundation. This chapter prepares oscillations, waves, quantum mechanics, coupled systems, and rotations.

2.1 Complex numbers as rotations

A complex number $z = a + \mathrm{i}b$ has conjugate $z^* = a - \mathrm{i}b$ and squared magnitude $|z|^2 = zz^* = a^2 + b^2$. Euler's identity,

$$\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta, \tag{2.1}$$

turns oscillation into rotation in a plane. Multiplying by $\mathrm{e}^{\mathrm{i}\theta}$ rotates a complex number by θ without changing its magnitude. Physical displacements may be obtained by taking a real part; in quantum theory the complex amplitude itself carries relative-phase information.

2.2 Vectors, matrices, and bases

A vector is a geometric object whose components depend on a chosen basis. In two Cartesian dimensions, $\boldsymbol{v} = (v_x, v_y)$ has magnitude $\sqrt{v_x^2 + v_y^2}$. The dot product $\boldsymbol{u} \cdot \boldsymbol{v} = u_x v_x + u_y v_y$ measures alignment. A matrix represents a linear transformation: $A(\alpha\boldsymbol{u} + \beta\boldsymbol{v}) = \alpha A\boldsymbol{u} + \beta A\boldsymbol{v}$.

The rotation matrix

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{2.2}$$

changes vector components while preserving lengths. Matrix multiplication represents successive transformations; order generally matters.

An eigenvector keeps its direction under a linear transformation:

$$A\boldsymbol{v} = \lambda \boldsymbol{v}. \tag{2.3}$$

In coupled oscillations, eigenvectors are mode shapes and eigenvalues determine squared frequencies. In quantum mechanics, eigenvectors of an observable represent states with definite measurement outcomes.

Worked example 2.1 (Two normal directions)**.** For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, the vectors $(1, 1)$ and $(1, -1)$ are eigenvectors with eigenvalues 3 and 1. Adding or subtracting two coupled equations finds these same in-phase and out-of-phase directions.

Exercises

Write $3 - 4\mathrm{i}$ in polar form by finding its magnitude and angle. Multiply $R(\pi/2)$ by $(1, 0)$. Verify directly the two eigenvector statements in the example.

Answers are in Appendix A.

Chapter 3

Fields, partial derivatives, and vector calculus

Foundation. This chapter prepares fluids, electromagnetism, diffusion, wave equations, and geometry.

A scalar field assigns a number to each point, such as $T(x, y, z, t)$. A vector field assigns an arrow, such as fluid velocity $\boldsymbol{v}(\boldsymbol{r}, t)$. A partial derivative varies one coordinate while holding the others fixed.

The gradient $\boldsymbol{\nabla} f$ points toward fastest increase of a scalar field. Divergence measures local outward flux, curl measures local circulation, and the Laplacian measures how a value differs from its neighbors:

$$\boldsymbol{\nabla} f = (\partial_x f, \partial_y f, \partial_z f), \tag{3.1}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{A} = \partial_x A_x + \partial_y A_y + \partial_z A_z, \tag{3.2}$$

$$\nabla^2 f = \partial_x^2 f + \partial_y^2 f + \partial_z^2 f. \tag{3.3}$$

For a small box, positive divergence means more of the field leaves than enters. The divergence theorem and Stokes's theorem connect local derivatives to boundaries:

$$\int_V \boldsymbol{\nabla} \cdot \boldsymbol{A} \, \mathrm{d}V = \oint_{\partial V} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{S}, \qquad \int_S (\boldsymbol{\nabla} \times \boldsymbol{A}) \cdot \mathrm{d}\boldsymbol{S} = \oint_{\partial S} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{\ell}. \tag{3.4}$$

These theorems explain why integral conservation laws and local differential equations are two descriptions of the same bookkeeping.

Worked example 3.1 (Reading a divergence)**.** For $\boldsymbol{A} = (ax, ay, az)$, $\boldsymbol{\nabla} \cdot \boldsymbol{A} = 3a$. If $a > 0$, arrows grow outward in every direction, so a small closed surface has positive net outward flux.

Exercises

Find $\boldsymbol{\nabla} f$ and $\nabla^2 f$ for $f = x^2 + y^2 + z^2$. Find the divergence of $\boldsymbol{A} = (-y, x, 0)$ and describe why zero divergence does not mean the field is zero.

Answers are in Appendix A.

Part II

First Steps and Newtonian Mechanics

Chapter 4

What a physical model is

Physics often starts with an ordinary question: how long will a walk take, or where will a rolling ball stop? To answer, we decide what to measure, describe what matters for that question, and use the description to make a prediction. Then we compare the prediction with what happens. That description is a *model*. We will make a very simple one here before adding calculus or forces.

4.1 Quantities, numbers, and units

Suppose you want to know how long it will take to walk across a room. You can measure the room's length and estimate your walking speed. A length of 6 m and a speed of 1.2 m/s are different kinds of quantities. Each needs both a number and a unit; writing "6" alone leaves the reader guessing what was measured.

A unit conversion changes how we report a quantity without changing the quantity itself. For example, 3.0 m = 300 cm and 2.0 min = 120 s. When we multiply or divide quantities, their units travel through the arithmetic. Dividing distance by speed gives

$$\frac{\text{m}}{\text{m/s}} = \text{s}, \tag{4.1}$$

so the result has the unit of time that the question asks for. This unit check can catch a wrong operation before we rely on its number.

Worked example 4.1 (A walk across a room)**.** A room is about 6 m long and you expect to walk at about 1.2 m/s. Time is distance divided by speed, so $t = (6\,\text{m})/(1.2\,\text{m/s}) = 5\,\text{s}$. This is an estimate of total time, not a claim that your speed stays exactly constant during every step. If a measurement gives a very different time, we should ask whether the room, speed estimate, or model was wrong.

Before doing precise arithmetic, make a rough estimate. If someone walks at around a meter per second, crossing a room several meters wide should take several seconds. An answer of several hours would signal a unit conversion or arithmetic error.

4.2 Graphs and simple relationships

A graph places one measured quantity along a horizontal axis and another along a vertical axis. For the walking example, place elapsed time t horizontally and distance x vertically. A straight line that rises by the same amount each second represents constant speed. Its steepness, or slope, is distance change divided by time change. A steeper line means faster motion.

For a relationship $y = Cx$, the constant C sets the steepness of a straight line through the origin. If x doubles, y doubles. A horizontal line means the plotted quantity stays constant. A curve means the steepness changes, so the rate of change is not constant. The next chapter will show how to measure the slope of a curve at one moment.

4.3 Choose a model and find its limits

The room calculation treated walking speed as constant. That choice gave a useful estimate because short changes in speed did not matter much to the question. For a sprinter who accelerates from rest, or a crowded hallway where someone repeatedly stops, the same model may be poor. A model is valuable when its assumptions are clear enough that we know when to improve it.

We also choose a direction when position or motion can have a sign. If right is positive, a displacement left is negative. The sign then tells us direction; it does not mean the measured distance is impossible. In later chapters, we will draw arrows when one signed direction is not enough.

After any calculation, check four things. Do the units match the quantity asked for? Does a positive or negative sign fit the chosen direction? Does the answer have a reasonable size? Which effect did the model leave out? If an experiment disagrees, these questions help us decide whether the arithmetic or the physical assumption needs attention.

Exercises

A friend walks 12 m in 10 s. Estimate the average speed and give its units. Name one reason this average might not describe the friend's speed at every moment. Sketch a distance-versus-time graph for steady walking, then one for a walk that begins slowly and speeds up.

Answers are in Appendix A.

Chapter 5

Change and accumulation

To describe motion, we need to answer two different questions. How quickly is position changing at a particular moment? How much did position change over an interval? A graph makes the first question a slope and the second an area. We will build each idea from ordinary arithmetic before using calculus notation.

5.1 Position, displacement, and average velocity

Choose a straight track and call one direction positive. The number $x(t)$ tells us the object's position at time t relative to an origin. If the object is at $x = 2\,\mathrm{m}$ at one time and $x = 10\,\mathrm{m}$ four seconds later, its displacement is $10 - 2 = 8\,\mathrm{m}$. Displacement includes direction through its sign. The distance traveled might be larger if the object turned around along the way.

Average velocity is displacement divided by elapsed time:

$$\bar{v} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}. \tag{5.1}$$

For the positions above, $\bar{v} = 8\,\mathrm{m}/4\,\mathrm{s} = 2\,\mathrm{m/s}$ in the positive direction. On a position-versus-time graph, the same calculation is the slope of the straight line joining the two plotted points. The line tells us an average over the interval; it does not reveal exactly how the object moved inside it.

Worked example 5.1 (Equal averages, different trips)**.** Two walkers both begin at $x = 0$ and end at $x = 8\,\mathrm{m}$ after four seconds. One walks steadily at $2\,\mathrm{m/s}$. The other waits two seconds and then walks at $4\,\mathrm{m/s}$ for two seconds. Their average velocities are both $2\,\mathrm{m/s}$, but their position graphs have different slopes during the trip. To describe motion at one instant, we need a smaller time interval.

5.2 Velocity at an instant

Suppose the position graph curves. Choose a time t and measure the slope between that time and a nearby time $t + \Delta t$:

$$\frac{x(t + \Delta t) - x(t)}{\Delta t}. \tag{5.2}$$

As Δt becomes smaller, this average describes a shorter piece of the motion. If the slopes approach one value, that value is the instantaneous velocity $v(t)$, or derivative of position:

$$v(t) = \lim_{\Delta t \to 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} = \frac{\mathrm{d}x}{\mathrm{d}t}. \tag{5.3}$$

The limit means the value approached by the nearby slopes; we are not dividing by zero. On the graph it is the slope of the tangent line at that point.

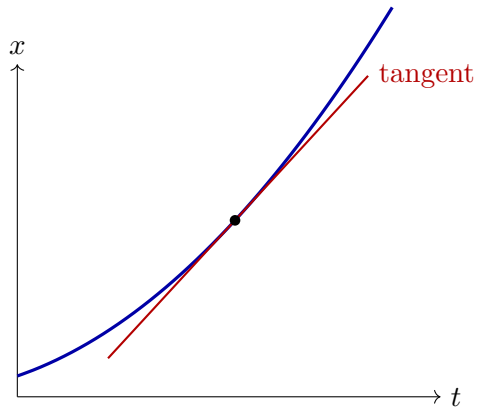


Figure 5.1: Instantaneous velocity is the slope of the tangent to a position–time graph. The graph itself gives position; its steepness gives velocity.

Worked example 5.2 (Finding one slope from nearby times)**.** Use numerical time values in seconds and suppose the numerical position in meters is $x(t) = t^2$. At $t = 2$, the slope over the following interval Δt is $[(2 + \Delta t)^2 - 2^2]/\Delta t = 4 + \Delta t$. Restoring the units, the slope approaches $4\,\mathrm{m/s}$ as the interval shrinks. The calculation gives the velocity at that moment even though the average over a longer interval is different.

A positive velocity means position is increasing along the chosen axis; a negative one means it is decreasing. Speed is the size $|v|$ and has no direction. These are different answers when the object reverses direction.

5.3 Acceleration: a change in velocity

Velocity itself can change. Acceleration $a(t)$ measures that change per unit time. Just as velocity came from a position graph's slope, acceleration comes from a velocity graph's slope:

$$a(t) = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}. \tag{5.4}$$

For a straight-line motion with constant acceleration a, the velocity changes by a each second. If $v(0) = 1\,\mathrm{m/s}$ and $a = 2\,\mathrm{m/s^2}$, then after one second $v = 3\,\mathrm{m/s}$ and after two seconds $v = 5\,\mathrm{m/s}$. The unit $\mathrm{m/s^2}$ means a change of meters per second of velocity each second.

Acceleration is not the same as speed. A cart moving right can have a negative acceleration while slowing down. A car moving around a bend can have acceleration even at constant speed because its direction changes; the next chapter will use vectors to describe that case.

5.4 Adding velocity to find displacement

Now reverse the question. If we know velocity at each time, how do we find displacement? Over a short interval of duration Δt where velocity changes little, displacement is approximately $v\Delta t$. Add many such short contributions. As the intervals shrink, the sum becomes an integral:

$$x(t) - x(t_0) = \int_{t_0}^{t} v(s)\,\mathrm{d}s. \tag{5.5}$$

The integration variable s simply labels times between the starting time t_0 and the final time t. The integral is signed area under a velocity-versus-time graph. Area above the time axis adds positive displacement; area below it adds negative displacement.

Worked example 5.3 (Area under a simple velocity graph)**.** A cart moves at $3\,\mathrm{m/s}$ for two seconds, then at $-1\,\mathrm{m/s}$ for one second. The first rectangle under the velocity graph contributes $3(2) = 6\,\mathrm{m}$; the second contributes $(-1)(1) = -1\,\mathrm{m}$. Total displacement is $5\,\mathrm{m}$, while total distance traveled is $6 + 1 = 7\,\mathrm{m}$.

For constant velocity v, the area is a rectangle of height v and width $t - t_0$, so $x(t) = x(t_0) + v(t - t_0)$. This is also our first differential equation: the local rule $\dot{x} = v$ needs the starting position $x(t_0)$ to select one particular motion.

Exercises

A cyclist's position changes from $2\,\mathrm{m}$ to $14\,\mathrm{m}$ in $4\,\mathrm{s}$. Find the average velocity. Sketch a position graph with that average but a steeper slope near the end. If $v = -2\,\mathrm{m/s}$ for three seconds, what is the signed displacement and what distance is traveled?

Answers are in Appendix A.

Chapter 6

Vectors and first motion models

A signed position works well on a straight track. On a map, one number cannot say both how far east and how far north an object lies. We need an arrow with components in each direction. Once we can add and measure those arrows, the slope and area ideas from the previous chapter will describe motion in a plane.

6.1 Arrows and components

Start at an origin on a map. Choose x positive to the right and y positive upward. An arrow from the origin to the point $(3, 4)\,\mathrm{m}$ is a position vector. We write it as $\boldsymbol{r} = 3\hat{\boldsymbol{i}} + 4\hat{\boldsymbol{j}}$ in meters, where $\hat{\boldsymbol{i}}$ and $\hat{\boldsymbol{j}}$ are unit arrows along the axes. The numbers 3 and 4 are the components of $\boldsymbol{r}$. They depend on our choice of axes; the actual arrow between the two points does not.

To add two arrows, add their horizontal components and their vertical components separately. A $3\,\mathrm{m}$ step east followed by a $4\,\mathrm{m}$ step north gives displacement $(3, 4)\,\mathrm{m}$. The straight-line length of that displacement is

$$|\boldsymbol{r}| = \sqrt{3^2 + 4^2} = 5\,\mathrm{m}. \tag{6.1}$$

The distance actually walked is $3 + 4 = 7\,\mathrm{m}$. This is the same difference between path length and displacement we saw on a straight track.

A vector can point left or down; its corresponding components are negative. For example, adding $(3, 4)$ and $(-1, 2)$ gives $(2, 6)$. Draw the arrows head to tail to check that the answer points in the direction you expect. In three dimensions we add a third component z in the same way: $\boldsymbol{r} = x\hat{\boldsymbol{i}} + y\hat{\boldsymbol{j}} + z\hat{\boldsymbol{k}}$.

6.2 Velocity and acceleration in two directions

At each time t, let $\boldsymbol{r}(t)$ be the position arrow. Its change over a short interval is a displacement arrow. Divide that arrow by the elapsed time and shrink the interval to find velocity:

$$\boldsymbol{v}(t) = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}t}\hat{\boldsymbol{i}} + \frac{\mathrm{d}y}{\mathrm{d}t}\hat{\boldsymbol{j}}. \tag{6.2}$$

The derivative acts on each component. If horizontal position changes steadily but vertical position changes faster and faster, horizontal velocity is constant while vertical velocity changes. Acceleration is the rate of change of the velocity arrow:

$$\boldsymbol{a}(t) = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = a_x\hat{\boldsymbol{i}} + a_y\hat{\boldsymbol{j}}. \tag{6.3}$$

The velocity arrow points along the instantaneous motion; the acceleration arrow points in the direction velocity is changing. They do not have to point the same way. A ball moving upward can have downward acceleration while it slows down.

6.3 A ball launched horizontally

Imagine releasing a ball from the edge of a table with an initial horizontal velocity v_{x0} and no initial vertical velocity. Choose x to the right and y upward. Measurements near Earth's surface show an approximately constant downward acceleration with magnitude $g \approx 9.8\,\mathrm{m/s^2}$ when air resistance is small. For now we use that measured acceleration, so $a_x = 0$ and $a_y = -g$. The next chapter explains the force behind it. We can use the one-dimensional ideas separately on each axis.

Since horizontal acceleration is zero, horizontal velocity stays at v_{x0} and horizontal position changes by $v_{x0}t$. Vertical velocity starts at zero and changes by $-g$ each second, so $v_y(t) = -gt$. The area under that straight-line velocity graph is a negative triangle of area $gt^2/2$. If the ball starts at (x_0, y_0), then

$$x(t) = x_0 + v_{x0}t, \qquad y(t) = y_0 - \tfrac{1}{2}gt^2. \tag{6.4}$$

These are two coordinate descriptions of one curved path. The horizontal motion continues while the ball falls. A faster horizontal launch changes where it lands, but in this simple model it does not change how quickly it loses height.

Worked example 6.1 (A short launch)**.** A ball leaves a table 5 m above the floor at 3 m/s horizontally. Use $g = 10\,\mathrm{m/s^2}$. The vertical drop after one second is $gt^2/2 = 5\,\mathrm{m}$, so it reaches the floor then. During the same second it moves $3(1) = 3\,\mathrm{m}$ horizontally. Just before landing, its velocity components are $(3, -10)\,\mathrm{m/s}$. The negative vertical component means downward.

The formulas depend on constant downward acceleration and negligible drag. With substantial air resistance, the force and acceleration change with velocity, so we would need a different equation.

6.4 Direction and a first vector diagram

For a moving ball, draw the position arrow from the chosen origin, the velocity arrow tangent to its path, and the acceleration arrow downward. Watching all three at once reveals why a curved path does not require acceleration to point along the path. If you later add drag, record which arrows change compared with the ideal model.

The dot product gives a way to ask how much one vector points along another. For vectors $\boldsymbol{A} = (A_x, A_y)$ and $\boldsymbol{B} = (B_x, B_y)$, it is $\boldsymbol{A} \cdot \boldsymbol{B} = A_x B_x + A_y B_y$. A zero dot product means perpendicular directions. We will use it when a force acts at an angle to a displacement. The cross product, needed later for rotation, will be introduced with torque.

Exercises

An object begins with horizontal velocity $3\,\mathrm{m/s}$ and zero vertical velocity. With vertical acceleration $-10\,\mathrm{m/s^2}$, find its velocity components after $2\,\mathrm{s}$. Sketch the velocity and acceleration arrows. Explain why the horizontal component did not change in this model.

Answers are in Appendix A.

Chapter 7

First forces and oscillations

The motion chapters told us where an object goes once its acceleration is known. This chapter asks where that acceleration comes from. We will begin with a single force that stays nearly constant, then add a second force from a surface. At the end we will meet a spring, whose force changes as the object moves.

7.1 A force changes motion

Imagine a small cart on a level track. Push it to the right and it speeds up to the right. Push it to the left while it is moving right and it slows down. A force is a push or pull with a direction, so we draw it as an arrow. If several forces act, add their arrows to find the *net force*. Forces of equal size in opposite directions cancel.

For an object with constant mass m, Newton's second law connects net force to acceleration $\boldsymbol{a}$:

$$\boldsymbol{F}_{\text{net}} = m\boldsymbol{a}. \tag{7.1}$$

Mass measures how hard it is to change the object's velocity. A newton is the force that gives a 1 kg mass an acceleration of $1\,\text{m/s}^2$. If the net force is zero, the acceleration is zero; the object may still be moving at constant velocity.

Before using the equation, draw the object alone. Add one arrow for each force acting *on* it, choose a positive direction, and translate the arrows into signed numbers. This is a force diagram. It separates the physical choice of forces from the algebra used afterward.

Worked example 7.1 (A horizontal push)**.** A 2 kg cart is pushed right with 8 N while a resistive force of 2 N acts left. Choose right as positive. The net force is $8 - 2 = 6\,\text{N}$, so $a = F_{\text{net}}/m = 3\,\text{m/s}^2$ right. If the forces had equal sizes, the cart's acceleration would be zero, but a moving cart would not instantly stop.

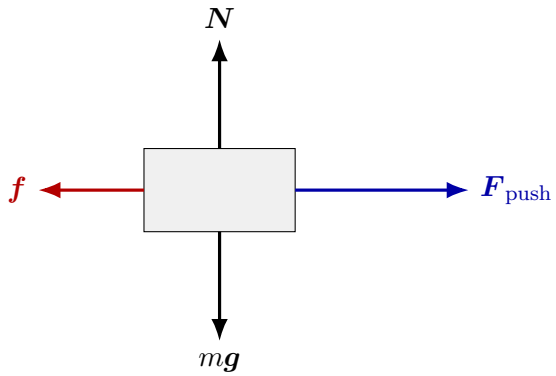


Figure 7.1: A free-body diagram contains forces acting on the chosen object. The horizontal imbalance determines horizontal acceleration; the vertical forces cancel in this example.

7.2 Gravity near Earth

Let a small ball fall near Earth's surface. Earth's gravitational pull on it points downward. Over a modest change in height, we can treat the pull as having nearly constant magnitude mg, where $g \approx 9.8\,\mathrm{m/s^2}$. The value g is an acceleration; multiplying it by mass gives a force.

Choose upward as the positive vertical direction. If we ignore air resistance while the ball is in flight, its only force is gravity, so

$$F_{\mathrm{net,z}} = -mg, \qquad ma_z = -mg, \qquad a_z = -g. \tag{7.2}$$

The minus sign follows from our chosen direction. The mass cancels in this model, so a heavy ball and a light ball released together have the same acceleration if air resistance is negligible. In everyday air, a feather behaves differently because another force matters.

If a ball starts at rest and falls for one second, its velocity changes downward by about $9.8\,\mathrm{m/s}$. The distance fallen during that second is about $4.9\,\mathrm{m}$: it begins slowly and speeds up, so multiplying its final speed by the whole time would overestimate the distance. A later mechanics chapter will calculate this distance by adding the changing velocity over time.

7.3 What changes when the ball rests on a table?

Gravity still pulls a ball downward after it lands on a table. The ball does not keep accelerating through the table because the table pushes up. That upward contact force is called the *normal force*; normal means perpendicular to the surface. For a ball at rest on a level table, zero vertical acceleration means the upward normal force balances the downward weight: $N - mg = 0$.

The equality $N = mg$ belongs to this particular situation. If the support moves up or down with acceleration, or if the surface is tilted, we must draw the forces again and solve the new balance. The force diagram tells us which rule is justified.

7.4 A different force: the spring

Now attach a cart to a spring on a horizontal track. Mark the cart's equilibrium position, where the spring is neither stretched nor compressed. Call the cart's displacement from that position x, and choose right as positive. Pull the cart to the right: the spring pulls it left. Push the cart to the left: the spring pushes it right. The direction of the spring force therefore changes when x changes sign.

For a spring used within its elastic range, a useful measured rule is Hooke's law:

$$F_x = -kx, \tag{7.3}$$

where $k > 0$ is stiffness. A stiffer spring needs more force for the same displacement. The unit of k is newtons per meter. The minus sign encodes the observed return toward equilibrium: positive displacement gives negative force, and negative displacement gives positive force.

If the cart is released from one side, the spring accelerates it toward equilibrium. When it reaches equilibrium it is still moving, so it passes through; on the other side the spring force reverses and slows it. This repeated motion is an oscillation. Its acceleration is not constant because the spring force changes with position. We will derive the full mathematical motion after practicing Newton's law with constant forces.

7.5 Energy as a preview

A moving cart has kinetic energy, which grows with its speed. A raised ball or stretched spring can store energy that later becomes motion. In the falling-ball model, gravitational potential energy decreases as kinetic energy increases. In the spring model, stored spring energy is greatest at the turning points, while kinetic energy is greatest near equilibrium.

For now, use energy as a check on the story: if a ball falls without significant drag, it should speed up; if a spring cart reaches a turning point, its speed should be zero there. The later mechanics chapter will calculate these changes from work done by a force and derive when mechanical energy is conserved.

7.6 Checkpoint before the deeper pass

You should now be able to choose one object, draw the forces acting on it, choose signs, and use their sum to find its acceleration. You should also be able to explain why gravity is approximately constant over a short fall, why a table can balance weight, and why the direction of a spring force changes as the cart crosses equilibrium.

Exercises

A 2 kg cart experiences a net horizontal force of 6 N. Find its acceleration and direction. A ball at rest on a level table has weight 5 N; draw the two vertical forces on it. Finally, for a spring cart displaced left of equilibrium, state the direction of the spring force and explain your sign choice.

Answers are in Appendix A.

Chapter 8

Newtonian motion and conservation laws

The first chapters taught us how to describe motion. Now we want to predict it. We will start with one object and one force, then add contact forces and finally forces that change during motion. Only after those examples will we look for shortcuts using energy and momentum.

8.1 What a force diagram tells us

Put a cart on a straight track and choose right as the positive direction. A force is a push or pull on the cart. Forces are vectors, so a push to the right and a push of the same size to the left add to zero. We write their sum as $\boldsymbol{F}_{\text{net}}$. The word *net* matters: the cart responds to the sum of all forces acting on it, not to one selected force.

Newton's first law says that, in an inertial frame, an object with zero net force keeps a constant velocity. It may remain at rest, or it may keep moving in a straight line. An inertial frame is one in which this rule works without adding fictitious forces. A laboratory fixed to Earth is often a good approximation for short, ordinary motions, although Earth's rotation can matter in more precise problems.

For an object whose mass m stays constant, Newton's second law is

$$\boldsymbol{F}_{\text{net}} = m\boldsymbol{a}, \tag{8.1}$$

where $\boldsymbol{a}$ is the rate of change of velocity. Force has the SI unit newton, with $1\,\text{N} = 1\,\text{kg}\,\text{m/s}^2$. If the same net force acts on twice the mass, the acceleration is half as large. If all forces cancel, the acceleration is zero, which agrees with the first law.

To use the equation, first choose the object. Then draw only forces *on* that object, choose axes, and add force components along each axis. Newton's third law concerns a different pair of arrows: if object A pushes object B, B pushes A with an equal and opposite force. The two third-law forces act on different objects, so they do not cancel in one object's force diagram.

Worked example 8.1 (One cart, two horizontal forces)**.** A 2 kg cart is pushed right with 8 N, while a resistive force of 2 N acts left. With right positive, $F_{\text{net,x}} = 8 - 2 = 6\,\text{N}$. Equation (8.1) gives $a_x = 6/2 = 3\,\text{m/s}^2$. If the push and resistance were both 8 N, the net force and acceleration would be zero. The cart could still move at constant velocity.

8.2 One force at a time: falling near Earth

For a small object close to Earth's surface, gravity pulls downward with approximately constant magnitude mg, where $g \approx 9.8\,\text{m/s}^2$. This is a model with a limited range: far from Earth, the strength and direction of gravity change. For now, choose an upward vertical coordinate z. The gravitational force component is $F_{g,z} = -mg$ because it points opposite to positive z.

If the object is released and air resistance is small enough to ignore, gravity is the only force. Newton's law becomes

$$ma_z = -mg, \qquad a_z = -g. \tag{8.2}$$

The mass cancels. In this ideal model, two objects released with the same initial velocity have the same acceleration even if their masses differ. Air resistance can make their observed motions different; we will add it only after solving the simpler case.

Acceleration tells us how velocity changes. Since $a_z = \mathrm{d}v_z/\mathrm{d}t = -g$, a short time $\mathrm{d}t$ changes the velocity by $-g\mathrm{d}t$. Add those changes from $t = 0$ to a later time t:

$$v_z(t) - v_{z0} = \int_0^t (-g)\,\mathrm{d}s = -gt, \qquad v_z(t) = v_{z0} - gt. \tag{8.3}$$

Here v_{z0} is the velocity at the starting time. To find position, add the velocity over time. Because the velocity changes, we must use the expression we just found:

$$z(t) - z_0 = \int_0^t v_z(s)\,\mathrm{d}s = \int_0^t (v_{z0} - gs)\,\mathrm{d}s \tag{8.4}$$

$$= v_{z0}t - \tfrac{1}{2}gt^2. \tag{8.5}$$

Thus $z(t) = z_0 + v_{z0}t - gt^2/2$. The factor $1/2$ comes from adding a velocity that changes linearly with time. The starting position z_0 and velocity v_{z0} are needed because the law by itself describes many possible falls.

Worked example 8.2 (A ball released from a balcony)**.** Release a ball from rest at height 20 m above the ground. Set $z_0 = 20\,\text{m}$, $v_{z0} = 0$, and use $g = 10\,\text{m/s}^2$ for an estimate. The model gives $z(t) = 20 - 5t^2$ in meters when t is in seconds. The ball reaches $z = 0$ when $t^2 = 4\,\text{s}^2$, so the positive time is 2 s. Its velocity then is $v_z = -10(2) = -20\,\text{m/s}$: the negative sign means downward. The prediction assumes no significant air resistance and a balcony low enough that g is effectively constant.

We have derived a constant-acceleration formula for one particular force. The same two integration steps work for any constant net force. They do not apply unchanged when the force varies with position or velocity.

8.3 Adding a surface: normal force and friction

Place the ball on a horizontal table. Gravity still pulls down, but the table pushes up on the ball. This contact force is called the *normal force*; normal means perpendicular to the surface. If the ball has no vertical acceleration, vertical force balance gives $N - mg = 0$, hence $N = mg$. This equality follows from this situation's force balance. On an accelerating elevator or an inclined surface, N generally has a different value.

A surface can also resist motion along itself. In a simple dry-friction model, the magnitude of kinetic friction while two surfaces slide is $F_k = \mu_k N$, where μ_k is a dimensionless coefficient. The friction force points opposite the relative sliding direction. Static friction is different: before sliding begins, its magnitude adjusts as needed up to a maximum $\mu_s N$. The inequality $|F_s| \le \mu_s N$ is a limit, not a command to set static friction equal to its maximum every time.

Worked example 8.3 (A box that has not started moving)**.** A 5 kg box rests on a level floor. Take $g = 10\,\mathrm{m/s^2}$, so $N = 50\,\mathrm{N}$. Suppose $\mu_s = 0.4$. Static friction can oppose a horizontal push up to $\mu_s N = 20\,\mathrm{N}$. If you push with 10 N, friction is 10 N in the opposite direction and the box stays still. If you push with 25 N, this static model cannot keep the box at rest. After motion begins, use a kinetic-friction model with its own coefficient.

This example shows why we determine forces from a specific situation. A list of force formulas does not tell us which forces act, what direction they have, or whether a contact constraint fixes their size.

8.4 When the force changes: a spring

So far the net force was constant or was found from a static balance. A spring introduces a new kind of question. Push a cart attached to a spring away from its equilibrium position. The farther the cart is displaced, the stronger the spring pulls it back. Near equilibrium, many springs follow Hooke's law,

$$F_x = -kx, \tag{8.6}$$

where x is displacement from equilibrium and $k > 0$ is the spring stiffness. If $x > 0$, the force is negative; if $x < 0$, the force is positive. In both cases it points toward equilibrium. The unit of k is N/m because force equals stiffness times displacement.

Newton's law now reads $m\ddot{x} = -kx$. Unlike the falling-ball example, acceleration is not constant: it changes whenever x changes. Rearranging gives

$$\ddot{x} + \frac{k}{m}x = 0. \tag{8.7}$$

To see what sort of function can satisfy this rule, recall that the second derivative of a cosine is the negative of the original cosine times the square of its frequency. Try $x(t) = A\cos(\omega t + \phi)$. Its second derivative is $-\omega^2 x(t)$. Substitution into (8.7) works when $\omega^2 = k/m$, or $\omega = \sqrt{k/m}$. The

constants A and ϕ are fixed by the starting position and velocity. This is a check of the proposed solution, not a claim that every changing-force problem can be solved by guessing a cosine.

A spring therefore adds a new step to our method: write the force in terms of position, use Newton's law to form a differential equation, and test a solution against both that equation and the initial conditions. More complicated restoring forces may only behave like $-kx$ close to equilibrium.

8.5 Work: force acting through a distance

Sometimes we need only a final speed, not a position at every time. Imagine the falling ball again. Gravity acts over the distance traveled, and the ball speeds up. We call the contribution of a force along a displacement *work*. For a small displacement $\mathrm{d}\boldsymbol{r}$, the work is $\mathrm{d}W = \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r}$. The dot product keeps the component of force along the motion. A perpendicular force does no work over that displacement.

For a finite path, add the small contributions:

$$W = \int_{\text{path}} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r}. \tag{8.8}$$

Work is measured in joules, with $1\,\mathrm{J} = 1\,\mathrm{N\,m}$. A $5\,\mathrm{N}$ force pointing along a straight $2\,\mathrm{m}$ displacement does $10\,\mathrm{J}$ of work. The same force pointing directly opposite does $-10\,\mathrm{J}$.

The connection to speed follows from Newton's law. For a constant mass, take the dot product of $m\dot{\boldsymbol{v}} = \boldsymbol{F}_{\text{net}}$ with $\boldsymbol{v}$. Since $\boldsymbol{v} \cdot \dot{\boldsymbol{v}} = \frac{1}{2}\mathrm{d}(v^2)/\mathrm{d}t$, we get

$$\boldsymbol{F}_{\text{net}} \cdot \boldsymbol{v} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\tfrac{1}{2}mv^2\right). \tag{8.9}$$

The quantity $K = \frac{1}{2}mv^2$ is kinetic energy. Multiply by $\mathrm{d}t$, use $\boldsymbol{v}\mathrm{d}t = \mathrm{d}\boldsymbol{r}$, and add along the path:

$$W_{\text{net}} = K_f - K_i = \Delta K. \tag{8.10}$$

This work–kinetic-energy theorem applies even when the net force changes during the motion, provided we compute its work correctly.

Worked example 8.4 (Speed after falling through a height)**.** A ball descends through height h with negligible air resistance. Gravity and displacement both point down, so gravity does positive work mgh. Equation (8.10) gives $mgh = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$, hence $v_f^2 = v_i^2 + 2gh$. We obtain the final speed without first finding how long the fall took. The formula says nothing by itself about the direction of velocity or the path followed.

8.6 Potential energy and conservation

For gravity near Earth, lifting an object by height h requires work against gravity. We can store this bookkeeping in a gravitational potential energy $V_g = mgz$, where z is measured upward. If the

object moves upward, gravity does negative work and V_g increases by the same amount. If gravity is the only force doing work, the sum $K + V_g$ remains constant.

The spring has its own potential energy. Because its force grows linearly with displacement, stretching it from $x = 0$ to x requires

$$V_s(x) - V_s(0) = \int_0^x ks \, \mathrm{d}s = \tfrac{1}{2}kx^2. \tag{8.11}$$

Choose $V_s(0) = 0$. The spring's stored energy is positive whether it is stretched or compressed. At a turning point the cart is momentarily at rest and energy is stored in the spring. At equilibrium the spring energy is least and the cart's speed is greatest.

A force is called conservative when its work between two points depends only on those endpoints. We can then define a potential energy V so that its work is $-\Delta V$. Combining this with (8.10) gives

$$\Delta K = -\Delta V, \qquad E = K + V = \text{constant} \tag{8.12}$$

when no nonconservative force does work and the potential has no explicit time dependence. In several dimensions, the local relation is $\boldsymbol{F} = -\boldsymbol{\nabla} V$. Friction usually converts mechanical energy into thermal energy, so $K + V$ for the moving object decreases even though total energy is still conserved.

Power tells us how quickly work is done. Its instantaneous value is $P = \mathrm{d}W/\mathrm{d}t = \boldsymbol{F} \cdot \boldsymbol{v}$, measured in watts, with $1\,\mathrm{W} = 1\,\mathrm{J/s}$.

8.7 Momentum and impulse

Energy answers many speed questions. Momentum is useful when a force acts for a short time, as in a kick or collision. For a constant-mass particle, momentum is $\boldsymbol{p} = m\boldsymbol{v}$, and Newton's second law is $\boldsymbol{F}_{\text{net}} = \mathrm{d}\boldsymbol{p}/\mathrm{d}t$. Add the force over a time interval:

$$\boldsymbol{J} = \int_{t_i}^{t_f} \boldsymbol{F}_{\text{net}} \, \mathrm{d}t = \boldsymbol{p}_f - \boldsymbol{p}_i. \tag{8.13}$$

The vector $\boldsymbol{J}$ is impulse. Its direction matters. A force opposite the initial velocity decreases the momentum in that direction; a sufficiently large impulse can reverse it.

Worked example 8.5 (A short push)**.** A $0.5\,\mathrm{kg}$ cart starts at rest. A nearly constant $4\,\mathrm{N}$ force acts right for $0.25\,\mathrm{s}$. The impulse is $J = (4)(0.25) = 1\,\mathrm{N\,s}$ to the right. The final momentum is $1\,\mathrm{kg\,m/s}$, so the final speed is $1/0.5 = 2\,\mathrm{m/s}$. We did not need the cart's position during the push.

For several interacting objects, add their individual momenta. Forces that the objects exert on each other cancel in pairs in the total, so only external forces change the system's total momentum. This is why we must say what is included in the system before claiming momentum is conserved.

8.8 A collision, step by step

Consider two carts moving on a straight track. Cart 1 has mass m_1 and initial velocity v_1; cart 2 has mass m_2 and initial velocity v_2. Suppose they stick after colliding. During a brief collision, assume the total external impulse is negligible. The total momentum just before and after must then agree:

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f. \tag{8.14}$$

Solve this one equation for their shared final velocity:

$$v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}. \tag{8.15}$$

The velocities are signed according to the direction chosen for the track. Notice that we have not assumed kinetic energy is conserved. Sticking usually converts some kinetic energy into deformation, sound, and thermal energy.

Worked example 8.6 (Two carts stick)**.** A $1\,\mathrm{kg}$ cart moves right at $4\,\mathrm{m/s}$ toward a stationary $1\,\mathrm{kg}$ cart. Equation (8.15) gives $v_f = 2\,\mathrm{m/s}$ right. Before the collision, $K_i = \tfrac{1}{2}(1)(4^2) = 8\,\mathrm{J}$. Afterward, $K_f = \tfrac{1}{2}(2)(2^2) = 4\,\mathrm{J}$. The missing $4\,\mathrm{J}$ of kinetic energy has changed form; momentum remains $4\,\mathrm{kg\,m/s}$ throughout the isolated collision.

For a more general one-dimensional collision, it is helpful to separate motion of the center of mass from motion of the carts relative to each other. Let $M = m_1 + m_2$, $V = (m_1 v_1 + m_2 v_2)/M$, and $u = v_1 - v_2$. Solving these definitions gives $v_1 = V + (m_2/M)u$ and $v_2 = V - (m_1/M)u$. Substitution into the kinetic energy yields

$$K = \tfrac{1}{2} M V^2 + \tfrac{1}{2} \mu u^2, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}. \tag{8.16}$$

The quantity μ is called reduced mass. In a sticking collision the final relative velocity is zero, so the initial relative-motion term is the kinetic energy converted into other forms. In an elastic collision, by definition, total kinetic energy is conserved as well as total momentum.

8.9 Rotation: angular momentum and torque

Linear momentum tracks motion along a line. To study motion around a chosen point, we also need the position relative to that point. Define angular momentum by $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$, where $\times$ is the cross product. Its magnitude depends on the component of momentum perpendicular to $\boldsymbol{r}$. Motion directly toward or away from the point has zero angular momentum about it.

A force can change angular momentum through torque $\boldsymbol{\tau} = \boldsymbol{r} \times \boldsymbol{F}$. Differentiate the definition of $\boldsymbol{L}$:

$$\frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \boldsymbol{v} \times m\boldsymbol{v} + \boldsymbol{r} \times \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} \tag{8.17}$$

$$= \boldsymbol{0} + \boldsymbol{r} \times \boldsymbol{F} = \boldsymbol{\tau}. \tag{8.18}$$

The first cross product vanishes because the two factors point in the same direction. If a force always points directly toward a fixed center, its torque about that center is zero and angular momentum is conserved.

For motion in a plane under such a central force, a short displacement sweeps an area $\mathrm{d}A = \frac{1}{2}|\boldsymbol{r} \times \boldsymbol{v}|\mathrm{d}t$. Because $L = m|\boldsymbol{r} \times \boldsymbol{v}|$, the swept area per time is $\mathrm{d}A/\mathrm{d}t = L/(2m)$. This explains why a planet moves faster near the Sun: with a shorter radius, it must travel farther along its orbit in the same time to sweep an equal area.

8.10 A changing force from motion: drag

Return to a falling object, but now allow air or fluid resistance. At sufficiently slow speeds in some viscous flows, a useful model is drag opposite the velocity with magnitude bv, where $b > 0$. Choose down as positive. Gravity contributes $+mg$ and drag contributes $-bv$, so

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = mg - bv. \tag{8.19}$$

Here the acceleration changes as the object speeds up. It is therefore incorrect to put a fixed acceleration into the earlier constant-acceleration formulas.

Start by finding the speed at which the two forces balance: $mg - bv_\infty = 0$, hence $v_\infty = mg/b$. This is the terminal speed in this model. To see how the object approaches it, set $w = v - v_\infty$. Equation (8.19) becomes $\mathrm{d}w/\mathrm{d}t = -(b/m)w$. A quantity whose rate of decrease is proportional to itself decays exponentially:

$$w(t) = w(0)\mathrm{e}^{-bt/m}, \qquad v(t) = v_\infty + (v_0 - v_\infty)\mathrm{e}^{-bt/m}. \tag{8.20}$$

The time scale is $\tau = m/b$. After one τ, the difference between the actual speed and terminal speed has shrunk by a factor e. At higher speeds, a quadratic drag model may be more suitable; its coefficient depends on shape and flow conditions.

8.11 An optional extension: a rocket changes mass

The previous examples kept the object's mass constant. A rocket expels fuel, so the rocket alone is an open system. We must count momentum carried away by the exhaust. Let the rocket have mass m and speed v, and let the exhaust leave backward at speed u relative to the rocket. Over a short interval, the rocket loses mass, so $\mathrm{d}m < 0$ and the expelled mass is $-\mathrm{d}m > 0$.

With no external force, momentum before and after the short interval gives, to first order in small changes,

$$mv = (m + \mathrm{d}m)(v + \mathrm{d}v) + (-\mathrm{d}m)(v - u), \tag{8.21}$$

$$0 = m\mathrm{d}v + u\mathrm{d}m. \tag{8.22}$$

Divide by m and integrate from initial mass m_i to final mass $m_f < m_i$:

$$v_f - v_i = -u \int_{m_i}^{m_f} \frac{\mathrm{d}m}{m} = u \ln \frac{m_i}{m_f}. \tag{8.23}$$

The speed increases as mass is expelled. Applying $\mathrm{d}(mv)/\mathrm{d}t = 0$ to the rocket alone would omit the exhaust momentum and give a wrong result. Gravity and drag can be added later as external forces; the equation above describes an ideal rocket in their absence.

Check the chain of reasoning

Before moving on, try to tell the story of each model without looking at the equations: identify the object and forces, choose a sign convention, write Newton's law, and decide whether the force stays constant. Then choose a tool that answers the question being asked: integrate acceleration for a trajectory, calculate work for a speed, or calculate impulse for a momentum change.

Exercises

A ball is thrown upward at $10\,\mathrm{m/s}$ with $g = 10\,\mathrm{m/s^2}$. Using upward positive, find its velocity after one second and its height change during that second. A $2\,\mathrm{kg}$ cart receives a $6\,\mathrm{N\,s}$ impulse to the left: find its velocity change. Finally, explain why the constant-acceleration formula cannot describe a falling object approaching terminal speed.

Answers are in Appendix A.

Part III

Classical Mechanics

Chapter 9

Gravitation, orbits, and rigid bodies

The conserved quantities from Newtonian mechanics now meet geometry. Orbits use energy and angular momentum to reduce a two-body problem, while rigid bodies show why a single rotational coordinate is sometimes not enough. These are the first places where the shape of a system becomes part of its dynamics. Gravity and rotation bring the conservation laws together. An orbit is a continuously falling trajectory, shaped by both attraction and angular momentum. A rotating body is a collection of moving pieces whose geometry matters as much as its total mass. We will simplify both problems by identifying what is conserved and choosing coordinates that expose it.

9.1 Distance from a center

The gravity that made a ball fall near Earth also holds a satellite in orbit. To describe a satellite, choose the center of Earth as an origin. Its separation from that center is r, and $\hat{\boldsymbol{r}}$ is the unit arrow pointing outward. Gravity points inward, opposite to $\hat{\boldsymbol{r}}$. This language says what direction a force has before we choose an x or y axis.

For a spherical source, every point at the same distance r is equivalent. A sphere of radius r has area $4\pi r^2$, which will matter when we later ask how a field spreads. A thin spherical shell of thickness $\mathrm{d}r$ has volume approximately $4\pi r^2 \mathrm{d}r$. We need no further coordinate formulas yet; the orbit calculation below uses only r and an angle in its orbital plane.

A potential energy $V(r)$ tells us how energy changes as we move radially. If a small outward displacement is $\mathrm{d}r$, the radial force does work $F_r \mathrm{d}r = -\mathrm{d}V$, so

$$F_r = -\frac{\mathrm{d}V}{\mathrm{d}r}. \tag{9.1}$$

This is the radial version of the work–energy idea from the previous chapter. When we later allow a potential to vary in several directions, its corresponding derivative is called the gradient, $\boldsymbol{\nabla} V$.

9.2 Newtonian gravity and potential energy

The near-Earth force mg was a good approximation over a short fall. To discuss a satellite, the separation from Earth changes enough that we need a law valid at different distances. For two small masses, Newton's gravitational law says the attractive force has magnitude GMm/r^2. Here M is the source mass, m is the moving mass, r is their separation, and G is the gravitational constant. With the outward unit vector $\hat{\boldsymbol{r}}$, the force is

$$\boldsymbol{F} = -\frac{GMm}{r^2}\hat{\boldsymbol{r}}. \tag{9.2}$$

The negative sign says the force points inward. Doubling separation makes the force one quarter as large. Close to Earth's surface, where r changes little compared with Earth's radius R, we may set $g \approx GM/R^2$ and recover $F \approx mg$.

Now ask how much work gravity does when a mass moves radially. From the previous chapter, $F_r = -\mathrm{d}V/\mathrm{d}r$. Choose gravitational potential energy to be zero infinitely far away, where this isolated source's force approaches zero. Then

$$\frac{\mathrm{d}V}{\mathrm{d}r} = \frac{GMm}{r^2}, \tag{9.3}$$

$$V(r) - V(\infty) = \int_{\infty}^{r} \frac{GMm}{s^2}\,\mathrm{d}s = -\frac{GMm}{r}. \tag{9.4}$$

Potential energy is negative at finite r under this convention. Moving the mass outward raises V toward zero, so energy must be supplied unless the mass already has enough kinetic energy. The zero point is a choice; differences in potential energy are what enter work and conservation equations.

Worked example 9.1 (Comparing two orbital distances)**.** At radius r, the potential energy is $-GMm/r$. At radius $2r$, it is $-GMm/(2r)$, which is higher by $GMm/(2r)$. The force magnitude at $2r$ is one quarter of its value at r. These are different comparisons: force describes the local pull, while a potential difference describes the accumulated work between locations.

A gravitational field describes force per unit test mass, $\boldsymbol{g} = \boldsymbol{F}/m$. Similarly, the potential per unit mass is $\Phi = V/m$. The radial relation becomes $g_r = -\mathrm{d}\Phi/\mathrm{d}r$. When a potential depends on all three spatial directions, the vector of its spatial rates of change is the gradient $\boldsymbol{\nabla}\Phi$, and $\boldsymbol{g} = -\boldsymbol{\nabla}\Phi$.

For an extended source, imagine dividing its mass into many small pieces. A piece at $\boldsymbol{r}'$ with mass $\mathrm{d}M = \rho(\boldsymbol{r}')\mathrm{d}^3r'$ contributes $-G\mathrm{d}M/|\boldsymbol{r} - \boldsymbol{r}'|$ to the potential at $\boldsymbol{r}$. Adding all pieces gives

$$\Phi(\boldsymbol{r}) = -G\int \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}^3r'. \tag{9.5}$$

Here ρ is mass per volume and d^3r' is a small source volume. We add potentials as scalars; then a spatial derivative gives the field. For a spherically symmetric body, contributions outside it combine as though all its mass were at the center. Inside a spherical shell, contributions cancel to give zero net field. Those symmetry results require summing the source contributions, not treating every interior point as an exterior point-mass problem.

9.3 The two-body problem and effective potential

The two masses both move, but their separation is what determines their mutual force. Separate out the center of mass and describe the relative motion in its orbital plane using r and ϕ. Conservation of angular momentum then lets us remove $\dot{\phi}$ from the energy, reducing the radial problem to motion in an effective one-dimensional potential. In the two-body reduction, μ is reduced mass, ℓ angular-momentum magnitude, E total mechanical energy, and e orbital eccentricity.

Separate center-of-mass motion from relative motion. The relative particle has reduced mass μ and potential $-Gm_1m_2/r$. With angular momentum magnitude ℓ,

$$E = \tfrac{1}{2}\mu\dot{r}^2 + V_{\text{eff}}(r), \qquad V_{\text{eff}} = -\frac{Gm_1m_2}{r} + \frac{\ell^2}{2\mu r^2}. \tag{9.6}$$

A circular orbit is an extremum of V_{eff}; a local minimum gives radial stability. The orbit is a conic,

$$r(\phi) = \frac{p}{1 + e\cos\phi}, \quad p = \frac{\ell^2}{\mu G m_1 m_2}, \quad e^2 = 1 + \frac{2E\ell^2}{\mu(Gm_1m_2)^2}. \tag{9.7}$$

Bound orbits have $E < 0$ and $0 \le e < 1$. For semimajor axis a,

$$E = -\frac{Gm_1m_2}{2a}, \qquad \tau^2 = \frac{4\pi^2 a^3}{G(m_1 + m_2)}. \tag{9.8}$$

These are Kepler's laws in dynamical form. For a light satellite around mass M, $v_c = \sqrt{GM/r}$ and $v_{\text{esc}} = \sqrt{2GM/r}$.

Worked example 9.2 (Why an orbiting astronaut is weightless)**.** For a circular orbit, $v^2/r = GM/r^2$. Both spacecraft and astronaut accelerate gravitationally together. A scale reads the contact force, which is nearly zero; gravity itself is not zero. Small differences in gravity across the spacecraft produce tidal accelerations.

Write $\kappa = Gm_1m_2$. The relative kinetic energy is $T = \tfrac{1}{2}\mu(\dot{r}^2 + r^2\dot{\phi}^2)$ and the angular momentum is $\ell = \mu r^2\dot{\phi}$. Substitution gives the centrifugal contribution $\ell^2/(2\mu r^2)$ in V_{eff}. It accounts for transverse kinetic energy, not an additional gravitational interaction.

To obtain the orbit shape, use $u = 1/r$ and primes for derivatives with respect to ϕ. Angular momentum gives $\dot{\phi} = \ell u^2/\mu$, so $\dot{r} = -(\ell/\mu)u'$ and $\ddot{r} = -(\ell^2/\mu^2)u^2u''$. The radial force equation becomes

$$\mu(\ddot{r} - r\dot{\phi}^2) = -\kappa u^2, \tag{9.9}$$

$$u'' + u = \frac{\mu\kappa}{\ell^2}, \tag{9.10}$$

$$u(\phi) = \frac{\mu\kappa}{\ell^2}[1 + e\cos\phi]. \tag{9.11}$$

We chose $\phi = 0$ at closest approach. Taking the reciprocal produces the conic equation. Substituting u and u' into

$$E = \frac{\ell^2}{2\mu}(u'^2 + u^2) - \kappa u = \frac{\mu\kappa^2}{2\ell^2}(e^2 - 1) \tag{9.12}$$

explains the relation between energy and eccentricity. Negative energy means $e < 1$, a bound ellipse. For a circular satellite orbit, equating mv^2/r to GMm/r^2 gives v_c; setting total energy to zero gives v_{esc}. These two speeds answer different physical questions.

9.4 Why an axis changes the moment of inertia

A rotating body's response depends on the axis we choose. Imagine a long rod. Spinning it around its long axis keeps every mass element close to that axis; tumbling it end over end moves many elements much farther. The rod's mass is unchanged, but its resistance to the two rotations is different.

For a chosen unit axis $\hat{\boldsymbol{n}}$, a small mass element $\mathrm{d}m$ at perpendicular distance $r_\perp$ contributes $r_\perp^2\,\mathrm{d}m$ to the moment of inertia. Adding all the elements gives $I_{\hat{n}} = \int r_\perp^2\,\mathrm{d}m$. The square makes distant mass count strongly. We will derive this expression again from rotational kinetic energy in the next section.

When the axis can point in any direction, a single value of I cannot describe every rotation. We collect the information in an inertia tensor I_{ij}. Its components change when the coordinate axes rotate, but the physical body and its rotational energy do not. Special directions, called principal axes, make the tensor diagonal. Along each one, angular momentum and angular velocity point in the same direction; for a general axis they need not.

Worked example 9.3 (Two masses and two axes)**.** Place two equal masses m at $x = \pm a$. About the z axis through their midpoint, each mass is distance a away, so $I_z = 2ma^2$. About the x axis joining them, each mass lies on the axis and the idealized point-mass contribution is zero. The comparison shows why quoting "the moment of inertia" without an axis is incomplete.

9.5 Rigid-body rotation

Two objects with equal mass can resist angular acceleration differently because their mass is distributed differently around the axis. Begin with each piece's speed, add its kinetic energy, and let the coefficient of $\omega^2/2$ define the moment of inertia. In three dimensions the axis itself can change, which is why a tensor replaces a single number. In rotation, I_{ij} is the inertia tensor, $\boldsymbol{\omega}$ angular velocity, and $\boldsymbol{L}$ angular momentum; tensor components depend on axes, while the physical rotation does not.

The inertia tensor about an origin is

$$I_{ij} = \int \rho(\boldsymbol{r})(r^2\delta_{ij} - x_i x_j)\mathrm{d}V, \quad \boldsymbol{L} = I\boldsymbol{\omega}, \quad T_{\mathrm{rot}} = \tfrac{1}{2}\boldsymbol{\omega}\cdot I\boldsymbol{\omega}. \tag{9.13}$$

Along a principal axis, $L = I\omega$. The parallel-axis theorem is $I = I_{\mathrm{CM}} + Md^2$. In body-fixed principal axes, Euler's equations include

$$I_1\dot{\omega}_1 + (I_3 - I_2)\omega_2\omega_3 = \tau_1, \tag{9.14}$$

with cyclic permutations. Torque-free rotation about the largest or smallest principal inertia is stable to small perturbations; rotation about the intermediate one is unstable.

Rolling without slipping imposes $v_{\mathrm{CM}} = R\omega$. A body descending height h satisfies $mgh = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I_{\mathrm{CM}}v^2/R^2$ if dissipative losses vanish. Static friction can enforce rolling without dissipating mechanical energy on a stationary rigid surface.

For rotation around a fixed axis, a mass element at perpendicular distance $r_\perp$ moves with speed $v = \omega r_\perp$. Thus

$$T = \int \tfrac{1}{2} v^2 \mathrm{d}m = \tfrac{1}{2}\omega^2 \int r_\perp^2 \mathrm{d}m, \qquad I = \int r_\perp^2 \mathrm{d}m. \tag{9.15}$$

For a rolling object write $I_{\mathrm{CM}} = \alpha m R^2$. Conservation of energy then gives

$$mgh = \tfrac{1}{2} m v^2 (1 + \alpha), \qquad v = \sqrt{\frac{2gh}{1+\alpha}}. \tag{9.16}$$

A larger α leaves more of the available energy in rotation and less in center-of-mass translation. This lets you compare shapes before inserting their particular moments of inertia.

9.6 Accelerated frames

If you describe motion from a turning platform, the coordinate axes move while the object moves. An apparent acceleration can therefore come from the frame itself. The safe way to find the extra terms is to differentiate vectors while explicitly accounting for the changing basis.

For axes rotating with $\boldsymbol{\Omega}$ and origin acceleration $\boldsymbol{A}$, the equation of motion includes fictitious forces

$$\boldsymbol{F}_{\mathrm{fict}} = -m\boldsymbol{A} - 2m\boldsymbol{\Omega} \times \boldsymbol{v}' - m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r}) - m\dot{\boldsymbol{\Omega}} \times \boldsymbol{r}. \tag{9.17}$$

These are translational, Coriolis, centrifugal, and Euler forces. They account for using a noninertial frame; they do not describe additional interactions.

For any vector $\boldsymbol{A}$, the rotating-frame derivative obeys

$$\left(\frac{\mathrm{d}\boldsymbol{A}}{\mathrm{d}t}\right)_{\mathrm{inertial}} = \left(\frac{\mathrm{d}\boldsymbol{A}}{\mathrm{d}t}\right)_{\mathrm{rotating}} + \boldsymbol{\Omega} \times \boldsymbol{A}. \tag{9.18}$$

Apply this rule first to position and then to velocity. The result is

$$\boldsymbol{a} = \boldsymbol{A} + \boldsymbol{a}' + 2\boldsymbol{\Omega} \times \boldsymbol{v}' + \dot{\boldsymbol{\Omega}} \times \boldsymbol{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r}). \tag{9.19}$$

Insert this into $m\boldsymbol{a} = \boldsymbol{F}$ and move every frame term to the force side. That produces the fictitious forces above, including their signs. The factor two in the Coriolis term comes from two differentiations involving the rotating basis.

Exercises

If an orbiting body moves farther from the central mass, how do the gravitational force magnitude and potential energy change? Explain which conserved quantity helps reduce the two-body problem.

Answers are in Appendix A.

Chapter 10

Oscillations, coupled systems, and chaos

The spring has appeared as a first model, an energy system, and a rotating body's local approximation. We now ask how the pattern changes when several oscillators interact or when the restoring law is nonlinear. The normal-mode idea introduced here will later reappear in waves, molecules, circuits, fields, and quantum theory. Oscillations recur throughout physics because a smooth potential looks approximately quadratic near a stable minimum. This makes the spring much more than one mechanical example: it is a model for small departures from equilibrium. We will first solve that model, then ask what changes when energy leaks away, a driver supplies energy, or nonlinearity becomes too large to ignore.

10.1 Can units predict a pendulum period?

Imagine predicting how long a pendulum takes to swing. You could begin by solving its equation of motion, but you already know something valuable: the answer must have the dimensions of time. Our first task is to use that requirement to narrow the possible answers. Keep the distinction between a dimension and a unit in mind: a distance can be measured in meters or kilometers, while remaining a length in either case. M, L, T denote the base dimensions mass, length, and time; square brackets such as $[v]$ mean "dimensions of." The release angle θ is dimensionless, so it can change the numerical factor in a period formula.

A physical quantity is a number times a unit. Its dimension is independent of the chosen unit system. The mechanical base dimensions are mass M, length L, and time T; SI also includes electric current, temperature, amount of substance, and luminous intensity. For example,

$$[v] = LT^{-1}, \quad [a] = LT^{-2}, \quad [F] = MLT^{-2}, \quad [E] = ML^2T^{-2}. \tag{10.1}$$

Only quantities with equal dimensions may be added. Arguments of exponentials, logarithms, and trigonometric functions are dimensionless. Dimensional consistency is necessary but does not determine dimensionless coefficients.

Worked example 10.1 (Pendulum scaling)**.** Assume a small-angle pendulum period depends only on length ℓ and gravitational acceleration g. Writing $\tau = C\ell^a g^b$ gives $a + b = 0$ and $-2b = 1$, so $\tau = C\sqrt{\ell/g}$. Dynamics supplies $C = 2\pi$. At finite amplitude, the release angle is another dimensionless variable, and C depends on it.

Let us unpack the pendulum argument rather than treating the exponents as a guess. Because $[\ell] = L$ and $[g] = LT^{-2}$, the proposed expression has dimensions

$$[\ell^a g^b] = L^a (LT^{-2})^b = L^{a+b} T^{-2b}. \tag{10.2}$$

Matching this with $[\tau] = L^0 T^1$ gives two equations, one for each independent dimension. From $-2b = 1$ we get $b = -1/2$; then $a + b = 0$ gives $a = 1/2$. We have learned that quadrupling the length doubles the period and quadrupling gravity halves it. What we have not learned is the numerical factor C. That missing information is exactly what a dynamical model must supply. Later, the small-angle equation of motion will complete the argument.

10.2 Simple and damped oscillators

Locate the equilibrium before solving the motion. At x_0, the force vanishes, so $V'(x_0) = 0$. Taylor expansion leaves $V''(x_0)(x - x_0)^2/2$ as the leading changing part of the potential. A positive curvature makes the force point back toward equilibrium, which is the physical origin of harmonic motion. In the equations below, x denotes displacement from that equilibrium. ω_0 is the undamped natural angular frequency, β the damping rate, F_0 the drive amplitude, and ω the drive frequency.

Near a smooth stable equilibrium, $V(x) \simeq V(x_0) + \frac{1}{2}k(x - x_0)^2$. Thus

$$\ddot{x} + \omega_0^2 x = 0, \quad \omega_0 = \sqrt{k/m}, \quad x = A\cos(\omega_0 t + \phi). \tag{10.3}$$

With linear damping and a sinusoidal drive,

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos\omega t. \tag{10.4}$$

The free underdamped solution has envelope $\mathrm{e}^{-\beta t}$ and frequency $\omega_d = \sqrt{\omega_0^2 - \beta^2}$. Critical damping occurs at $\beta = \omega_0$; overdamping has two real decay rates.

The steady-state displacement is $A(\omega)\cos(\omega t - \delta)$, where

$$A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}, \quad \delta = \operatorname{atan2}(2\beta\omega, \omega_0^2 - \omega^2). \tag{10.5}$$

For weak damping, $Q = \omega_0/(2\beta)$ and the resonance width is approximately 2β. The displacement peak occurs at $\sqrt{\omega_0^2 - 2\beta^2}$ if that quantity is real; it is not exactly ω_0 for finite damping.

For the free damped problem try $x = \mathrm{e}^{st}$. Each time derivative supplies a factor s, turning the differential equation into

$$s^2 + 2\beta s + \omega_0^2 = 0, \qquad s = -\beta \pm \sqrt{\beta^2 - \omega_0^2}. \tag{10.6}$$

Complex roots describe oscillation with decay, repeated roots mark critical damping, and distinct real roots give overdamping. This is why comparing β with ω_0 classifies the motion.

For a steady sinusoidal drive, try $x = \mathrm{Re}(X\mathrm{e}^{\mathrm{i}\omega t})$. Substitution gives

$$(\omega_0^2 - \omega^2 + 2\mathrm{i}\beta\omega)X = \frac{F_0}{m}, \qquad X = \frac{F_0/m}{\omega_0^2 - \omega^2 + 2\mathrm{i}\beta\omega}. \tag{10.7}$$

The magnitude of X is the displacement amplitude, while its argument is the phase relative to the drive. Taking the magnitude and angle of this one complex fraction gives both formulas above. The full solution also contains a transient determined by initial conditions; damping removes that transient at late times.

10.3 Coupled modes and beats

Watch what the coupling spring does in two special motions. If the masses move together by the same amount, its length does not change. If they move in opposite directions, it stretches twice as much as either displacement. These observations suggest adding and subtracting the equations before attempting a general solution. In coupled systems, x_1, x_2 are displacements and q_+, q_- are collective coordinates. An eigenvector gives a mode shape; its eigenvalue gives the associated squared frequency.

Two equal masses attached to fixed walls by springs k and coupled by a spring κ obey

$$m\ddot{x}_1 = -(k+\kappa)x_1 + \kappa x_2, \quad m\ddot{x}_2 = \kappa x_1 - (k+\kappa)x_2. \tag{10.8}$$

The coordinates $q_+ = x_1 + x_2$ and $q_- = x_1 - x_2$ oscillate with $\omega_+ = \sqrt{k/m}$ and $\omega_- = \sqrt{(k+2\kappa)/m}$. General motion superposes these modes. Nearby frequencies yield beats through $\cos\omega_1 t + \cos\omega_2 t = 2\cos[(\omega_1-\omega_2)t/2]\cos[(\omega_1+\omega_2)t/2]$.

Adding gives $m\ddot{q}_+ = -kq_+$; subtracting gives $m\ddot{q}_- = -(k+2\kappa)q_-$. Each collective coordinate is now an independent oscillator. Once their initial values are known, solve them and reconstruct

$$x_1 = \frac{q_+ + q_-}{2}, \qquad x_2 = \frac{q_+ - q_-}{2}. \tag{10.9}$$

The normal coordinates are useful because they follow motions the system can sustain without continually changing their shape. Beats arise when nearby mode frequencies slowly move into and out of phase.

10.4 Linear algebra and normal modes

The two-mass example exposed motions that keep their shape: both masses move together or oppositely. For many masses, we can find such patterns systematically with a matrix. An eigenvector is a pattern that a linear operation changes only by a scale factor; its eigenvalue records that factor.

An eigenvector obeys $A\boldsymbol{v} = \lambda\boldsymbol{v}$. For real symmetric matrices, eigenvalues are real and an orthonormal eigenbasis exists. Hermitian complex matrices have the analogous property. Diagonalization separates coupled degrees of freedom.

For a quadratic mechanical system,

$$T = \tfrac{1}{2}\dot{\boldsymbol{q}}^{\mathsf{T}} M \dot{\boldsymbol{q}}, \qquad V = \tfrac{1}{2}\boldsymbol{q}^{\mathsf{T}} K \boldsymbol{q}, \tag{10.10}$$

the mode equation is $K\boldsymbol{a} = \omega^2 M\boldsymbol{a}$. Positive M and positive-definite K imply real nonzero frequencies and stable small oscillations. Zero eigenvalues often correspond to a symmetry; negative stiffness eigenvalues indicate instability.

For constant matrices M and K, the mechanical equations are $M\ddot{\boldsymbol{q}} + K\boldsymbol{q} = 0$. Try a motion in which every coordinate has the same frequency but a different amplitude:

$$\boldsymbol{q}(t) = \boldsymbol{a}\cos(\omega t + \phi) \quad \Longrightarrow \quad (K - \omega^2 M)\boldsymbol{a} = 0. \tag{10.11}$$

A nonzero amplitude vector exists only when $\det(K - \omega^2 M) = 0$. Solve this equation for the allowed squared frequencies, then find the associated amplitude ratios. Finally, superpose the modes to meet the initial conditions. This sequence separates the properties of the system from the particular way you set it moving.

10.5 When the small-angle approximation fails

The spring equation is linear because force is proportional to displacement. A real pendulum has a restoring factor $\sin\theta$ rather than exactly θ. Close to its lowest point, the difference is small. To judge how small, compare the first correction with the leading term rather than relying on the vague claim that an angle "looks small."

With θ measured in radians, the Taylor expansion is

$$\sin\theta = \theta - \frac{\theta^3}{6} + O(\theta^5). \tag{10.12}$$

Replacing $\sin\theta$ by θ drops a term of size about $\theta^3/6$. Relative to the leading term, its size is about $\theta^2/6$. At $\theta = 0.1$, that is about 0.0017, or 0.17%, for the instantaneous restoring factor. At a larger amplitude the correction grows; its effect on an entire period requires following the motion, not just checking one point. This gives the next section a concrete reason to study nonlinear behavior.

10.6 Nonlinearity and deterministic chaos

The small-angle pendulum allowed us to replace $\sin\theta$ by θ. Retaining the full sine changes more than a numerical coefficient: sums of solutions no longer generally solve the equation. For a driven system, we therefore study trajectories and their stability rather than expecting every motion to be a sum of independent sinusoids.

Nonlinear systems need not obey superposition. A driven pendulum, $\ddot{\theta} + 2\beta\dot{\theta} + \omega_0^2 \sin\theta = f\cos\omega t$, can show periodic, quasiperiodic, or chaotic motion. Chaos is sensitive dependence on initial conditions, often quantified locally by $|\delta x(t)| \sim |\delta x(0)|e^{\lambda t}$ with positive largest Lyapunov exponent.

A phase portrait displays trajectories; a Poincare section samples them once per driving period. Determinism does not guarantee long-term practical predictability. Exponential separation is valid only while perturbations remain small; bounded trajectories eventually saturate the separation.

A useful starting point for stability is to write a nearby trajectory as $\theta + \delta\theta$. Subtract the original pendulum equation and retain terms linear in the separation:

$$\delta\ddot{\theta} + 2\beta\delta\dot{\theta} + \omega_0^2 \cos\theta(t)\, \delta\theta = 0. \tag{10.13}$$

The coefficient now changes along the reference trajectory. This equation explains how two solutions obeying the same deterministic law can separate. Chaos is not implied merely by nonlinearity; it requires analyzing the actual dynamics, parameters, and long-time perturbation growth.

Exercises

Two identical oscillators are coupled by a spring. Sketch an in-phase and an out-of-phase motion. Explain why their frequencies can differ.

Answers are in Appendix A.

Chapter 11

Lagrangian and Hamiltonian mechanics

Newton's laws taught us to follow forces component by component. That method remains valuable, but constraints and changing coordinates can make it cumbersome. The action reformulates the same mechanics as a statement about paths; Hamiltonian mechanics will then reorganize those paths into a phase-space flow. Newton's laws remain valid, but constrained systems can make their component form cumbersome. A pendulum moves along a circle; an angle describes that motion more naturally than two coordinates and an unknown tension. Analytical mechanics lets us choose such coordinates first and obtain their equations systematically. It also reveals why symmetries and conservation laws appear together.

11.1 From a pendulum to a new coordinate

A pendulum bob is free to move along a circular arc, but the rod keeps it at a fixed distance ℓ from the pivot. We could describe the bob with two Cartesian coordinates and then add a constraint equation. Its angle θ from the downward vertical is simpler: one number identifies every allowed position. A coordinate chosen to describe an allowed motion is called a *generalized coordinate*. Its time derivative $\dot{\theta}$ measures how quickly that coordinate changes.

If the bob has mass m, it moves at speed $v = \ell\dot{\theta}$, so its kinetic energy is $T = \frac{1}{2}m\ell^2\dot{\theta}^2$. Taking potential energy zero at the bottom, its height above that point is $\ell(1-\cos\theta)$ and its gravitational potential energy is $V = mg\ell(1-\cos\theta)$. The difference

$$L(\theta,\dot{\theta}) = T - V = \tfrac{1}{2}m\ell^2\dot{\theta}^2 - mg\ell(1-\cos\theta) \tag{11.1}$$

is the Lagrangian for this model. It has units of energy. The rod tension does not appear in this expression because the angle already respects the fixed-length constraint.

11.2 What stationary action means

The Lagrangian assigns a number to each moment along a proposed motion. Add those contributions between two fixed times to assign a number to the whole proposed path:

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\,\mathrm{d}t. \tag{11.2}$$

This is the action. Its unit is energy times time. The principle of stationary action says that the physical path makes the first-order change of S vanish when compared with nearby paths that have the same starting and ending positions. Stationary means that the first-order change is zero; the action need not be a minimum.

We can derive the equation this rule imposes without guessing it. Let $q(t)$ be a proposed path and $\eta(t)$ a small change in its shape. Write a nearby path as $q_\epsilon(t) = q(t) + \epsilon\eta(t)$. Keep the endpoints fixed, so $\eta(t_1) = \eta(t_2) = 0$. Differentiate the action with respect to ϵ at zero:

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \dot{q}}\dot{\eta} \right) \mathrm{d}t. \tag{11.3}$$

The dot on η is a time derivative. Integrate the second term by parts so that both terms contain η rather than its derivative:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}}\dot{\eta}\,\mathrm{d}t = \left[\frac{\partial L}{\partial \dot{q}}\eta \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}} \right) \eta\,\mathrm{d}t \tag{11.4}$$

$$= -\int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}} \right) \eta\,\mathrm{d}t. \tag{11.5}$$

The boundary term vanishes because η is zero at both endpoints. The remaining integral can vanish for every smooth interior change η only if its coefficient is zero. We obtain the Euler–Lagrange equation:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0.} \tag{11.6}$$

For several independent generalized coordinates q_i, the same equation holds for each i.

Worked example 11.1 (Recovering the pendulum equation)**.** For the pendulum Lagrangian above, $\partial L/\partial\dot{\theta} = m\ell^2\dot{\theta}$ and $\partial L/\partial\theta = -mg\ell\sin\theta$. The Euler–Lagrange equation becomes $m\ell^2\ddot{\theta} + mg\ell\sin\theta = 0$, or $\ddot{\theta} + (g/\ell)\sin\theta = 0$. Near the downward position, $|\theta| \ll 1$ in radians and $\sin\theta \approx \theta$, so the small-angle motion satisfies $\ddot{\theta} + (g/\ell)\theta = 0$. Its frequency is $\omega_0 = \sqrt{g/\ell}$ and its period is $2\pi\sqrt{\ell/g}$. We have now found the numerical factor that dimensional reasoning alone could not supply.

The action method gives the same pendulum motion that a careful force calculation would give. Its advantage grows when constraints or coordinates make force components cumbersome. We will keep track of its assumptions: the simple form $L = T - V$ applies to the conservative mechanical models used here, while other systems can require different Lagrangians or generalized forces.

11.3 Constraints and multipliers

Not every variation is physically allowed. A bead constrained to a surface must satisfy $f(q) = 0$, so its trial paths cannot move freely in every coordinate direction. Introduce a multiplier $\lambda(t)$ and vary the augmented action

$$S_*[q, \lambda] = \int [L(q, \dot{q}, t) + \lambda(t) f(q, t)] \, \mathrm{d}t. \tag{11.7}$$

Variation with respect to λ returns the constraint. Variation with respect to q_i gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \lambda \frac{\partial f}{\partial q_i}. \tag{11.8}$$

The multiplier is the generalized constraint force required to keep the path on the allowed surface. It should be solved from the constraint and the equations of motion, rather than discarded as an algebraic nuisance. In constrained problems, λ is a multiplier whose value becomes the generalized constraint force.

The same idea handles constrained equilibrium. To maximize entropy subject to normalization and fixed mean energy, add multipliers to the objective; to minimize a potential subject to a geometric condition, add a multiplier to the potential. In each case the multiplier measures the response of the optimum to a small change in the imposed constraint. This common structure links mechanics, thermodynamics, and optimization.

11.4 Symmetry and Noether's theorem

A symmetry means that a change in our description or configuration leaves the relevant action unchanged. If sliding the whole experiment in space changes nothing, there is no preferred location. If shifting the starting time changes nothing, there is no preferred clock reading. Conservation laws give these statements a precise dynamical form.

A continuous symmetry of the action implies a conserved quantity, with appropriate boundary qualifications. Time translation yields energy, spatial translation momentum, and rotation angular momentum. A cyclic coordinate obeys $\partial L/\partial q_i = 0$, so its canonical momentum $p_i = \partial L/\partial \dot{q}_i$ is constant.

If L has no explicit time dependence, $E_L = \sum_i \dot{q}_i p_i - L$ is conserved. This equals $T + V$ for usual time-independent mechanical coordinates and velocity-independent potentials; more general Lagrangians require using the definition.

The energy statement can be checked directly. Write $p_i = \partial L/\partial \dot{q}_i$ and differentiate $E_L = \sum_i \dot{q}_i p_i - L$:

$$\dot{E}_L = \sum_i \dot{q}_i \left(\dot{p}_i - \frac{\partial L}{\partial q_i} \right) - \frac{\partial L}{\partial t} = -\frac{\partial L}{\partial t}. \tag{11.9}$$

The Euler–Lagrange equations remove the bracketed terms. Thus absence of explicit time dependence makes E_L constant, even though L usually changes along the motion through $q(t)$ and $\dot{q}(t)$. Distinguishing explicit dependence from dependence through the trajectory is the key to using this result correctly.

11.5 Hamiltonian dynamics and phase space

The Lagrangian describes a state using coordinates and velocities. Hamiltonian mechanics replaces velocities by momenta. This is useful because position and momentum form natural pairs, each with its own first-order evolution equation. Instead of following a curve through configuration space, we follow a flow through phase space. Later, $p_i = \partial L/\partial \dot{q}_i$ is canonical momentum and H is the Hamiltonian, not automatically identical to kinetic plus potential energy in every theory.

When velocities can be expressed in terms of momenta, define

$$H(q,p,t) = \sum_i p_i \dot{q}_i - L, \qquad \dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{11.10}$$

A state is a point in phase space (q,p). The Poisson bracket is

$$\{f,g\} = \sum_i \left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i} \right), \qquad \dot{f} = \{f,H\} + \frac{\partial f}{\partial t}. \tag{11.11}$$

Hamiltonian flow preserves phase-space volume (Liouville's theorem). Canonical transformations preserve the Poisson structure. For periodic one-dimensional motion, an action variable $J = (2\pi)^{-1} \oint p \mathrm{d}q$ is approximately conserved under sufficiently slow parameter changes away from separatrices.

The transformation explains Hamilton's equations. Begin with $H = \sum_i p_i \dot{q}_i - L$ and take its differential:

$$\mathrm{d}H = \sum_i (\dot{q}_i \mathrm{d}p_i + p_i \mathrm{d}\dot{q}_i) - \sum_i \left(\frac{\partial L}{\partial q_i}\mathrm{d}q_i + \frac{\partial L}{\partial \dot{q}_i}\mathrm{d}\dot{q}_i \right) - \frac{\partial L}{\partial t}\mathrm{d}t \tag{11.12}$$

$$= \sum_i (\dot{q}_i \mathrm{d}p_i - \dot{p}_i \mathrm{d}q_i) - \frac{\partial L}{\partial t}\mathrm{d}t. \tag{11.13}$$

The velocity differentials cancel because $p_i = \partial L/\partial \dot{q}_i$. Comparing coefficients with the differential of $H(q,p,t)$ gives Hamilton's equations. For $H = p^2/(2m) + kx^2/2$, they read $\dot{x} = p/m$ and $\dot{p} = -kx$; differentiating the first recovers the spring equation. The new formalism has reorganized the same motion.

11.6 Fields and continuum action

A field gives a dynamical variable at each position. You can picture it as the limit of a very large set of coupled coordinates, with the sum over sites replaced by an integral. The variation now includes changes in both the field and its derivatives, but integration by parts performs the same job as in mechanics.

For fields $\phi_a(x)$ with Lagrangian density $\mathcal{L}(\phi_a, \partial_\mu \phi_a, x)$,

$$S = \int \mathcal{L}\, \mathrm{d}^4 x, \qquad \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) - \frac{\partial \mathcal{L}}{\partial \phi_a} = 0. \tag{11.14}$$

The mechanical sum over coordinates becomes a continuum of local degrees of freedom. This is the bridge from oscillators to elastic media, electromagnetism, and quantum fields.

For a stretched string, take the action $S = \int \mathrm{d}t\mathrm{d}x\, \mathcal{L}$ with

$$\mathcal{L} = \tfrac{1}{2}\mu(\partial_t y)^2 - \tfrac{1}{2}\mathcal{T}(\partial_x y)^2. \tag{11.15}$$

The first term is kinetic energy per length; the second is the increase of elastic energy at small slope. Varying y with the appropriate boundary terms fixed gives

$$\partial_t(\mu\partial_t y) + \partial_x(-\mathcal{T}\partial_x y) = 0, \qquad \mu\partial_t^2 y = \mathcal{T}\partial_x^2 y. \tag{11.16}$$

This is our first example of a wave equation arising from an action. Each point supplies inertia, while spatial bending supplies a restoring force.

Exercises

For a simple pendulum, identify the coordinate that describes its motion and the two energies in its Lagrangian. Why must the endpoints of the trial paths be fixed when deriving its equation?

Answers are in Appendix A.

Part IV

Continuum Physics, Waves, and Optics

Chapter 12

Elasticity, fluids, and transport

Until now, the objects were a few particles or coordinates. We now let the number of degrees of freedom become enormous and replace the list with fields. The same conservation logic survives: what changes inside a region must be explained by what crosses its boundary or is created within it. When tracking every atom is unnecessary, we describe matter using fields such as density, velocity, and temperature. The guiding method is to balance what enters and leaves a small region, then relate the fluxes and forces to local fields. Conservation laws supply the bookkeeping; material properties supply the response.

12.1 A stretched rod: stress and strain

Take a rod of original length L and cross-sectional area A. Pull its ends with equal and opposite forces of magnitude F. If the rod extends by ΔL, its fractional extension $\epsilon = \Delta L/L$ is called strain. It has no unit: it compares a change in length with the original length. The force per area $\sigma = F/A$ is stress, measured in pascals, or newtons per square meter.

For sufficiently small reversible stretches of many solids, stress is approximately proportional to strain:

$$\sigma = Y\epsilon, \tag{12.1}$$

where Y is Young's modulus. This is a measured material relation, not a consequence of Newton's laws alone. Substitute the definitions to find

$$\frac{F}{A} = Y\frac{\Delta L}{L}, \qquad F = \frac{YA}{L}\Delta L. \tag{12.2}$$

The whole rod acts like a spring with stiffness $k = YA/L$. A thicker rod is stiffer because more material shares the load; a longer rod is easier to stretch because the same fractional strain produces a larger extension.

Worked example 12.1 (Comparing two rods)**.** Two rods have the same material and length, but one has twice the cross-sectional area. Under the same pulling force, $\Delta L = FL/(YA)$ shows that the wider rod extends half as much. The comparison assumes both remain in the small, elastic regime. A permanent deformation or fracture would require a different material model.

12.2 From one rod to a displacement field

A three-dimensional solid can stretch differently at different places. Describe how each point moves with a displacement vector $\boldsymbol{u}(x, y, z, t)$. The symbol u_x, for example, is its x component. A partial derivative such as $\partial_x u_x$ asks how that component changes as we move in the x direction while holding the other coordinates fixed. For a uniformly stretched rod, this derivative is the local version of $\Delta L/L$.

A rigid translation moves every point equally and has zero displacement gradient. A small rigid rotation has gradients but does not change shape. To isolate deformation, take the symmetric part of the gradients:

$$\epsilon_{ij} = \tfrac{1}{2}(\partial_i u_j + \partial_j u_i). \tag{12.3}$$

The indices i, j label coordinate directions. For $i = j = x$, this says $\epsilon_{xx} = \partial_x u_x$, the familiar one-dimensional stretch. Mixed components describe shear. Repeated indices in later equations are summed over x, y, z; for instance, $\epsilon_{kk} = \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}$ measures the leading fractional volume change.

Stress also needs two direction labels in three dimensions: one for the force component and one for the orientation of the surface receiving it. Its components form σ_{ij}. For a homogeneous isotropic solid in the small-strain regime, one common material law is

$$\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}. \tag{12.4}$$

Here δ_{ij} is one when $i = j$ and zero otherwise; λ and μ are Lamé elastic constants. The symbol μ here means shear modulus, not mass per length or reduced mass. This formula is the three-dimensional partner of the rod's $\sigma = Y\epsilon$, with enough information to describe both volume change and shear.

To predict motion, apply Newton's law to a small volume of material. Differences in stress across its faces produce a net force. If ρ is mass per volume and f_i is an additional force per volume, the local balance is

$$\rho \partial_t^2 u_i = \partial_j \sigma_{ij} + f_i. \tag{12.5}$$

The partial time derivative is acceleration at a fixed location. A solution needs initial displacement and velocity plus conditions at the solid's boundaries. For example, a clamped end has zero displacement, while a free surface has specified traction. Only after choosing those conditions can we calculate a particular deformation or elastic wave.

12.3 Hydrostatics and buoyancy

Picture a thin horizontal slab of stationary fluid. Pressure below pushes upward; pressure above pushes downward; gravity supplies the slab's weight. Equilibrium requires the upward pressure excess to support that weight. Pressure must therefore decrease as height increases.

A static fluid cannot sustain shear stress. Pressure can vary with position. The vector $\boldsymbol{\nabla} p$ collects its spatial rates of change; in a vertical column it reduces to the derivative $\mathrm{d}p/\mathrm{d}z$ in the vertical

direction. Hydrostatic balance is $\boldsymbol{\nabla} p = \rho \boldsymbol{g}$. For constant density with upward z, $p(z) = p(z_0) - \rho g(z - z_0)$. Integrating pressure over a submerged object's surface gives a buoyant force equal to the weight of displaced fluid.

Surface tension γ is energy per area and force per length. Across a curved interface,

$$\Delta p = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right). \tag{12.6}$$

A spherical liquid droplet has $\Delta p = 2\gamma/R$; a thin soap bubble has two interfaces and approximately $4\gamma/R$.

For slab area A and thickness $\mathrm{d}z$, vertical balance gives

$$p(z)A - p(z + \mathrm{d}z)A - \rho g A \mathrm{d}z = 0, \qquad \frac{\mathrm{d}p}{\mathrm{d}z} = -\rho g. \tag{12.7}$$

Integrate only after deciding whether ρ can be treated as constant. To understand buoyancy, imagine replacing the submerged object by fluid of the same shape. That fluid would be in equilibrium, so the surrounding pressure forces must balance its weight. The same pressure forces act on the object, giving the weight of displaced fluid as the buoyant force.

For a spherical droplet, a small increase $\mathrm{d}R$ requires pressure work $\Delta p \, \mathrm{d}V$ and increases surface energy by $\gamma \mathrm{d}A$. Since $\mathrm{d}V = 4\pi R^2 \mathrm{d}R$ and $\mathrm{d}A = 8\pi R \mathrm{d}R$, equality gives $\Delta p = 2\gamma/R$. Geometry explains the inverse-radius dependence.

12.4 Counting mass in a small region

Choose a fixed box inside a flowing fluid. Let $\rho(\boldsymbol{r}, t)$ be mass per volume and $\boldsymbol{v}(\boldsymbol{r}, t)$ the local fluid velocity. The mass in a small volume V is $\int_V \rho \, \mathrm{d}V$. It can change when mass crosses the box's boundary. If $\mathrm{d}\boldsymbol{S}$ points outward from a small surface patch, then $\rho \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{S}$ is the mass flowing outward through that patch per unit time.

Add all patches and use conservation of mass:

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_V \rho \, \mathrm{d}V = - \oint_{\partial V} \rho \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{S}. \tag{12.8}$$

The minus sign says outward flow decreases the mass inside. This equation makes sense before we choose any coordinates or a special fluid model.

To turn the whole-box balance into a rule at each point, examine a tiny rectangular box. The difference between the x flow leaving one face and entering the opposite face, divided by its volume, approaches $\partial_x(\rho v_x)$. Add the y and z face pairs. The sum is called the divergence of the mass flux:

$$\boldsymbol{\nabla} \cdot (\rho \boldsymbol{v}) = \partial_x(\rho v_x) + \partial_y(\rho v_y) + \partial_z(\rho v_z). \tag{12.9}$$

Likewise, $\partial_t \rho$ measures density change at a fixed location. Since the box can be arbitrarily small, (12.8) gives the local continuity equation

$$\partial_t \rho + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{v}) = 0. \tag{12.10}$$

Integrating this local rule over a larger volume reverses the argument: internal face fluxes cancel, leaving only its outside boundary. That is the physical content of the divergence theorem.

12.5 Following a fluid parcel

A second viewpoint rides along with one small parcel of fluid. A field value may change because time passes at one place or because the parcel enters a region where the field has a different value. The chain rule combines both effects. For a scalar field $f(\boldsymbol{r}, t)$,

$$\frac{Df}{Dt} = \partial_t f + \boldsymbol{v} \cdot \boldsymbol{\nabla} f. \tag{12.11}$$

The vector $\boldsymbol{\nabla} f = (\partial_x f, \partial_y f, \partial_z f)$ points toward fastest increase of f. For velocity itself, the parcel's acceleration is $D\boldsymbol{v}/Dt = \partial_t \boldsymbol{v} + (\boldsymbol{v} \cdot \boldsymbol{\nabla})\boldsymbol{v}$. A steady flow has $\partial_t \boldsymbol{v} = 0$ at fixed locations but can still accelerate a parcel when it passes from a wide pipe into a narrow one.

Newton's law for a parcel balances pressure forces, viscosity, and gravity. For an incompressible Newtonian fluid of constant density and dynamic viscosity η, it becomes

$$\rho\left(\partial_t \boldsymbol{v} + (\boldsymbol{v} \cdot \boldsymbol{\nabla})\boldsymbol{v}\right) = -\boldsymbol{\nabla} p + \eta \nabla^2 \boldsymbol{v} + \rho \boldsymbol{g}, \qquad \boldsymbol{\nabla} \cdot \boldsymbol{v} = 0. \tag{12.12}$$

Here p is pressure, $\boldsymbol{g}$ gravitational acceleration, and the Laplacian $\nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2$ measures how neighboring values differ. These are the Navier–Stokes equations under the stated material and incompressibility assumptions. They are a model to solve with initial and boundary data, not a single formula for every flow.

If viscosity is negligible, flow is steady, density is constant, and we follow one streamline, projecting the force equation along the motion gives

$$\mathrm{d}\left(\tfrac{1}{2} v^2\right) = -\frac{\mathrm{d}p}{\rho} - g\,\mathrm{d}z. \tag{12.13}$$

Adding the small changes along that streamline yields Bernoulli's relation $p + \rho v^2/2 + \rho g z = \text{constant}$ there. The pressure, motion, and height terms all have units of energy per volume. The constant need not be the same on different streamlines without additional conditions.

12.6 Viscosity and dimensionless parameters

Viscosity transfers momentum between neighboring layers moving at different speeds. To decide whether it dominates a flow, estimate the inertial term as $\rho U^2/L$ and the viscous term as $\eta U/L^2$. Their ratio is the Reynolds number. This estimate is a practical way to choose which terms deserve attention before solving a flow equation.

The Reynolds number $\mathrm{Re} = \rho U L/\eta$ compares inertia with viscosity. Low Reynolds number implies creeping flow; large Reynolds number permits inertial instabilities but does not by itself specify a universal transition threshold. For steady fully developed laminar flow through a circular pipe,

$$v(r) = \frac{\Delta p}{4\eta L}(R^2 - r^2), \qquad Q_V = \frac{\pi R^4 \Delta p}{8\eta L}. \tag{12.14}$$

These results assume a Newtonian incompressible fluid and no slip at the wall. Stokes drag on an isolated sphere at very low Reynolds number is $F = 6\pi\eta R v$.

Worked example 12.2 (Pipe radius matters)**.** Holding pressure drop, length, and viscosity fixed, doubling a pipe's radius multiplies laminar volume flow by $2^4 = 16$. This strong dependence is a viscous-flow result; applying it after the flow becomes turbulent is unjustified.

For a straight pipe, assume steady axial flow $v(r)$ with pressure decreasing uniformly by Δp over length L. The axial momentum equation reduces to

$$0 = \frac{\Delta p}{L} + \eta \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}v}{\mathrm{d}r}\right). \tag{12.15}$$

Multiply by r and integrate. Finite slope at the center removes the singular integration term, giving $v' = -\Delta p\, r/(2\eta L)$. Integrate again and impose the no-slip condition $v(R) = 0$ to obtain the parabolic profile. The volume flow is the sum over annular areas:

$$Q_V = \int_0^R v(r)\, 2\pi r \mathrm{d}r = \frac{\pi \Delta p}{2\eta L}\int_0^R (R^2 r - r^3)\mathrm{d}r = \frac{\pi R^4 \Delta p}{8\eta L}. \tag{12.16}$$

Two powers of radius come from area and two from the increased speed allowed by the gentler viscous gradient.

12.7 Diffusion, heat conduction, and random walks

A concentration gradient drives a particle flux from high concentration toward low concentration. Conservation then tells us how that flux changes the concentration. Heat conduction uses the same mathematical structure with a different stored quantity and a different transport coefficient.

Fick's law $\boldsymbol{j} = -D\boldsymbol{\nabla} n$ together with number conservation yields $\partial_t n = D\nabla^2 n$ for constant D. Fourier's heat law is $\boldsymbol{q} = -\kappa \boldsymbol{\nabla} T$, giving

$$\partial_t T = \alpha \nabla^2 T, \qquad \alpha = \frac{\kappa}{\rho c_p} \tag{12.17}$$

when material properties are constant and the appropriate mechanical conditions permit use of c_p. A diffusion Green function in d dimensions is

$$n(\boldsymbol{r}, t) = \frac{N}{(4\pi D t)^{d/2}} \exp\left(-\frac{r^2}{4Dt}\right), \qquad \left\langle r^2 \right\rangle = 2dDt. \tag{12.18}$$

Diffusion distance grows as $\sqrt{t}$, whereas ballistic travel grows as t.

Combine $\partial_t n + \boldsymbol{\nabla}\cdot\boldsymbol{j} = 0$ with $\boldsymbol{j} = -D\boldsymbol{\nabla} n$:

$$\partial_t n = \boldsymbol{\nabla}\cdot(D\boldsymbol{\nabla} n) = D\nabla^2 n \tag{12.19}$$

for spatially constant D. The minus sign in the flux law makes peaks flatten rather than sharpen. In one dimension, an independent random walk of step length a every Δt has $\left\langle x^2 \right\rangle = Na^2 = (a^2/\Delta t)t$. Comparing with $2Dt$ identifies $D = a^2/(2\Delta t)$. The continuum coefficient thus measures the cumulative effect of microscopic steps, while the diffusion equation describes their smooth large-scale distribution.

12.8 A diffusion problem with fixed boundaries

The diffusion law now gives us a specific question: a quantity is held at zero at both ends of an interval, but begins with a nonzero profile inside. How does that profile relax? The boundaries select the spatial shapes that are allowed. We can solve one shape at a time and then add them to represent the starting profile.

Let us solve one separated problem explicitly. On $0 < x < L$, hold the ends of a diffusing field at zero, and put $u = XT$ into $\partial_t u = D\partial_x^2 u$. Wherever division is allowed,

$$\frac{\dot{T}}{DT} = \frac{X''}{X} = -k^2. \tag{12.20}$$

One side depends only on time and the other only on position, so both must be constant. We choose a negative constant because the zero-end boundary conditions call for sines. The two ordinary equations give

$$X_n(x) = \sin\frac{n\pi x}{L}, \qquad T_n(t) = \mathrm{e}^{-D(n\pi/L)^2 t}, \qquad n = 1, 2, \ldots. \tag{12.21}$$

Thus $u(x,t) = \sum_n b_n X_n(x) T_n(t)$, with b_n found by expanding the initial profile in a sine series. We did not guess an entire complicated solution. We solved for simple allowed pieces and used the initial condition to decide how much of each piece to include.

Exercises

Draw a small box within a flowing fluid. Label the mass entering and leaving it. What extra information is needed beyond mass conservation to predict the flow?

Answers are in Appendix A.

Chapter 13

Waves, sound, and interference

The continuum description gave us fields that change from place to place. A wave is what happens when a local change communicates with neighboring regions. The oscillator language returns, spread across space; later, the same mathematics will describe light, quantum amplitudes, and numerical grids. A wave lets a disturbance travel without requiring the material to travel with it. Each part of the medium responds locally and influences its neighbors. We will derive the wave equation from that interaction, then use boundary conditions and superposition to explain standing waves, packets, and interference.

13.1 From a string to a wave equation

Take a short string segment of length Δx. Its mass is $\mu\Delta x$. At small slope, the vertical component of tension is approximately $\mathcal{T}\partial_x y$. The two endpoints generally have different slopes, so their vertical forces do not cancel. That slope difference is the source of transverse acceleration. $y(x,t)$ is displacement, μ is linear mass density, $\mathcal{T}$ tension, k wave number, ω angular frequency, and $v = \omega/k$ phase speed.

For small transverse displacement $y(x,t)$ of a string with constant tension $\mathcal{T}$ and linear mass density μ, transverse force balance on a short segment yields

$$\mu\partial_t^2 y = \mathcal{T}\partial_x^2 y, \qquad v = \sqrt{\mathcal{T}/\mu}. \tag{13.1}$$

The general one-dimensional solution is $f(x-vt)+g(x+vt)$. A sinusoidal wave has phase $kx-\omega t+\phi$, wavelength $\lambda = 2\pi/k$, period $\tau = 2\pi/\omega$, and phase speed ω/k.

Newton's law for the segment reads

$$\mu\Delta x\,\partial_t^2 y = \mathcal{T}[\partial_x y(x+\Delta x,t) - \partial_x y(x,t)] \simeq \mathcal{T}\partial_x^2 y\,\Delta x. \tag{13.2}$$

Cancel Δx and take its limit to obtain the wave equation. To find the speed, try $y = A\cos(kx-\omega t)$. Its second derivatives are $-\omega^2 y$ and $-k^2 y$, so $\omega^2 = (\mathcal{T}/\mu)k^2$. A fixed phase obeys $kx-\omega t =$ constant and moves at $\omega/k = \sqrt{\mathcal{T}/\mu}$. More tension speeds up communication along the string; more inertia slows it down.

13.2 Energy, impedance, and boundaries

To find transported energy, look at the work one string segment does on its neighbor. To find allowed standing waves, look at what the endpoints permit. These are different questions about the same displacement field, and both can be answered without introducing a new wave law.

For a sinusoidal traveling string wave of displacement amplitude A,

$$\langle P\rangle = \tfrac{1}{2}\mu\omega^2 A^2 v. \tag{13.3}$$

Waves carry energy even when material elements only oscillate locally. Reflection and transmission are fixed by boundary conditions. At a fixed string end the displacement changes sign on reflection; at a free end it does not.

Fixed ends at $x = 0, L$ allow $k_n = n\pi/L$, so $f_n = nv/(2L)$ for positive integers n. Standing waves are superpositions of counterpropagating waves; their nodes are permanently at zero displacement.

The instantaneous power passing to the right is $P = -\mathcal{T}(\partial_x y)(\partial_t y)$. For $y = A\cos(kx - \omega t)$, this becomes $\mathcal{T}k\omega A^2 \sin^2(kx - \omega t)$. Average the squared sine over a cycle and use $k = \omega/v$ and $\mathcal{T} = \mu v^2$ to recover $\langle P\rangle = \mu v \omega^2 A^2/2$.

For fixed endpoints, a separated spatial mode is $X = C\sin kx + D\cos kx$. The condition $X(0) = 0$ sets $D = 0$; $X(L) = 0$ then requires $\sin kL = 0$. Nontrivial modes therefore have $kL = n\pi$. Discrete frequencies come from the boundaries, even though a freely traveling wave could have any wavelength.

Worked example 13.1 (The first two notes of a fixed string)**.** A string is fixed at both ends, has length $L = 1\,\mathrm{m}$, and carries waves at $v = 100\,\mathrm{m/s}$. The lowest standing wave fits half a wavelength into the string, so $\lambda_1 = 2L = 2\,\mathrm{m}$ and $f_1 = v/\lambda_1 = 50\,\mathrm{Hz}$. The next mode fits a full wavelength, giving $f_2 = 100\,\mathrm{Hz}$. Both use the same string and wave speed; the boundary decides which patterns can persist.

13.3 Complex numbers and Fourier analysis

A sine changes into a cosine when differentiated; a cosine changes back into a sine with a minus sign. Complex exponentials package that cycle into one operation. We use them because they simplify the work, and take the real part when reconstructing a real displacement or field. Fourier analysis extends the same idea: build a complicated shape from simple oscillating pieces. A complex exponential records oscillation compactly; the measured displacement or field is real.

Euler's identity $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ converts sinusoidal differentiation into multiplication by $\mathrm{i}\omega$. Physical fields are real parts of complex representations. With the convention

$$\widetilde{f}(k) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-\mathrm{i}kx}\mathrm{d}x, \qquad f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widetilde{f}(k)\mathrm{e}^{\mathrm{i}kx}\mathrm{d}k, \tag{13.4}$$

spatial differentiation becomes multiplication by $\mathrm{i}k$. Fourier methods separate linear, translation-invariant differential equations into algebraic equations. Sharp localization requires a broad spectrum.

The Fourier differentiation rule follows from integration by parts. For a sufficiently decaying differentiable function,

$$\widetilde{f'}(k) = \int f'(x)\mathrm{e}^{-\mathrm{i}kx}\mathrm{d}x \tag{13.5}$$

$$= \Big[f(x)\mathrm{e}^{-\mathrm{i}kx}\Big]_{-\infty}^{\infty} + \mathrm{i}k\int f(x)\mathrm{e}^{-\mathrm{i}kx}\mathrm{d}x = \mathrm{i}k\widetilde{f}(k). \tag{13.6}$$

The boundary term vanishes under the stated assumption. A second derivative consequently becomes $-k^2\widetilde{f}$. That minus sign will turn a diffusion equation into exponential decay and a wave equation into oscillation. Keeping track of the transform convention makes these signs predictable.

13.4 Dispersion and wave packets

A single infinite sinusoid has no identifiable beginning or localized center. To describe a pulse, combine nearby wave numbers. Their phases remain close in some regions and cancel in others, creating an envelope. The speed of that envelope need not equal the speed of the individual crests inside it. A positive k and ω define the direction of the phase $kx - \omega t$; reversing that sign reverses propagation.

A dispersion relation specifies $\omega(k)$. Phase and group speeds are

$$v_p = \frac{\omega}{k}, \qquad v_g = \frac{\mathrm{d}\omega}{\mathrm{d}k}. \tag{13.7}$$

The group velocity tracks a narrow packet envelope under suitable weak-distortion conditions. If $\omega''(k) \neq 0$, a packet generally spreads. In strongly absorbing or anomalously dispersive media, the group velocity need not be the information velocity.

Add two equal-amplitude waves with nearby frequencies and wave numbers. The trigonometric sum becomes

$$2A\cos\left(\frac{\Delta k\, x - \Delta\omega\, t}{2}\right)\cos(\bar{k}x - \bar{\omega}t). \tag{13.8}$$

The slowly varying first factor moves at $\Delta\omega/\Delta k$. Taking a narrow-band limit gives $v_g = \mathrm{d}\omega/\mathrm{d}k$, while the rapid carrier moves at $\bar{\omega}/\bar{k}$. If all wave numbers have $\omega = vk$, both speeds equal v and an arbitrary packet can retain its shape in the ideal model.

13.5 Sound and the Doppler effect

Sound is a sequence of small compressions and rarefactions. A compression raises the pressure, pressure gradients accelerate the fluid, and the resulting motion changes the density. The relevant stiffness is how much the pressure changes when a fluid element is compressed under the conditions of the wave. In sound, ρ_0 is background density and v_s sound speed.

Linear pressure disturbances in a fluid travel at

$$v_s^2 = \left(\frac{\partial p}{\partial \rho}\right)_s = \frac{\gamma p}{\rho} \tag{13.9}$$

for an ideal gas with locally adiabatic compression. Plane-wave acoustic impedance is $Z = \rho v_s$, and intensity is $I = p_{\mathrm{rms}}^2/Z$. The sound intensity level is $10 \log_{10}(I/I_0)$ decibels for a specified reference I_0.

For collinear motion through a stationary medium, using positive observer speed toward the source and positive source speed toward the observer,

$$f_{\mathrm{obs}} = f_{\mathrm{src}} \frac{v_s + v_o}{v_s - v_{\mathrm{src}}}. \tag{13.10}$$

This classical formula depends on motion relative to the medium and differs from relativistic light Doppler shifts.

Around a uniform background ρ_0, p_0, use small perturbations $\delta\rho, \delta p$ and velocity u in one dimension. Linearized continuity and momentum balance give

$$\partial_t \delta\rho = -\rho_0 \partial_x u, \qquad \rho_0 \partial_t u = -\partial_x \delta p. \tag{13.11}$$

With $\delta p = v_s^2 \delta\rho$, take a time derivative of the first equation and substitute the second. You obtain $\partial_t^2 \delta\rho = v_s^2 \partial_x^2 \delta\rho$. For an adiabatic ideal-gas change, $p \propto \rho^\gamma$, so the derivative is $v_s^2 = \gamma p_0/\rho_0$.

For Doppler shifts, count crests. In one source period, a source moving toward the observer advances by $v_{\mathrm{src}}/f_{\mathrm{src}}$, shortening the forward wavelength to $(v_s - v_{\mathrm{src}})/f_{\mathrm{src}}$. The observer meets those crests at relative speed $v_s + v_o$. Dividing that speed by the wavelength gives the stated frequency ratio.

13.6 Superposition and coherence

Fields add before intensities do. Represent two monochromatic contributions by complex amplitudes A_1 and $A_2 \mathrm{e}^{\mathrm{i}\delta}$. The detector measures a quantity proportional to the squared magnitude of their sum. Expanding that square is the whole origin of the interference term.

For two coherent contributions with intensities I_1, I_2 and phase difference δ,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos\delta. \tag{13.12}$$

The interference term averages away when relative phase varies rapidly over the measurement. Coherence describes stable phase correlations, not merely equal wavelength.

With real nonnegative A_1, A_2 and a common intensity normalization,

$$|A_1 + A_2 \mathrm{e}^{\mathrm{i}\delta}|^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos\delta. \tag{13.13}$$

For equal individual intensities I_0, constructive interference gives $4I_0$ and destructive interference gives zero. Energy is redistributed across the interference pattern rather than created at the bright fringes. If the phase wanders, the measured cross term contains $\langle \cos\delta \rangle$; a uniformly wandering phase makes that average vanish.

Exercises

A string is made twice as long while its wave speed stays fixed. What happens to its lowest standing-wave frequency? Sketch the nodes of that mode.

Answers are in Appendix A.

Chapter 14

Geometrical and physical optics

Waves have shown us why phase matters. Optics gives us a useful change of scale: when wavelength is small, the wave description becomes a ray picture; when apertures become comparable to wavelength, the full interference structure returns. This will be our first sustained example of one theory yielding different approximations. Optics offers two complementary descriptions. Rays are useful when wavelength is small compared with the features that shape the light. Wave amplitudes are necessary when phase differences, apertures, or polarization matter. We will connect these descriptions by asking which information each approximation retains.

14.1 Rays, refraction, and imaging

A ray between two points does not generally choose the shortest geometric distance when it crosses different media. Its travel time depends on the refractive index along the route. Let the crossing point move slightly along the interface and require the first-order change in travel time to vanish. n is refractive index, θ an angle measured from the normal, f focal length, and s, s' object and image distances. The sign of an image distance records whether the image is real or virtual; it is not an arbitrary algebraic sign.

In the short-wavelength limit, Fermat's principle makes optical travel time stationary. Reflection has equal incidence and reflection angles; refraction obeys

$$n_1 \sin\theta_1 = n_2 \sin\theta_2. \tag{14.1}$$

For $n_1 > n_2$, total internal reflection occurs above $\theta_c = \arcsin(n_2/n_1)$. The transmitted side still contains an evanescent field near the interface.

For a thin paraxial lens in air, with real object distance $s > 0$ and real image distance $s' > 0$,

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}, \qquad m = -\frac{s'}{s}. \tag{14.2}$$

A virtual image has $s' < 0$. With radii positive when their centers of curvature lie to the right for left-to-right incident light, the thin lensmaker equation is $1/f = (n-1)(1/R_1 - 1/R_2)$. Large angles and finite apertures introduce aberrations.

Place the endpoints horizontal distance D apart and perpendicular distances h_1, h_2 from a flat interface. If the crossing point is at horizontal coordinate x, the travel time is

$$t(x) = \frac{n_1}{c}\sqrt{x^2 + h_1^2} + \frac{n_2}{c}\sqrt{(D-x)^2 + h_2^2}. \tag{14.3}$$

Setting its derivative to zero gives

$$n_1 \frac{x}{\sqrt{x^2 + h_1^2}} = n_2 \frac{D-x}{\sqrt{(D-x)^2 + h_2^2}}. \tag{14.4}$$

The ratios are the sines of the angles to the normal, giving Snell's law. For imaging, the paraxial thin-lens construction uses a central undeviated ray and a ray initially parallel to the axis. Similar triangles give $y'/y = -s'/s$ and $y' = y(1 - s'/f)$. Equating these expressions yields $1/f = 1/s + 1/s'$. The minus sign in magnification records an inverted image.

14.2 Diffraction as Fourier analysis

To understand a slit, add the amplitudes from every illuminated point across its width. At a distant observation angle, contributions from different points travel slightly different distances. Those path differences become phase differences. The diffraction pattern is the result of that continuous interference. In diffraction, λ is wavelength, a slit width, and d slit separation.

In Fraunhofer diffraction, the far-field complex amplitude is proportional to the Fourier transform of the aperture field. For a uniformly illuminated slit of width a,

$$I(\theta) = I(0)\left(\frac{\sin\beta}{\beta}\right)^2, \qquad \beta = \frac{\pi a \sin\theta}{\lambda}. \tag{14.5}$$

The first minima satisfy $a\sin\theta = \pm\lambda$. Two narrow slits separated by d yield maxima at $d\sin\theta = m\lambda$, modulated by each slit's diffraction envelope.

For a circular aperture of diameter D, the first dark ring is at approximately $1.22\lambda/D$ radians for small angles. The Rayleigh criterion is a useful resolution convention, not an absolute limit independent of noise or measurement method. Grating maxima obey $d\sin\theta = m\lambda$ at normal incidence.

Let x run across the slit and set $q = k\sin\theta$. Apart from common factors, the far-field amplitude is

$$\mathcal{A}(\theta) \propto \int_{-a/2}^{a/2} \mathrm{e}^{\mathrm{i}qx}\,\mathrm{d}x = \frac{2\sin(qa/2)}{q} = a\frac{\sin\beta}{\beta}. \tag{14.6}$$

At $\theta = 0$, take the limit $\sin\beta/\beta \to 1$. Divide by that central amplitude and square the magnitude to obtain the intensity profile. The first zero occurs when $\beta = \pi$, giving $a\sin\theta = \lambda$. A narrower slit spreads the pattern more widely because greater angular change is needed to accumulate the same phase difference across it.

14.3 Polarization and interfaces

An analyzer transmits the electric-field component along its axis. Projecting a field amplitude through angle θ multiplies it by $\cos\theta$; squaring to obtain intensity gives Malus's law. At a dielectric interface, reflection instead follows from matching the electromagnetic fields on both sides.

Polarization specifies the transverse electric field's direction and phase trajectory. Two equal orthogonal components with a $\pi/2$ relative phase produce circular polarization. An ideal linear analyzer obeys Malus's law, $I = I_0 \cos^2\theta$, for linearly polarized input.

At normal incidence between transparent nonmagnetic media,

$$R = \left|\frac{n_1 - n_2}{n_1 + n_2}\right|^2, \qquad T = 1 - R. \tag{14.7}$$

The transmitted intensity includes the impedance factor; it is not simply the square of the electric-field amplitude ratio. At oblique incidence, the Fresnel coefficients differ for the two linear polarizations. Brewster's angle satisfies $\tan\theta_B = n_2/n_1$ for the same ideal media.

Worked example 14.1 (Interference fringe spacing)**.** For slit separation d and screen distance $L \gg d$, with small angles, $y_m/L \simeq \sin\theta_m = m\lambda/d$. Thus adjacent bright fringes are separated by $\Delta y = \lambda L/d$. Decreasing slit width broadens the envelope; decreasing slit separation broadens the fringe spacing.

For normal incidence in transparent nonmagnetic media, tangential field continuity gives

$$E_i + E_r = E_t, \qquad n_1(E_i - E_r) = n_2 E_t. \tag{14.8}$$

The reflected magnetic field has the opposite sign because the wave travels in the opposite direction. Solving these two equations gives $r = E_r/E_i = (n_1 - n_2)/(n_1 + n_2)$ and $t = E_t/E_i = 2n_1/(n_1 + n_2)$. The energy fractions are $R = |r|^2$ and $T = (n_2/n_1)|t|^2$. Their sum is one, providing an immediate check on the amplitude calculation.

14.4 Lasers and optical amplification

A laser requires more than emission of light. Amplification must favor the same optical mode strongly enough to overcome losses, and pumping must replenish the energy given to that mode. Think of the resonator as repeatedly returning part of the field through the gain material: below threshold it fades; above threshold growth continues until saturation changes the gain.

Stimulated emission generates a photon in the stimulating mode. A laser combines gain, feedback, and a pumping process that sustains nonequilibrium populations. In a simple gain description, intensity grows as $I(z) = I(0)\mathrm{e}^{gz}$ before saturation. Laser threshold occurs when round-trip gain balances round-trip loss. An isolated two-level system cannot maintain population inversion under steady resonant pumping alone; practical gain schemes use additional levels or other mechanisms.

If a short distance $\mathrm{d}z$ increases intensity by a fraction $g\mathrm{d}z$, then $\mathrm{d}I = gI\mathrm{d}z$ and integration gives $I(z) = I(0)\mathrm{e}^{gz}$. For a cavity of length L, mirror intensity reflectivities R_1, R_2, and distributed intensity-loss coefficient α, a round trip multiplies intensity by

$$R_1 R_2 \mathrm{e}^{2(g-\alpha)L}. \tag{14.9}$$

Threshold sets this factor to one, giving $g_{\mathrm{th}} = \alpha + (2L)^{-1}\ln[1/(R_1R_2)]$. This elementary model shows how material gain, cavity length, and mirror losses enter the same balance.

Exercises

Draw a ray crossing from air into a slower medium and show how it bends. Explain when a wave description is needed to predict an optical effect.

Answers are in Appendix A.

Part V

Thermal and Statistical Physics

Chapter 15

Thermodynamics

The conservation laws of mechanics and fluids tracked energy locally. Thermodynamics asks what can be said when the microscopic details are deliberately hidden. We will keep the same discipline of defining a system and its boundary, but now add temperature, entropy, and the direction of spontaneous change. Thermodynamics asks what we can predict without following every molecule. Its strength comes from carefully defining a system, the energy transfers across its boundary, and the constraints on its states. Much of the mathematics is ordinary calculus; the subtlety is knowing which variables are held fixed and which processes can actually occur.

15.1 A system, its boundary, and temperature

Place a gas inside a cylinder with a movable piston. The gas is our system; the cylinder walls and piston form its boundary. We can warm the gas through a wall or compress it by pushing the piston. These actions change the gas's internal energy U, which includes microscopic kinetic and interaction energy. We do not need to track every molecule to measure a change in U.

When two bodies are placed in thermal contact and no net heat continues to flow between them, they are in thermal equilibrium. Temperature T is the quantity that agrees for systems in such mutual equilibrium. Pressure p measures the normal force the gas exerts per area of wall, and volume V describes the space available to it. For a simple gas at equilibrium, the quantities p, V, T are linked by an equation of state; they are not three independent knobs once the amount of gas is fixed.

15.2 The first law: keeping track of energy

Energy crosses the gas boundary in different ways. Heat Q is energy transferred because of a temperature difference. Work W_{by} is energy the gas transfers through organized motion of its boundary, such as pushing the piston. We choose $Q > 0$ for heat entering the gas and $W_{\text{by}} > 0$ for

work done by the gas. Conservation of energy then reads

$$\Delta U = Q - W_{\mathrm{by}}. \tag{15.1}$$

The signs are part of the model, not a fact to guess from a formula. If the gas receives 100 J of heat and pushes the piston with 40 J of work, its internal energy rises by 60 J. If it instead loses 100 J of heat with no work, $\Delta U = -100\,\mathrm{J}$.

Internal energy is a property of the state: once initial and final equilibrium states are fixed, ΔU is fixed. Heat and work describe the path between those states. A gas could reach the same final state by receiving more heat and doing more work, or receiving less of both. For a small change, we write $\mathrm{d}U = \delta Q - \delta W_{\mathrm{by}}$; the different symbols remind us that U has state values while Q and W label transfers.

15.3 Why piston work equals pressure times volume change

Suppose the piston has area A and moves outward a small distance $\mathrm{d}x$. The gas pushes it with force approximately $F = pA$ if the gas stays close enough to mechanical equilibrium for its pressure to be well defined throughout the move. Force times displacement gives work $\delta W_{\mathrm{by}} = F\mathrm{d}x = pA\mathrm{d}x$. Since the gas volume increases by $\mathrm{d}V = A\mathrm{d}x$,

$$\delta W_{\mathrm{by}} = p\mathrm{d}V. \tag{15.2}$$

Expansion has $\mathrm{d}V > 0$ and positive work by the gas. Compression has $\mathrm{d}V < 0$, so work by the gas is negative: the surroundings do work on it. For a finite quasistatic path, add the small pieces, $W_{\mathrm{by}} = \int_{V_i}^{V_f} p(V)\mathrm{d}V$. We need the pressure along the path, not only its endpoint values.

Worked example 15.1 (Two ways to raise internal energy)**.** First, heat a gas in a rigid container. It cannot change volume, so pressure-volume work is zero and $\Delta U = Q$. Next, allow the same gas to expand under a movable piston while heating it. If the same heat enters but the gas does positive work, equation (15.1) says its internal-energy increase is smaller. This is why specifying only the added heat does not determine the final temperature for every process.

A simple ideal-gas model obeys $pV = Nk_{\mathrm{B}}T$, where N is particle number and k_{B} is Boltzmann's constant. For a monatomic ideal gas under conditions where classical equipartition applies, $U = 3Nk_{\mathrm{B}}T/2$. We will derive the statistical reason later; here the formula tells us how a measured temperature change relates to internal-energy change. Other materials require different relations, so the first law itself is more general than this gas model.

15.4 Entropy and the second law

Energy conservation permits many processes that never occur spontaneously, such as heat flowing from a cold object into a hotter one without compensation. Entropy supplies the missing restriction.

To calculate an entropy change between equilibrium states, you may choose a reversible reference path even when the actual process was irreversible, because entropy itself is a state function.

For a reversible heat transfer, $\mathrm{d}S = \delta Q_{\mathrm{rev}}/T$. More generally,

$$\Delta S \geq \int \frac{\delta Q}{T_{\mathrm{boundary}}}, \qquad \Delta S_{\mathrm{isolated}} \geq 0. \tag{15.3}$$

Entropy can decrease locally if enough entropy is transferred elsewhere. The second law constrains total entropy production, not every subsystem's entropy separately.

For equilibrium neighboring states of a simple one-component system,

$$\boxed{\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}N.} \tag{15.4}$$

This fundamental relation identifies temperature, pressure, and chemical potential as derivatives of $U(S, V, N)$. It is a relation among state variables; it does not imply every actual process is reversible.

For an ideal gas with fixed particle number and constant C_V, use a reversible path and combine $\mathrm{d}U = C_V\mathrm{d}T$ with the first law:

$$T\mathrm{d}S = \mathrm{d}U + p\mathrm{d}V = C_V\mathrm{d}T + \frac{Nk_{\mathrm{B}}T}{V}\mathrm{d}V. \tag{15.5}$$

Divide by T and integrate to find

$$\Delta S = C_V \ln\frac{T_f}{T_i} + Nk_{\mathrm{B}} \ln\frac{V_f}{V_i}. \tag{15.6}$$

The formula depends only on endpoints. For insulated free expansion of an ideal gas, $T_f = T_i$ but $V_f > V_i$, so entropy increases even though no heat enters. The actual heat transfer is not equal to $T\mathrm{d}S$ along an irreversible process.

15.5 Heat capacities and ideal-gas processes

Heating at fixed volume and heating at fixed pressure ask different things of the added energy. At fixed volume, pressure-volume work vanishes. At fixed pressure, a gas usually expands as it warms, so some of the supplied energy leaves as work. That distinction explains why the two heat capacities differ.

Define total heat capacities $C_V = (\partial U/\partial T)_{V,N}$ and $C_p = (\partial H/\partial T)_{p,N}$, where $H = U + pV$. An ideal gas has $C_p - C_V = Nk_{\mathrm{B}}$ and $\gamma = C_p/C_V$. With constant heat capacities,

$$\text{isothermal:} \quad \Delta U = 0, \quad W_{\mathrm{by}} = Nk_{\mathrm{B}}T\ln(V_f/V_i), \tag{15.7}$$

$$\text{reversible adiabatic:} \quad pV^{\gamma} = \text{constant}, \quad TV^{\gamma-1} = \text{constant}. \tag{15.8}$$

An adiabatic process has no heat transfer; it is isentropic only if entropy production also vanishes.

Worked example 15.2 (Free expansion)**.** An ideal gas expands into a vacuum in a thermally insulated rigid container. There is no external work and no heat transfer, so $\Delta U = 0$ and $T_f = T_i$. Nevertheless $\Delta S = Nk_{\mathrm{B}} \ln(V_f/V_i) > 0$. The reversible adiabatic relation $pV^{\gamma} = \text{constant}$ does not apply to this irreversible process.

For an ideal gas, $H = U + pV = U + Nk_{\mathrm{B}}T$, so differentiation at fixed particle number gives $C_p = C_V + Nk_{\mathrm{B}}$. To derive the reversible adiabatic law, set $\delta Q = 0$ in the first law:

$$C_V \mathrm{d}T = -p\mathrm{d}V = -\frac{Nk_{\mathrm{B}}T}{V}\mathrm{d}V, \tag{15.9}$$

$$\frac{\mathrm{d}T}{T} = -\frac{Nk_{\mathrm{B}}}{C_V}\frac{\mathrm{d}V}{V} = -(\gamma - 1)\frac{\mathrm{d}V}{V}, \tag{15.10}$$

$$\ln T + (\gamma - 1)\ln V = \text{constant}. \tag{15.11}$$

Exponentiation gives $TV^{\gamma-1} = \text{constant}$; using $T = pV/(Nk_{\mathrm{B}})$ gives $pV^{\gamma} = \text{constant}$. The use of gas pressure in the work term requires the reversible mechanical path assumed here. Setting heat to zero alone would not justify this derivation.

15.6 Engines and refrigerators

Track one complete cycle so that the working substance returns to its initial state. Its net changes of internal energy and entropy are then zero. The reservoirs, however, have changed. Energy conservation determines the work; their total entropy change limits how large that work can be.

A cyclic engine absorbs Q_h from a hot reservoir and rejects magnitude Q_c to a cold reservoir. Its efficiency is $\eta = W/Q_h = 1 - Q_c/Q_h$. The entropy inequality gives

$$\eta \le 1 - \frac{T_c}{T_h}, \tag{15.12}$$

with equality for a reversible Carnot engine. A refrigerator's coefficient of performance is $\mathrm{COP} = Q_c/W \le T_c/(T_h - T_c)$. These ratios use absolute temperatures.

The hot reservoir loses entropy Q_h/T_h and the cold reservoir gains Q_c/T_c. Consequently

$$\Delta S_{\text{total}} = -\frac{Q_h}{T_h} + \frac{Q_c}{T_c} \ge 0 \quad \Longrightarrow \quad \frac{Q_c}{Q_h} \ge \frac{T_c}{T_h}. \tag{15.13}$$

Insert this into $\eta = 1 - Q_c/Q_h$ to obtain the Carnot bound. The result does not depend on whether the working substance is a gas or something else. It follows from the reservoir temperatures and the second law, with equality only when the total entropy production vanishes.

15.7 Thermodynamic potentials and Maxwell relations

Choose a thermodynamic potential by asking what the surroundings control. A bath fixes temperature more naturally than entropy; an external load may fix pressure more naturally than volume. A

Legendre transform changes the natural variables so that the potential matches those controls. The extra terms have a purpose: they cancel the differentials we no longer want to use as independent inputs.

Legendre transforms exchange natural variables:

$$H = U + pV, \qquad \mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p + \mu\mathrm{d}N, \tag{15.14}$$

$$F = U - TS, \qquad \mathrm{d}F = -S\mathrm{d}T - p\mathrm{d}V + \mu\mathrm{d}N, \tag{15.15}$$

$$G = U - TS + pV, \qquad \mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}N. \tag{15.16}$$

At fixed T, V, N, equilibrium minimizes Helmholtz free energy F; at fixed T, p, N, it minimizes Gibbs free energy G, subject to the available states and other constraints. Equality of mixed partial derivatives gives, for example,

$$\left(\frac{\partial S}{\partial V}\right)_{T,N} = \left(\frac{\partial p}{\partial T}\right)_{V,N}. \tag{15.17}$$

Such Maxwell relations turn hard-to-measure derivatives into accessible ones.

At fixed N, start with $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$ and define $F = U - TS$. The product rule gives

$$\mathrm{d}F = \mathrm{d}U - T\mathrm{d}S - S\mathrm{d}T = -S\mathrm{d}T - p\mathrm{d}V. \tag{15.18}$$

Thus $S = -(\partial F/\partial T)_V$ and $p = -(\partial F/\partial V)_T$. Differentiate each with respect to the other variable. Equality of mixed derivatives of a smooth $F(T, V)$ yields

$$\left(\frac{\partial S}{\partial V}\right)_T = -\frac{\partial^2 F}{\partial V \partial T} = -\frac{\partial^2 F}{\partial T \partial V} = \left(\frac{\partial p}{\partial T}\right)_V. \tag{15.19}$$

Rather than memorizing a disconnected list of Maxwell relations, derive the needed one from the differential of the appropriate potential.

15.8 Phase equilibrium and stability

When liquid and vapor coexist, neither phase can lower the total Gibbs energy by taking particles from the other. Their chemical potentials must match. Following that equality as temperature and pressure change gives the slope of the coexistence curve.

Coexisting phases share T, p, and chemical potential for each exchanged species. For a first-order transition, the coexistence curve obeys the Clapeyron equation

$$\frac{\mathrm{d}p}{\mathrm{d}T} = \frac{\Delta s}{\Delta v} = \frac{\ell}{T\Delta v}, \tag{15.20}$$

where s, v, ℓ are entropy, volume, and latent heat per particle (or all per mole). Stability requires positive appropriate heat capacities and positive isothermal compressibility, $\kappa_T = -V^{-1}(\partial V/\partial p)_T$.

A critical point ends a first-order coexistence line and can exhibit large fluctuations and diverging correlation length. The third law, in its usual perfect-crystal formulation, gives entropy approaching zero for a unique ground state as $T \to 0$; ground-state degeneracy can produce residual entropy.

For a pure phase, the chemical potential obeys $\mathrm{d}\mu = -s\mathrm{d}T + v\mathrm{d}p$, with s and v taken per particle. Along coexistence, $\mathrm{d}\mu_1 = \mathrm{d}\mu_2$, so

$$(v_2 - v_1)\mathrm{d}p = (s_2 - s_1)\mathrm{d}T. \tag{15.21}$$

A reversible phase conversion at temperature T transfers latent heat per particle $\ell = T(s_2 - s_1)$. Combining the two statements produces the Clapeyron equation. The sign of the volume change matters: it determines whether increasing pressure raises or lowers the transition temperature for a positive latent heat.

Exercises

A gas expands while being heated. State how you would account for heat, work, and internal-energy change, including your sign convention.

Answers are in Appendix A.

Chapter 16

Statistical mechanics

Thermodynamics gave us macroscopic rules without asking what each molecule was doing. Statistical mechanics now opens that box carefully. We will introduce probability here because we need it to connect microscopic possibilities with measurable averages. Thermodynamics told us which macroscopic changes are possible. Statistical mechanics asks why those rules emerge from microscopic states. The strategy is to assign probabilities consistently with the system's surroundings, count the states, and average the quantities we can measure. The partition function will become a compact way to perform many of those calculations at once.

16.1 Probabilities for microscopic states

A gas can have a definite temperature without our knowing the position and velocity of every molecule. We need a way to describe several possible microscopic states at once. Give each state i a probability p_i. Each probability is between zero and one, and the probabilities of all mutually exclusive possibilities add to one:

$$p_i \geq 0, \qquad \sum_i p_i = 1. \tag{16.1}$$

These are bookkeeping rules for possible outcomes, not new forces. For a system that can have energy E_i in state i, the mean energy is the probability-weighted sum $\langle E \rangle = \sum_i p_i E_i$. A mean need not equal the energy seen in any single observation.

Worked example 16.1 (A two-level system)**.** Suppose a system has energy 0 with probability 3/4 and energy Δ with probability 1/4. The probabilities normalize because $3/4 + 1/4 = 1$. Its mean energy is $(3/4)0 + (1/4)\Delta = \Delta/4$. Repeating the preparation many times would make the average observed energy approach this value, even though each individual measurement returns either 0 or Δ.

Probabilities also describe fluctuations. The variance of energy is the mean squared departure from its mean:

$$\sigma_E^2 = \sum_i p_i (E_i - \langle E \rangle)^2 = \left\langle E^2 \right\rangle - \langle E \rangle^2 . \tag{16.2}$$

For the example, $\langle E^2 \rangle = \Delta^2/4$ and $\sigma_E^2 = 3\Delta^2/16$. This tells us how widely individual observations spread around the average. Later we will connect such microscopic fluctuations to macroscopic response.

When a quantity such as position can vary continuously, probabilities over individual labels become a density $p(x)$. The probability of a narrow interval is approximately $p(x)\mathrm{d}x$, normalization becomes $\int p(x)\mathrm{d}x = 1$, and an average becomes $\langle f \rangle = \int f(x)p(x)\mathrm{d}x$. A density may have units of inverse length; the probability of an interval is dimensionless.

16.2 Microstates, macrostates, and ensembles

A gas can have the same pressure and temperature while its molecules occupy enormously many different positions and momenta. A macrostate records the coarse information; a microstate specifies the microscopic configuration. An ensemble is a probability model for such microstates, chosen to reflect what the system can exchange with its surroundings. i labels a microstate, E_i its energy, p_i its probability, and Z the partition function that normalizes the weights.

A macrostate groups many microscopic configurations. Boltzmann's entropy is $S = k_\mathrm{B} \ln \Omega$ for equally likely accessible microstates. For general probabilities,

$$S = -k_\mathrm{B} \sum_i p_i \ln p_i. \tag{16.3}$$

The microcanonical ensemble fixes E, V, N. The canonical ensemble fixes T, V, N by exchanging energy with a reservoir. The grand canonical ensemble fixes T, V, μ by exchanging energy and particles. Ensemble equivalence typically emerges for large ordinary systems away from qualifications such as long-range interactions.

If there are Ω equally probable states, set $p_i = 1/\Omega$ in the general entropy expression. Then

$$S = -k_\mathrm{B} \sum_{i=1}^{\Omega} \frac{1}{\Omega} \ln \frac{1}{\Omega} = k_\mathrm{B} \ln \Omega. \tag{16.4}$$

The logarithm also makes entropy additive for independent systems: their numbers of states multiply, $\Omega_{AB} = \Omega_A \Omega_B$, so $S_{AB} = S_A + S_B$. This is a useful bridge between microscopic counting and macroscopic thermodynamics.

16.3 The canonical partition function

Consider a small system exchanging energy with a much larger reservoir. If the system occupies a state of energy E_i, the reservoir has that much less energy available. The probability of the state is proportional to the number of reservoir microstates compatible with the remaining energy. This is the physical source of the Boltzmann factor. The parameter $\beta = 1/(k_\mathrm{B}T)$ increases as temperature falls. Averages use the probabilities p_i; they are not automatically the value of a single microstate.

A reservoir's multiplicity gives probability proportional to $\mathrm{e}^{-\beta E_i}$, where $\beta = 1/(k_{\mathrm{B}}T)$. Define

$$Z = \sum_i \mathrm{e}^{-\beta E_i}, \quad p_i = \frac{\mathrm{e}^{-\beta E_i}}{Z}, \quad F = -k_{\mathrm{B}}T \ln Z, \quad U = -\frac{\partial \ln Z}{\partial \beta}. \tag{16.5}$$

Further derivatives yield

$$p = k_{\mathrm{B}}T \left(\frac{\partial \ln Z}{\partial V} \right)_{T,N}, \qquad \left\langle (\Delta E)^2 \right\rangle = k_{\mathrm{B}}T^2 C_V. \tag{16.6}$$

The fluctuation relation links microscopic variability to a macroscopic response.

Worked example 16.2 (Two-level system)**.** For one particle with nondegenerate energies 0 and $\Delta > 0$, $Z = 1 + \mathrm{e}^{-\beta\Delta}$ and

$$U = \frac{\Delta}{\mathrm{e}^{\beta\Delta} + 1}, \qquad C_V = k_{\mathrm{B}}(\beta\Delta)^2 \frac{\mathrm{e}^{\beta\Delta}}{(1 + \mathrm{e}^{\beta\Delta})^2}. \tag{16.7}$$

At low temperature the excited state is rarely occupied. At high temperature both states become nearly equally populated, so $U \to \Delta/2$ and the heat capacity again tends to zero.

Write the reservoir entropy as $S_R(E_{\mathrm{tot}} - E_i)$ and expand about E_{tot}:

$$S_R(E_{\mathrm{tot}} - E_i) \simeq S_R(E_{\mathrm{tot}}) - \left(\frac{\partial S_R}{\partial E_R} \right) E_i = S_R(E_{\mathrm{tot}}) - \frac{E_i}{T}. \tag{16.8}$$

Because reservoir multiplicity is proportional to $\mathrm{e}^{S_R/k_{\mathrm{B}}}$, the relative weight is $\mathrm{e}^{-\beta E_i}$. Summing all weights gives Z, which normalizes the probabilities. Now differentiate:

$$-\frac{\partial \ln Z}{\partial \beta} = -\frac{1}{Z}\frac{\partial Z}{\partial \beta} = \frac{\sum_i E_i \mathrm{e}^{-\beta E_i}}{Z} = \langle E \rangle . \tag{16.9}$$

For a temperature-independent spectrum, a second derivative gives $\partial_\beta^2 \ln Z = \left\langle E^2 \right\rangle - \langle E \rangle^2$. Since $\mathrm{d}\beta/\mathrm{d}T = -1/(k_{\mathrm{B}}T^2)$, this becomes $\left\langle (\Delta E)^2 \right\rangle = k_{\mathrm{B}}T^2 C_V$. Differentiating a normalization sum has revealed both the mean energy and its fluctuations.

16.4 A second route to the same probabilities

The reservoir argument gave us the Boltzmann weight. We can reach the same result by asking which probabilities make the least extra assumptions when we know only that they add to one and have a specified mean energy U. The measure of spread over possible states is the dimensionless Shannon entropy $S_0 = -\sum_i p_i \ln p_i$; physical entropy is $k_{\mathrm{B}} S_0$.

We want to maximize S_0 while enforcing $\sum_i p_i = 1$ and $\sum_i p_i E_i = U$. Introduce constants α and β to keep those constraints during the variation:

$$\mathcal{F} = -\sum_i p_i \ln p_i - \alpha \left(\sum_i p_i - 1 \right) - \beta \left(\sum_i p_i E_i - U \right). \tag{16.10}$$

For each p_i, the derivative of $-p_i \ln p_i$ is $-(\ln p_i + 1)$. Setting the first-order change to zero gives $-(\ln p_i + 1) - \alpha - \beta E_i = 0$, or $p_i = C\mathrm{e}^{-\beta E_i}$. Normalization sets $C = 1/Z$, with $Z = \sum_i \mathrm{e}^{-\beta E_i}$. The temperature relation identifies $\beta = 1/(k_\mathrm{B}T)$. The calculation shows exactly which information selected the distribution: normalization, mean energy, and no further constraint.

If we know another conserved average, we must add another constraint and its multiplier. If we know less, we cannot claim the same probabilities. This is why the ensemble always reflects what a system can exchange with its surroundings.

16.5 Classical ideal gas and equipartition

For a dilute noninteracting gas, position and momentum can be integrated separately. Every location in the container contributes equally, while large momenta are suppressed by their kinetic-energy Boltzmann factor. The Gaussian integral from calculus therefore becomes the basic counting tool for the gas.

For N identical dilute particles with negligible interactions,

$$Z_N = \frac{1}{N!}\left(\frac{V}{\lambda_T^3}\right)^N, \qquad \lambda_T = \frac{h}{\sqrt{2\pi m k_\mathrm{B} T}}. \tag{16.11}$$

The $N!$ factor removes classical overcounting of permutations of identical particles. The Maxwell–Boltzmann regime requires $n\lambda_T^3 \ll 1$, where $n = N/V$. Differentiating $\ln Z$ yields $pV = Nk_\mathrm{B}T$ and $U = 3Nk_\mathrm{B}T/2$.

Each independent quadratic term contributes $k_\mathrm{B}T/2$ under classical equilibrium conditions. The speed density is

$$f(v) = 4\pi\left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{3/2} v^2 \mathrm{e}^{-mv^2/(2k_\mathrm{B}T)}, \quad v \geq 0. \tag{16.12}$$

It gives $v_\mathrm{rms} = \sqrt{3k_\mathrm{B}T/m}$, distinct from the mean and most probable speeds.

For one spinless particle, the classical phase-space partition integral is

$$Z_1 = \frac{1}{h^3}\int_V \mathrm{d}^3 r \int \mathrm{d}^3 p\, \mathrm{e}^{-\beta(p_x^2+p_y^2+p_z^2)/(2m)} \tag{16.13}$$

$$= \frac{V}{h^3}\left(\frac{2\pi m}{\beta}\right)^{3/2} = \frac{V}{\lambda_T^3}. \tag{16.14}$$

For N identical particles, use $Z_N = Z_1^N/N!$. Then $\ln Z_N$ contains $N\ln V - (3N/2)\ln\beta$ plus terms independent of V and β. Its derivatives immediately give $p = Nk_\mathrm{B}T/V$ and $U = 3N/(2\beta)$. Each of the three quadratic momentum components contributes equally to that energy.

16.6 Quantum statistics

The new ingredient is the allowed occupation of one quantum state. A fermionic state can contain zero or one particle; a bosonic state can contain any nonnegative integer. Counting those possibil-

ities with the same energy and particle-exchange weight produces the two distribution functions. Their different denominators encode a physical difference in allowed states.

For noninteracting particles occupying single-particle levels ϵ_j, mean occupations are

$$\bar{n}_j = \frac{1}{\mathrm{e}^{\beta(\epsilon_j-\mu)}-1} \quad \text{(bosons)}, \qquad \bar{n}_j = \frac{1}{\mathrm{e}^{\beta(\epsilon_j-\mu)}+1} \quad \text{(fermions)}. \tag{16.15}$$

Bosons allow multiple occupancy of a single state; fermions obey Pauli exclusion. Both approach Maxwell–Boltzmann statistics when occupations are small. For fermions at $T = 0$, levels below the Fermi energy are filled. For a uniform three-dimensional ideal Bose gas with one internal state,

$$T_c = \frac{2\pi\hbar^2}{mk_\mathrm{B}}\left[\frac{n}{\zeta(3/2)}\right]^{2/3}, \quad \frac{N_0}{N} = 1 - \left(\frac{T}{T_c}\right)^{3/2} \quad (T < T_c). \tag{16.16}$$

This thermodynamic-limit result depends on dimension, dispersion, and confinement; it is not a universal formula for all traps or interacting systems.

For one level of energy ϵ, put $x = \mathrm{e}^{-\beta(\epsilon-\mu)}$. Its weight at occupation n is x^n. Fermions give $Z_\mathrm{level} = 1 + x$, while bosons give the geometric series $Z_\mathrm{level} = \sum_{n=0}^{\infty} x^n = 1/(1-x)$ for $x < 1$. In either case,

$$\bar{n} = x\frac{\mathrm{d}}{\mathrm{d}x}\ln Z_\mathrm{level}. \tag{16.17}$$

Thus $\bar{n}_F = x/(1+x)$ and $\bar{n}_B = x/(1-x)$, which are the distributions above. When $x \ll 1$, both reduce to x: exclusion and enhanced multiple occupancy matter little when a typical state is almost always empty.

16.7 Blackbody radiation

An equilibrium cavity contains electromagnetic standing-wave modes. Count how many modes lie in a small frequency interval, then multiply by the average thermal energy of each mode. This separates the problem into geometry, which determines the density of modes, and statistics, which determines their occupation.

Photons have two transverse polarizations and equilibrium chemical potential zero. Counting modes and using Bose statistics gives spectral energy density per angular frequency,

$$u_\omega(T) = \frac{\hbar\omega^3}{\pi^2c^3}\frac{1}{\mathrm{e}^{\hbar\omega/(k_\mathrm{B}T)}-1}. \tag{16.18}$$

Integration yields $u = aT^4$ and isotropic pressure $p = u/3$. A black surface emits flux $j^\star = \sigma T^4$, where

$$a = \frac{\pi^2k_\mathrm{B}^4}{15\hbar^3c^3}, \qquad \sigma = \frac{ac}{4}. \tag{16.19}$$

The spectral peak shifts inversely with temperature, but its numerical location depends on whether the spectrum is expressed per wavelength or per frequency. These peaks cannot be converted merely by $\lambda = c/\nu$ because the density's Jacobian changes.

Including two polarizations, the number of modes per volume in a spherical shell of wave-vector space is

$$\frac{\mathrm{d}N_{\mathrm{modes}}}{V} = 2\frac{4\pi k^2 \mathrm{d}k}{(2\pi)^3} = \frac{\omega^2 \mathrm{d}\omega}{\pi^2 c^3}. \tag{16.20}$$

Multiply by photon energy $\hbar\omega$ and mean Bose occupation $1/(\mathrm{e}^{\beta\hbar\omega} - 1)$ to obtain u_ω. For the total thermal energy density, set $x = \hbar\omega/(k_{\mathrm{B}}T)$:

$$u = \frac{(k_{\mathrm{B}}T)^4}{\pi^2\hbar^3 c^3}\int_0^\infty \frac{x^3}{\mathrm{e}^x - 1}\mathrm{d}x. \tag{16.21}$$

The integral is a pure number, so the T^4 law is already established before evaluating it. Its value $\pi^4/15$ fixes the coefficient. Excluding vacuum zero-point energy here is appropriate because we are calculating the thermal radiation energy.

16.8 Interactions, order, and critical behavior

Independent-particle counting cannot describe every material: neighbors can favor one another's alignment. A free-energy model expresses this competition between order and thermal disorder in terms of a collective variable. To find the predicted equilibrium, locate stationary points and then check which are minima.

The Ising model $H = -J\sum_{\langle ij\rangle} s_i s_j - h\sum_i s_i$, with $s_i = \pm 1$, illustrates collective order. An order parameter such as magnetization distinguishes phases. A schematic Landau free energy is

$$F(m) = F_0 + a(T - T_c)m^2 + bm^4, \qquad a, b > 0. \tag{16.22}$$

Minimization predicts $m = 0$ above T_c and $m^2 = a(T_c - T)/(2b)$ below it. Fluctuations can invalidate these mean-field exponents, especially in low dimensions. Universality groups very different systems by shared symmetry, dimension, and interaction range.

For the stated Landau model,

$$\frac{\mathrm{d}F}{\mathrm{d}m} = 2a(T - T_c)m + 4bm^3 = 2m[a(T - T_c) + 2bm^2]. \tag{16.23}$$

Above T_c, only $m = 0$ is a minimum. Below T_c, the curvature there becomes negative and two minima appear at $m = \pm\sqrt{a(T_c - T)/(2b)}$. The model respects $m \to -m$, but either ordered equilibrium chooses one sign. This is the elementary picture of spontaneous symmetry breaking; the quantitative role of fluctuations requires more than minimizing a uniform polynomial.

Exercises

A two-level system has a low-energy state and a high-energy state. Describe the likely occupations at low and high temperature and relate them to the mean energy.

Answers are in Appendix A.

Part VI

Electricity and Magnetism

Chapter 17

Electrostatics, potentials, and matter

The continuum chapters taught us to think in fields and fluxes; thermodynamics taught us to respect boundaries and state variables. Electromagnetism combines both habits. We begin with the simplest field, electrostatics, before allowing charge to move and electric and magnetic fields to evolve together. The electric field describes what a test charge would experience at each location. Potential describes the same electrostatic situation through energy per charge. We will move between these descriptions according to the geometry: symmetry often makes fields simple, while superposition and boundary-value problems often favor potential.

17.1 Flux through a closed surface

Before calculating an electric field, we need a way to count how much of it passes through a boundary. Imagine many short arrows representing a field $\boldsymbol{E}$ on a small flat patch of area $\mathrm{d}A$. Choose a unit arrow $\hat{\boldsymbol{n}}$ perpendicular to the patch. The contribution $\boldsymbol{E} \cdot \hat{\boldsymbol{n}}\, \mathrm{d}A$ is positive when the field points through the patch in the chosen outward direction, negative when it points inward, and zero when it runs along the patch.

Add the contributions over a closed surface to get its flux:

$$\Phi_E = \oint_{\partial V} \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S}, \qquad \mathrm{d}\boldsymbol{S} = \hat{\boldsymbol{n}}\, \mathrm{d}A. \tag{17.1}$$

The symbol ∂V means the boundary of a volume V. For a radial field of constant magnitude $E(r)$ on a sphere of radius r, every surface arrow points in the same direction as the outward normal. The integral then reduces to $\Phi_E = E(r) 4\pi r^2$.

Flux is a property of the chosen field and surface; it is not itself the number of field lines in a drawing. Its usefulness will become clear when we learn Gauss's law: the total outward electric flux is set by the charge enclosed. The same kind of boundary accounting later describes mass flowing out of a fluid region.

17.2 A field from one charge

A test charge placed near a source charge feels an electric force. Instead of recalculating the force for every test charge, define the electric field $\boldsymbol{E}$ at a point as force per unit positive test charge: $\boldsymbol{F} = q_{\text{test}}\boldsymbol{E}$. The field describes the source's influence throughout space even before we place the test charge there.

For a point source charge q at the origin, Coulomb's law gives

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{q}{4\pi\varepsilon_0 r^2}\hat{\boldsymbol{r}}, \tag{17.2}$$

where r is distance from the charge, $\hat{\boldsymbol{r}}$ points outward, and $\varepsilon_0 = \varepsilon_0$ is vacuum permittivity. A positive source produces outward field arrows; a negative source reverses them. The test charge's own sign then determines whether its force follows or opposes the field.

Several source charges contribute fields that add as vectors. This superposition works for the electrostatic vacuum model. When sources are spread through a region, we replace a sum of separate charges by an integral over small charge elements.

17.3 Gauss's law from a centered sphere

Place a sphere of radius r around one positive point charge. Symmetry makes the field radial and equal in magnitude everywhere on that sphere. The outward flux introduced above is therefore $E(r)4\pi r^2$. Substitute Coulomb's field:

$$\Phi_E = \frac{q}{4\pi\varepsilon_0 r^2}(4\pi r^2) = \frac{q}{\varepsilon_0}. \tag{17.3}$$

The r^2 in the surface area cancels the $1/r^2$ in the field. The same result holds for any closed surface enclosing the charge, even when the field is not uniform on that surface. For several charges, flux contributions add. The resulting Gauss law is

$$\oint_{\partial V} \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S} = \frac{Q_{\text{enc}}}{\varepsilon_0}, \tag{17.4}$$

where Q_{enc} is the total charge inside the chosen surface. Charges outside can alter the local field on the surface, but their net contribution to its closed-surface flux is zero.

Gauss's law is always a constraint on the field. It directly solves for field magnitude only when symmetry also tells us its direction and variation over a useful surface. For a uniformly charged solid sphere of radius R and total charge Q, a centered sphere at $r < R$ encloses the fraction $Q(r^3/R^3)$. Thus $E(r)4\pi r^2 = Q(r^3/R^3)/\varepsilon_0$, or

$$E(r) = \frac{Qr}{4\pi\varepsilon_0 R^3} \quad (r < R). \tag{17.5}$$

Outside, $E(r) = Q/(4\pi\varepsilon_0 r^2)$. Inside, enclosed charge grows like r^3 while area grows like r^2, so the field grows like r.

17.4 From a whole surface to a local rule

Imagine shrinking a closed box around one point. The outward flow of a vector field per unit box volume approaches its divergence $\boldsymbol{\nabla} \cdot \boldsymbol{E}$. Since enclosed charge per volume approaches charge density ρ, Gauss's law becomes the local equation $\boldsymbol{\nabla} \cdot \boldsymbol{E} = \rho/\varepsilon_0$. This is the same whole-region to local-region step used for fluid mass conservation.

In electrostatics, moving a test charge around a closed path returns it to its starting position without net electric work. The circulation $\oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{\ell}$ is therefore zero. Locally we write this as $\boldsymbol{\nabla} \times \boldsymbol{E} = 0$, where curl measures circulation per small area with an orientation. This statement changes when magnetic fields vary with time; we will revisit it in Maxwell's equations.

17.5 Potential as energy per charge

For an electrostatic field, define electric potential ϕ as potential energy per unit charge. A small positive charge moving along a small displacement $\mathrm{d}\boldsymbol{r}$ changes its potential energy by $-q\boldsymbol{E} \cdot \mathrm{d}\boldsymbol{r}$. Dividing by q gives $\mathrm{d}\phi = -\boldsymbol{E} \cdot \mathrm{d}\boldsymbol{r}$. In several dimensions, the spatial derivative vector of ϕ is its gradient, so

$$\boldsymbol{E} = -\boldsymbol{\nabla}\phi. \tag{17.6}$$

The field points toward decreasing potential for a positive test charge. Potential differences are physical; adding the same constant to ϕ everywhere changes no field.

Combine this relation with the local Gauss law to obtain $\nabla^2\phi = -\rho/\varepsilon_0$, where $\nabla^2\phi = \boldsymbol{\nabla} \cdot \boldsymbol{\nabla}\phi$ is the Laplacian. This is Poisson's equation. In a charge-free region it reduces to $\nabla^2\phi = 0$, called Laplace's equation. To determine a particular potential, we must also specify boundary values or suitable behavior at infinity. The next sections use those conditions for conductors and other charge distributions.

17.6 An advanced boundary tool: Green functions

Gauss's law is powerful when symmetry tells us the direction and size of the field on a simple surface. A conductor with an irregular boundary usually lacks that symmetry. Then we solve the potential equation with specified boundary values. A Green function is a systematic way to build the potential from responses to individual small sources, while respecting the boundary.

First look at a one-dimensional analogue. Let a potential-like function $y(x)$ live on $0 < x < L$, with its value fixed to zero at both ends. Suppose its source is $s(x)$ and its equation is $-y''(x) = s(x)$. Ask for the response to a source concentrated at one point x'. Call that response $G(x, x')$:

$$-\frac{\partial^2 G}{\partial x^2} = \delta(x - x'), \qquad G(0, x') = G(L, x') = 0. \tag{17.7}$$

The symbol $\delta(x - x')$ represents a unit source concentrated at x': integrating it across any small interval containing x' gives one. Away from that point, $G'' = 0$, so G is a straight line on either

side. The two lines meet continuously at x' and vanish at the endpoints. Integrating the equation through x' shows that the slope changes by -1 there. These four requirements determine

$$G(x,x') = \frac{x_<\,(L - x_>)}{L}, \qquad x_< = \min(x,x'), \quad x_> = \max(x,x'). \tag{17.8}$$

Each small source element $s(x')\mathrm{d}x'$ contributes its own response. Because the equation is linear, we add them:

$$y(x) = \int_0^L G(x,x')s(x')\,\mathrm{d}x'. \tag{17.9}$$

The kernel above is tied to the *type and location* of the homogeneous boundary conditions. Nonzero endpoint values do not require a new kernel: first write $y = h + u$, where a simple function h has the desired endpoint values. Then u has zero endpoint values and can be found with the same G. Changing from fixed-value to fixed-derivative conditions, or changing the domain, generally changes the Green function. The same construction works for electrostatic potential in three dimensions, though its geometry is more involved.

17.7 Conductors and boundary conditions

If an electrostatic field persisted inside an ideal conductor, mobile charges would continue rearranging. Equilibrium is reached when their rearrangement cancels the interior field. At the surface, the field can change abruptly in its normal component because a surface charge is present. Thin integration regions let us calculate that jump without resolving microscopic detail.

In electrostatic equilibrium an ideal conductor has zero internal electric field and constant potential. Excess charge lies on its surface. Across an interface,

$$E_{2\perp} - E_{1\perp} = \sigma/\varepsilon_0, \qquad \boldsymbol{E}_{2\parallel} = \boldsymbol{E}_{1\parallel} \tag{17.10}$$

when σ denotes total surface charge in vacuum-form equations. The normal points from region 1 to 2. Boundary conditions follow by applying Gauss's law to a thin pillbox and the zero-curl law to a narrow loop.

A pillbox of face area A straddling a charged surface has flux $(E_{2\perp} - E_{1\perp})A$ as its height shrinks. The enclosed charge is σA, so Gauss's law gives the normal jump. A thin rectangular loop across the same interface has circulation $(E_{2\parallel} - E_{1\parallel})\ell$ in the corresponding limit. Electrostatic zero circulation forces tangential continuity. At a conductor bordering vacuum, $E_1 = 0$ therefore gives $E_{2\perp} = \sigma/\varepsilon_0$ and zero tangential field.

17.8 Multipoles and dipoles

From far away, a compact charge distribution is hard to distinguish from a point charge. If its total charge vanishes, however, that first approximation says nothing. Expand in source size divided by observation distance and retain the first term that survives the charge cancellations.

At distances large compared with a localized charge distribution, expand its potential:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q}{r} + \frac{\boldsymbol{p}\cdot\hat{\boldsymbol{r}}}{r^2} + \cdots\right], \qquad \boldsymbol{p} = \int \boldsymbol{r}'\rho(\boldsymbol{r}')\mathrm{d}^3r'. \tag{17.11}$$

The leading nonzero multipole dominates far away. An ideal permanent dipole in a slowly varying external field has $U = -\boldsymbol{p}\cdot\boldsymbol{E}$, torque $\boldsymbol{p}\times\boldsymbol{E}$, and force $\boldsymbol{\nabla}(\boldsymbol{p}\cdot\boldsymbol{E})$ when its moment is held fixed.

For $r' \ll r$, expand the inverse distance:

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} = \frac{1}{r}\left(1 - \frac{2\hat{\boldsymbol{r}}\cdot\boldsymbol{r}'}{r} + \frac{r'^2}{r^2}\right)^{-1/2} \simeq \frac{1}{r} + \frac{\hat{\boldsymbol{r}}\cdot\boldsymbol{r}'}{r^2} + \cdots. \tag{17.12}$$

Insert this into the source integral for ϕ. The first integral is total charge Q; the second is the dipole moment projected along $\hat{\boldsymbol{r}}$. For charges $+q$ and $-q$ separated by vector $\boldsymbol{d}$ pointing from negative to positive, $\boldsymbol{p} = q\boldsymbol{d}$. Their leading distant potential falls as $1/r^2$, faster than a monopole potential, because the net charge contribution cancels.

17.9 Capacitance and field energy

Charging a capacitor becomes progressively harder: each added piece of charge must move through the voltage already established by earlier charge. The energy is therefore the integral of a changing voltage, rather than final voltage times total charge. Begin with a nearly uncharged capacitor and assemble it incrementally.

For a two-conductor capacitor, $C = Q/\Delta\phi$. Its stored energy is

$$U = \frac{Q^2}{2C} = \frac{1}{2}C(\Delta\phi)^2, \qquad U_E = \frac{\varepsilon_0}{2}\int E^2\mathrm{d}V. \tag{17.13}$$

Parallel plates of area A, spacing $d \ll \sqrt{A}$, and vacuum between them have $C \simeq \varepsilon_0 A/d$, neglecting fringing. Capacitors add in parallel, while inverse capacitances add in series.

Worked example 17.1 (Energy and the battery constraint)**.** Insert a dielectric with relative permittivity κ into an ideal capacitor. If disconnected, Q stays fixed, $C \to \kappa C$, and stored energy falls by $1/\kappa$. If connected to a voltage source, voltage stays fixed and stored energy rises by κ; battery work must be included in the total energy accounting.

At intermediate charge q, the voltage is q/C. The required work is

$$U = \int_0^Q \frac{q}{C}\mathrm{d}q = \frac{Q^2}{2C}. \tag{17.14}$$

For wide parallel plates, $E = Q/(\varepsilon_0 A)$ and $\Delta\phi = Ed$, so $C = Q/\Delta\phi = \varepsilon_0 A/d$. Substituting into $U = Q\Delta\phi/2$ gives $U = (\varepsilon_0 E^2/2)Ad$. The plate volume Ad identifies $\varepsilon_0 E^2/2$ as the vacuum field-energy density. This route connects the circuit element to the field filling its interior.

17.10 Polarization and dielectric response

A dielectric need not transport free charge across the whole sample to respond to an electric field. Positive and negative charge can shift slightly within its microscopic constituents, producing many small dipoles. Polarization averages those dipoles into a field. The distinction between free and bound charge is a bookkeeping choice that makes material response easier to organize.

The polarization $\boldsymbol{P}$ is electric dipole moment per volume. Bound charges obey $\rho_b = -\boldsymbol{\nabla} \cdot \boldsymbol{P}$ and $\sigma_b = \boldsymbol{P} \cdot \hat{\boldsymbol{n}}$. Define

$$\boldsymbol{D} = \varepsilon_0 \boldsymbol{E} + \boldsymbol{P}, \qquad \boldsymbol{\nabla} \cdot \boldsymbol{D} = \rho_{\text{free}}. \tag{17.15}$$

For a linear isotropic dielectric, $\boldsymbol{P} = \varepsilon_0 \chi_e \boldsymbol{E}$ and $\boldsymbol{D} = \varepsilon \boldsymbol{E}$. Real materials can be anisotropic, nonlinear, dispersive, or lossy. The energy density $\frac{1}{2}\boldsymbol{E} \cdot \boldsymbol{D}$ applies to the usual linear nondispersive electrostatic case, not every material response.

Insert $\rho_{\text{total}} = \rho_{\text{free}} + \rho_b$ and $\rho_b = -\boldsymbol{\nabla} \cdot \boldsymbol{P}$ into Gauss's law:

$$\varepsilon_0 \boldsymbol{\nabla} \cdot \boldsymbol{E} = \rho_{\text{free}} - \boldsymbol{\nabla} \cdot \boldsymbol{P} \quad \Longrightarrow \quad \boldsymbol{\nabla} \cdot (\varepsilon_0 \boldsymbol{E} + \boldsymbol{P}) = \rho_{\text{free}}. \tag{17.16}$$

This motivates $\boldsymbol{D}$. For a homogeneous linear dielectric filling a parallel-plate capacitor, it gives $D = Q_{\text{free}}/A$ and $E = D/\varepsilon$, hence $C = \varepsilon A/d$. The constitutive relation is the extra material information needed to convert a free-charge constraint into an electric field.

Exercises

Sketch the electric field of a positive point charge. Explain how the field and potential differ in the information they provide.

Answers are in Appendix A.

Chapter 18

Currents, magnetism, and circuits

Electrostatics gave us fields created by charge at rest. The next step is motion: charge conservation becomes a continuity equation, magnetic force appears, and the field description can be reduced to circuits when the geometry permits. Allow charges to move and the description acquires currents and magnetic fields. Circuit models compress the field behavior into a few variables, but their laws still express charge conservation, induction, and energy transfer. We will use those connections to explain both steady response and the time-dependent approach to it.

18.1 Charge transport and resistance

Current counts charge crossing a surface per unit time. For carriers of number density n, charge q, and mean drift velocity $\boldsymbol{v}_d$, the current density is $\boldsymbol{J} = nq\boldsymbol{v}_d$. Negative carriers can drift opposite to the conventional current direction. In a uniform wire, relating this flow to the electric field yields the familiar resistance law. $\boldsymbol{J}$ is current density, I total current, $\boldsymbol{B}$ magnetic field, q particle charge, and $\boldsymbol{v}$ particle velocity.

Current is $I = \mathrm{d}Q/\mathrm{d}t = \int \boldsymbol{J} \cdot \mathrm{d}\boldsymbol{S}$. Local charge conservation is

$$\partial_t \rho + \boldsymbol{\nabla} \cdot \boldsymbol{J} = 0. \tag{18.1}$$

Ohmic response is $\boldsymbol{J} = \sigma \boldsymbol{E}$. A uniform wire has $R = L/(\sigma A)$ and voltage drop $V = IR$. Joule heating is $P = IV = I^2 R$. Ohm's law is a constitutive approximation; it is not a universal law of all components.

For uniform axial fields and current, $I = JA$ and $V = EL$. With $J = \sigma E$,

$$I = \sigma A \frac{V}{L} \quad \Longrightarrow \quad R = \frac{V}{I} = \frac{L}{\sigma A}. \tag{18.2}$$

A charge $\mathrm{d}Q$ loses potential energy $V\mathrm{d}Q$ crossing the resistor, so the rate is $P = V\mathrm{d}Q/\mathrm{d}t = VI$. Locally this energy transfer is $\boldsymbol{J} \cdot \boldsymbol{E}$ per volume. Integrating that local expression over the wire gives the same circuit power.

Worked example 18.1 (A resistor connected to a battery)**.** A 9 V ideal battery is connected across a 3 Ω resistor. If the resistor obeys Ohm's law, $I = V/R = 3\,\mathrm{A}$. Charge passes a cross section at three coulombs per second. The electrical power delivered to the resistor is $P = VI = 27\,\mathrm{W}$, which becomes thermal energy in this model. Doubling the resistance at the same voltage halves the current; it does not leave the current fixed.

18.2 Magnetic forces and steady fields

The cross product in the magnetic force immediately tells us its direction and its limitation: the force is perpendicular to velocity. A uniform magnetic field can therefore bend a trajectory without changing the particle's kinetic energy. Resolve velocity into parallel and perpendicular components to understand the resulting helix. In $q\boldsymbol{v} \times \boldsymbol{B}$, the cross product fixes the force direction by the right-hand rule.

The Lorentz force is

$$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}). \tag{18.3}$$

A magnetic field alone changes direction but not the speed of a point charge, since $\boldsymbol{v} \cdot (\boldsymbol{v} \times \boldsymbol{B}) = 0$. In a uniform field a nonrelativistic charge has gyroradius $r_L = mv_\perp/(|q|B)$ and cyclotron frequency $\omega_c = |q|B/m$.

For steady currents in vacuum,

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{r}') \times (\boldsymbol{r} - \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|^3} \mathrm{d}^3 r', \quad \oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{\ell} = \mu_0 I_{\mathrm{enc}}. \tag{18.4}$$

A long straight wire has $B = \mu_0 I/(2\pi r)$; a long ideal solenoid has $B \simeq \mu_0 n I$ internally. A small current loop has magnetic moment $\boldsymbol{m} = IA\hat{\boldsymbol{n}}$, torque $\boldsymbol{m} \times \boldsymbol{B}$, and energy $-\boldsymbol{m} \cdot \boldsymbol{B}$ in an external field.

The parallel component feels no magnetic force. In the perpendicular plane, equate force magnitude to the centripetal requirement:

$$|q| v_\perp B = \frac{m v_\perp^2}{r_L}, \qquad r_L = \frac{m v_\perp}{|q|B}, \qquad \omega_c = \frac{v_\perp}{r_L} = \frac{|q|B}{m}. \tag{18.5}$$

For the field of a long straight wire, cylindrical symmetry makes B constant and tangent around a centered circle. Ampere's law then reads $B(2\pi r) = \mu_0 I$, giving the $1/r$ result. In both examples, geometry does much of the work before the algebra begins.

18.3 Magnetized matter

A magnetized sample contains microscopic magnetic moments. Small current loops provide a useful picture of how neighboring contributions cancel in the interior while leaving an effective surface current. The magnetization field summarizes these moments per volume; it should not be confused with the magnetic field they help produce.

Magnetization $\boldsymbol{M}$ produces bound currents $\boldsymbol{J}_b = \boldsymbol{\nabla} \times \boldsymbol{M}$ and $\boldsymbol{K}_b = \boldsymbol{M} \times \hat{\boldsymbol{n}}$. Define $\boldsymbol{H} = \boldsymbol{B}/\mu_0 - \boldsymbol{M}$. A linear isotropic material can satisfy $\boldsymbol{B} = \mu \boldsymbol{H}$. Diamagnetism, paramagnetism, and ferromagnetism have different microscopic origins; ferromagnetic hysteresis requires more than a single constant permeability.

In magnetostatics, substitute $\boldsymbol{J}_{\text{total}} = \boldsymbol{J}_{\text{free}} + \boldsymbol{\nabla} \times \boldsymbol{M}$ into $\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J}_{\text{total}}$. Rearranging gives

$$\boldsymbol{\nabla} \times \left(\frac{\boldsymbol{B}}{\mu_0} - \boldsymbol{M} \right) = \boldsymbol{J}_{\text{free}}, \qquad \boldsymbol{\nabla} \times \boldsymbol{H} = \boldsymbol{J}_{\text{free}}. \tag{18.6}$$

This parallels the role of $\boldsymbol{D}$ in electrostatics: the auxiliary field places the free sources on one side, while a constitutive relation supplies the material response. A history-dependent magnet cannot be captured by one constant proportionality between $\boldsymbol{B}$ and $\boldsymbol{H}$.

18.4 Induction and inductance

A changing magnetic flux drives circulation of electric field, and moving a conductor through a magnetic field can also drive charge. The general loop law accounts for both. Its negative sign says that the induced response opposes the flux change; the energy needed to sustain that change must be supplied from somewhere.

For a stationary loop, Faraday's law is $\mathcal{E} = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{\ell} = -\mathrm{d}\Phi_B/\mathrm{d}t$. More generally a moving material loop obeys

$$\mathcal{E} = \oint (\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B}) \cdot \mathrm{d}\boldsymbol{\ell} = -\frac{\mathrm{d}}{\mathrm{d}t} \int_{S(t)} \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S}. \tag{18.7}$$

The minus sign expresses Lenz's law. A linear inductor has flux linkage LI, induced emf $-L\dot{I}$, and stored energy $LI^2/2$. Coupled circuits have mutual inductance terms. The vacuum magnetic energy density is $B^2/(2\mu_0)$.

For constant inductance, building the current requires an applied voltage $V = L\dot{I}$ to balance the induced emf. The supplied power is $P = VI = LI\dot{I}$, so

$$U(I) - U(0) = \int_0^I LI' \mathrm{d}I' = \tfrac{1}{2} L I^2. \tag{18.8}$$

The mathematical structure resembles capacitor charging, but the stored variable is now current. When current falls, the magnetic field returns energy to the circuit. An inductor therefore resists rapid changes of current rather than opposing current merely because it is nonzero.

18.5 Lumped circuits and transients

For each circuit, choose a charge or current variable and write a voltage balance with consistent signs. A resistor dissipates energy, a capacitor stores it electrically, and an inductor stores it magnetically. Those roles predict whether a response relaxes monotonically or oscillates before you

solve the equation. In circuits, V is voltage, R resistance, C capacitance, and L inductance; their units and signs follow the selected voltage and current directions.

Kirchhoff's current law expresses charge conservation at nodes when charge accumulation is negligible. The usual voltage law assumes a consistent lumped description, with changing magnetic flux represented by inductive elements.

An RC charging circuit obeys $R\dot{Q} + Q/C = V_0$, giving $Q = CV_0(1 - \mathrm{e}^{-t/(RC)})$ for an initially uncharged capacitor. An RL circuit has time constant L/R. A series RLC circuit satisfies

$$L\ddot{Q} + R\dot{Q} + Q/C = V(t), \tag{18.9}$$

mathematically identical to a driven damped oscillator.

With phasors proportional to $\mathrm{e}^{\mathrm{i}\omega t}$,

$$Z_R = R, \qquad Z_L = \mathrm{i}\omega L, \qquad Z_C = \frac{1}{\mathrm{i}\omega C}. \tag{18.10}$$

Series impedances add; parallel admittances add. Average power is $V_{\mathrm{rms}} I_{\mathrm{rms}} \cos\phi$. The lumped approximation becomes inadequate when circuit dimensions approach relevant signal wavelengths or propagation delays matter.

For RC charging, define the departure from final charge as $q = Q - CV_0$. Then $R\dot{q} + q/C = 0$, so $q = q(0)\mathrm{e}^{-t/(RC)}$. With $Q(0) = 0$, this reconstructs $Q = CV_0(1 - \mathrm{e}^{-t/(RC)})$. The same method solved linear drag: subtract the equilibrium value and solve for the decaying departure.

For a sinusoidal current or voltage, a derivative becomes multiplication by $\mathrm{i}\omega$. Thus $V_L = L\dot{I}$ gives $Z_L = \mathrm{i}\omega L$, while $I = C\dot{V}_C$ gives $Z_C = 1/(\mathrm{i}\omega C)$. Dividing the series RLC equation by L identifies

$$\omega_0^2 = \frac{1}{LC}, \qquad 2\beta = \frac{R}{L}. \tag{18.11}$$

All the oscillator methods now apply, with charge taking the role of displacement and inductance taking the role of inertia.

Exercises

In a simple resistor-capacitor discharge, describe how the voltage and current change with time. Identify the quantity that sets the time scale.

Answers are in Appendix A.

Chapter 19

Maxwell's equations and electromagnetic waves

An electric current creates a magnetic field, and a changing magnetic field can drive an electric current. These observations suggest a question: can changing electric and magnetic fields continue to produce one another after they leave their source? We will join the separate laws, check that they respect charge conservation, and then find out what travels through empty space.

19.1 The unified field equations

Gauss's electric law says that charge is a source of electric field. Gauss's magnetic law says there are no isolated magnetic poles in this model. Faraday's law says that a changing magnetic field drives a circulating electric field. Each of these was introduced in the preceding chapters. The remaining law says how currents create circulating magnetic fields, but it needs one extra term when electric fields change. Let ρ be charge per unit volume and $\boldsymbol{J}$ charge flow per unit area per unit time.

With matter represented by total charge and current densities, Maxwell's four equations are

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = \rho/\varepsilon_0, \qquad \boldsymbol{\nabla} \cdot \boldsymbol{B} = 0, \tag{19.1}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\partial_t \boldsymbol{B}, \qquad \boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \partial_t \boldsymbol{E}. \tag{19.2}$$

Why add $\mu_0 \varepsilon_0 \partial_t \boldsymbol{E}$ to the older current law? Charge can pile up in a small region, so the current flowing in need not equal the current flowing out at every instant. The local balance is $\partial_t \rho + \boldsymbol{\nabla} \cdot \boldsymbol{J} = 0$. The new term makes the magnetic law agree with this balance, including between the plates of a charging capacitor.

Take the divergence of the Ampere–Maxwell equation. Since the divergence of a curl is zero,

$$0 = \mu_0 \boldsymbol{\nabla} \cdot \boldsymbol{J} + \mu_0 \varepsilon_0 \partial_t (\boldsymbol{\nabla} \cdot \boldsymbol{E}) = \mu_0 (\boldsymbol{\nabla} \cdot \boldsymbol{J} + \partial_t \rho). \tag{19.3}$$

Without the displacement-current term, we would instead require $\boldsymbol{\nabla} \cdot \boldsymbol{J} = 0$ everywhere, excluding local charge accumulation. In a charging capacitor, current reaches the plates while the electric field

between them changes. Maxwell's added term lets those two descriptions belong to one consistent field law.

19.2 Potentials and gauge freedom

Potentials help solve the field equations because some constraints become identities. Writing $\boldsymbol{B}$ as a curl automatically makes its divergence zero. Once that is done, adding $-\partial_t \boldsymbol{A}$ to the electric field automatically accounts for Faraday induction. The price is that more than one pair of potentials describes the same measurable fields.

Write

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}, \qquad \boldsymbol{E} = -\boldsymbol{\nabla}\phi - \partial_t \boldsymbol{A}. \tag{19.4}$$

The fields are unchanged by $\boldsymbol{A} \to \boldsymbol{A} + \boldsymbol{\nabla}\chi$, $\phi \to \phi - \partial_t \chi$. In Lorenz gauge, $\boldsymbol{\nabla} \cdot \boldsymbol{A} + c^{-2}\partial_t \phi = 0$, the potentials satisfy sourced wave equations. The causal vacuum solutions are

$$\phi(\boldsymbol{r}, t) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\boldsymbol{r}', t - R/c)}{R} \mathrm{d}^3 r', \tag{19.5}$$

$$\boldsymbol{A}(\boldsymbol{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{r}', t - R/c)}{R} \mathrm{d}^3 r', \qquad R = |\boldsymbol{r} - \boldsymbol{r}'|. \tag{19.6}$$

Retarded time expresses finite propagation speed.

Check a gauge transformation directly:

$$\boldsymbol{B}' = \boldsymbol{\nabla} \times (\boldsymbol{A} + \boldsymbol{\nabla}\chi) = \boldsymbol{B}, \tag{19.7}$$

$$\boldsymbol{E}' = -\boldsymbol{\nabla}(\phi - \partial_t \chi) - \partial_t(\boldsymbol{A} + \boldsymbol{\nabla}\chi) = \boldsymbol{E}. \tag{19.8}$$

The cancellation uses commuting space and time derivatives for smooth χ. A gauge condition selects a convenient representative, not a new physical field. In the retarded solution, source distance R means a travel time R/c, so the field here and now depends on the source at $t - R/c$. This is how causality enters the integral explicitly.

19.3 Wave derivation and polarization

Set charge and current to zero in the region of interest. The fields may still be nonzero because disturbances can arrive from elsewhere. To obtain an equation for the electric field alone, take one more curl of Faraday's law and eliminate the magnetic field using the other curl equation. The constants ε_0 and μ_0 describe vacuum response, with $c = 1/\sqrt{\mu_0 \varepsilon_0}$.

In a source-free vacuum, take the curl of Faraday's law and use the curl-curl identity:

$$\nabla^2 \boldsymbol{E} - \frac{1}{c^2}\partial_t^2 \boldsymbol{E} = 0, \qquad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}. \tag{19.9}$$

The magnetic field obeys the same wave equation. For a plane wave, $\boldsymbol{E} \perp \boldsymbol{B} \perp \boldsymbol{k}$, $B = E/c$, and $\boldsymbol{B} = \hat{\boldsymbol{k}} \times \boldsymbol{E}/c$. Light is an electromagnetic wave.

Carry out that elimination without skipping the signs:

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{E}) = -\partial_t(\boldsymbol{\nabla}\times\boldsymbol{B}) = -\mu_0\varepsilon_0\partial_t^2\boldsymbol{E}, \tag{19.10}$$

$$\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{E}) - \nabla^2\boldsymbol{E} = -\mu_0\varepsilon_0\partial_t^2\boldsymbol{E}. \tag{19.11}$$

Since $\boldsymbol{\nabla}\cdot\boldsymbol{E} = 0$ here, both minus signs cancel. Comparing with $\partial_t^2 u = v^2\nabla^2 u$ identifies $c^2 = 1/(\mu_0\varepsilon_0)$. For a plane wave $\boldsymbol{E} = \boldsymbol{E}_0 \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}$, Gauss's law gives $\boldsymbol{k}\cdot\boldsymbol{E}_0 = 0$ and Faraday's law gives $\boldsymbol{B}_0 = (\boldsymbol{k}\times\boldsymbol{E}_0)/\omega$. Together with $\omega = ck$, these establish transversality and the field-amplitude ratio.

Worked example 19.1 (Checking the size and direction of a plane wave)**.** Suppose a vacuum electromagnetic wave has an electric-field peak $E_0 = 300\,\mathrm{V/m}$. Maxwell's wave relation gives $B_0 = E_0/c \approx 1.0\times10^{-6}\,\mathrm{T}$. If its electric field points in $+x$ and its magnetic field in $+y$, the energy-flow direction $\boldsymbol{E}\times\boldsymbol{B}$ is $+z$. The fields are perpendicular to each other and to the direction of travel; this is the transverse structure of the wave.

19.4 Energy, momentum, and radiation

A wave that heats an absorber or pushes a mirror must carry energy and momentum. To derive the energy balance, form the rate of change of squared field strengths. Dot the electric field into the Ampere–Maxwell equation and the magnetic field into Faraday's equation; their combination turns curls into a divergence. The Poynting vector $\boldsymbol{S} = \boldsymbol{E}\times\boldsymbol{B}/\mu_0$ points in the direction of electromagnetic energy transport; its sign is set by the field orientation.

Poynting's theorem is

$$\partial_t u + \boldsymbol{\nabla}\cdot\boldsymbol{S} = -\boldsymbol{J}\cdot\boldsymbol{E}, \quad u = \tfrac{1}{2}\varepsilon_0 E^2 + \frac{B^2}{2\mu_0}, \quad \boldsymbol{S} = \frac{1}{\mu_0}\boldsymbol{E}\times\boldsymbol{B}. \tag{19.12}$$

It states that decreasing field energy supplies outgoing flux and work on charges. Vacuum momentum density is $\boldsymbol{g} = \boldsymbol{S}/c^2$. Normal radiation pressure is I/c for full absorption and $2I/c$ for ideal reflection.

Accelerating charges radiate. For a nonrelativistic point charge in vacuum, the Larmor power is

$$P = \frac{q^2a^2}{6\pi\varepsilon_0 c^3}. \tag{19.13}$$

Near fields and radiation fields have different distance dependence: the radiative electric field scales as $1/r$, so its intensity scales as $1/r^2$. Radiation reaction requires care beyond inserting this power loss into every motion without checking consistency.

The needed vector identity is

$$\boldsymbol{\nabla}\cdot(\boldsymbol{E}\times\boldsymbol{B}) = \boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{E}) - \boldsymbol{E}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B}). \tag{19.14}$$

Substituting Maxwell's equations gives

$$\frac{1}{\mu_0}\boldsymbol{\nabla}\cdot(\boldsymbol{E}\times\boldsymbol{B}) = -\frac{1}{\mu_0}\boldsymbol{B}\cdot\partial_t\boldsymbol{B} - \varepsilon_0\boldsymbol{E}\cdot\partial_t\boldsymbol{E} - \boldsymbol{J}\cdot\boldsymbol{E}. \tag{19.15}$$

Use $\boldsymbol{E} \cdot \partial_t \boldsymbol{E} = \partial_t(E^2/2)$ and the analogous magnetic identity. Rearrangement yields Poynting's theorem. The right side is negative when fields do positive work on matter. For a vacuum plane wave, $S = \varepsilon_0 c E^2$; averaging a sinusoid gives $I = \varepsilon_0 c E_0^2/2$. Energy flux divided by c is incident momentum flux, which explains the pressure on an absorber.

19.5 Plasmas and electromagnetic response

If electrons move slightly relative to a nearly fixed positive background, charge separation creates a restoring electric field. The resulting collective oscillation resembles a spring, but its stiffness comes from electrostatics. This provides a simple route to the plasma frequency before considering more complicated kinetic or magnetic effects.

A plasma is a many-particle charged medium with collective behavior. With fixed ions and a cold electron fluid, small electron-density oscillations have frequency

$$\omega_p = \sqrt{\frac{n_e e^2}{m_e \varepsilon_0}}. \tag{19.16}$$

Thermal electrostatic screening has electron Debye length $\lambda_D = \sqrt{\varepsilon_0 k_{\mathrm{B}} T_e/(n_e e^2)}$ when other species' screening can be neglected. Multiple mobile species contribute to the inverse squared screening length. In an unmagnetized collisionless cold plasma, $\omega^2 = \omega_p^2 + c^2 k^2$ for transverse waves. Below ω_p, these waves are evanescent in the ideal model.

Magnetohydrodynamics treats a sufficiently conducting plasma as a fluid coupled to magnetic fields. Ideal MHD uses $\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B} = 0$ and predicts magnetic-flux transport with the fluid. Finite resistivity, kinetic effects, and reconnection can invalidate this ideal description.

In a simple uniform slab displacement model, displace the electron fluid by x. Surface charge separation gives a restoring-field magnitude $n_e e|x|/\varepsilon_0$, with the sign such that

$$m_e \ddot{x} = -\frac{n_e e^2}{\varepsilon_0} x \quad \Longrightarrow \quad \ddot{x} + \omega_p^2 x = 0. \tag{19.17}$$

The ions are treated as fixed and thermal pressure is neglected. The density dependence $\omega_p \propto \sqrt{n_e}$ reflects a stronger restoring field when more charge separates for the same displacement. In the stated transverse-wave dispersion relation, $k^2 = (\omega^2 - \omega_p^2)/c^2$ becomes negative below cutoff, producing exponential spatial decay rather than propagation in this ideal model.

Exercises

Name the pair of changing fields in an electromagnetic wave. Explain why the wave can travel through a vacuum and state one measurable property of it.

Answers are in Appendix A.

Part VII

Relativity and Gravitation

Chapter 20

Special relativity

Newtonian mechanics assumes that observers moving steadily relative to one another share a common time. But Maxwell's equations predict a definite speed for light in vacuum. Experiments show that inertial observers measure the same vacuum light speed, even when they move relative to the source. We will see what must change in our rules for comparing distance and time so both statements can hold.

20.1 Events, intervals, and Lorentz transformations

An event is a location and a time, such as a flash at a detector. Two observers can assign different coordinates to that event without disagreeing about what happened. Begin with transformations that are linear for uniform relative motion, and demand that light trajectories and the spacetime interval retain their form. x, t are coordinates assigned by one observer, primes denote another observer's coordinates, v relative frame speed, and c invariant light speed. The Lorentz factor γ is always at least one for $|v| < c$. The interval $\mathrm{d}s^2$ is invariant even though $\mathrm{d}x$ and $\mathrm{d}t$ separately change.

The laws of physics have the same form in inertial frames, and vacuum light speed is invariant. For a boost along x,

$$x' = \gamma(x - vt), \quad t' = \gamma(t - vx/c^2), \quad \gamma = (1 - v^2/c^2)^{-1/2}. \tag{20.1}$$

The invariant interval is $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}\boldsymbol{r}^2$. For a timelike worldline, proper time satisfies

$$\mathrm{d}\tau = \sqrt{-\mathrm{d}s^2}/c = \mathrm{d}t\sqrt{1 - v^2/c^2}. \tag{20.2}$$

Time dilation gives $\Delta t = \gamma\Delta\tau$ for a uniformly moving clock. Length contraction gives $L = L_0/\gamma$ when the endpoints are measured simultaneously in the observing frame. Simultaneity itself is frame dependent.

For a boost along x, consider $x' = A(x - vt)$ and $t' = A(t - vx/c^2)$. These send both $x = ct$ and $x = -ct$ into light trajectories of the same speed. Their interval satisfies

$$x'^2 - c^2t'^2 = A^2(1 - v^2/c^2)(x^2 - c^2t^2). \tag{20.3}$$

Invariance and continuity with the identity at $v = 0$ require $A = \gamma$. For a clock at rest in the primed frame, $\mathrm{d}x' = 0$ and $\mathrm{d}t' = \mathrm{d}\tau$, so invariance gives $\mathrm{d}\tau = \mathrm{d}t/\gamma$. For a moving rod, instead compare its endpoints at the same unprimed time: $\Delta t = 0$ gives $L_0 = \Delta x' = \gamma \Delta x = \gamma L$. Clock comparison and length measurement use different pairs of events; this is why specifying simultaneity matters.

20.2 Velocity addition, light cones, and causality

Velocity is a ratio of coordinate differences. If time transforms as well as distance, subtracting frame velocities is no longer sufficient. Derive the new rule by dividing the transformed displacement by the transformed time interval along the particle's path.

Collinear velocities transform as

$$u'_x = \frac{u_x - v}{1 - u_x v/c^2}. \tag{20.4}$$

A light signal stays at c. Timelike and null separated events can have a causal connection; spacelike separated events cannot exchange a signal at or below light speed. Spacelike event order may change under a boost without causing a causal contradiction.

For light from a source receding directly along the line of sight at speed v,

$$\frac{f_{\mathrm{obs}}}{f_{\mathrm{src}}} = \sqrt{\frac{1-\beta}{1+\beta}}, \qquad \beta = v/c. \tag{20.5}$$

This combines time dilation and changing propagation distance.

From the Lorentz transformation,

$$u'_x = \frac{\mathrm{d}x'}{\mathrm{d}t'} = \frac{\gamma(\mathrm{d}x - v\mathrm{d}t)}{\gamma(\mathrm{d}t - v\mathrm{d}x/c^2)} = \frac{u_x - v}{1 - u_x v/c^2}. \tag{20.6}$$

Putting $u_x = c$ makes numerator and denominator cancel to c. At low speeds the denominator approaches one, recovering $u'_x \simeq u_x - v$. For the receding light source, successive emissions separated by source proper time T_0 are separated in the observer frame by γT_0. The source moves farther away during that interval, adding travel delay $v\gamma T_0/c$. Therefore $T_{\mathrm{obs}} = \gamma T_0(1+\beta)$, whose reciprocal gives the stated Doppler factor.

20.3 Four-momentum and relativistic energy

The proper-time derivative provides a velocity that transforms consistently as a four-vector. Multiplying by invariant mass gives four-momentum. Its temporal component contains energy and its spatial components contain ordinary three-momentum. A single invariant then links quantities that different observers measure differently.

Four-velocity is $U^\mu = \mathrm{d}x^\mu/\mathrm{d}\tau$, with $U^\mu U_\mu = -c^2$. Four-momentum for a massive particle is

$$p^\mu = mU^\mu = (E/c, \boldsymbol{p}), \quad E = \gamma mc^2, \quad \boldsymbol{p} = \gamma m\boldsymbol{v}, \tag{20.7}$$

so

$$\boxed{E^2 = p^2c^2 + m^2c^4.} \tag{20.8}$$

Rest energy is mc^2 and kinetic energy is $(\gamma - 1)mc^2 \to mv^2/2$ for $v \ll c$. For a photon, $m = 0$ and $E = pc = h\nu$. The invariant mass of a multiparticle system includes kinetic and binding energy in its center-of-momentum frame.

Worked example 20.1 (Two photons can have nonzero system mass)**.** Two photons each have energy E and travel in opposite directions. Total momentum is zero and total energy is $2E$, so the system has invariant mass $M = 2E/c^2$, even though each photon is individually massless.

For a massive particle, substitute $E = \gamma mc^2$ and $p = \gamma mv$:

$$E^2 - p^2c^2 = \gamma^2 m^2 c^4 (1 - v^2/c^2) = m^2c^4. \tag{20.9}$$

The low-speed expansion makes the Newtonian limit explicit:

$$E = mc^2 + \tfrac{1}{2}mv^2 + \frac{3mv^4}{8c^2} + \cdots . \tag{20.10}$$

Rest energy is the leading term, and Newtonian kinetic energy is the first correction. For several particles, add their energies and momenta first, then evaluate $M^2c^4 = E_{\text{total}}^2 - c^2|\boldsymbol{p}_{\text{total}}|^2$. Adding individual rest masses alone would miss internal motion and binding.

20.4 Collisions and thresholds

At a production threshold, the final particles have the least possible energy compatible with their total momentum. This is easiest to describe in the center-of-momentum frame, where threshold means no relative motion among the products. Use an invariant to connect that simple condition to the laboratory setup.

Conserve total four-momentum, not Newtonian kinetic energy. For a projectile with total energy E_a and rest mass m_a striking a stationary target m_b, define the squared center-of-momentum energy

$$s = E_{\text{CM}}^2 = m_a^2c^4 + m_b^2c^4 + 2m_bc^2E_a. \tag{20.11}$$

At threshold, final particles have no relative motion in that frame, so $E_{\text{CM}} = \sum_f m_f c^2$. A fixed-target experiment spends much of its energy on center-of-momentum motion; colliders can use beam energy more efficiently for creating invariant mass.

For a stationary target, initial energy is $E_a + m_bc^2$ and initial momentum is $\boldsymbol{p}_a$. Hence

$$s = (E_a + m_bc^2)^2 - p_a^2c^2 \tag{20.12}$$

$$= (E_a^2 - p_a^2c^2) + 2m_bc^2E_a + m_b^2c^4. \tag{20.13}$$

Use the projectile mass-shell relation to obtain the expression above. Setting $s = (\sum_f m_f c^2)^2$ then gives

$$E_{a,\text{th}} = \frac{(\sum_f m_f)^2 - m_a^2 - m_b^2}{2m_b}c^2. \tag{20.14}$$

This is total projectile energy; subtract $m_a c^2$ for kinetic energy. Apply it to a kinematically and quantum-number-allowed channel, remembering that incident total energy cannot fall below the projectile's rest energy.

20.5 Electromagnetic covariance

An observer who moves relative to a charge distribution sees a different combination of charge density and current. It is therefore natural that the same physical electromagnetic situation splits into electric and magnetic parts differently for different observers. The transformation formulas below quantify that split; the invariant combinations provide a way to check it.

Electric and magnetic fields are components of a single antisymmetric field tensor. Their division depends on the observer. For a frame moving at $\boldsymbol{v}$ relative to the original frame,

$$\boldsymbol{E}'_{\parallel} = \boldsymbol{E}_{\parallel}, \qquad \boldsymbol{B}'_{\parallel} = \boldsymbol{B}_{\parallel}, \tag{20.15}$$

$$\boldsymbol{E}'_{\perp} = \gamma(\boldsymbol{E}_{\perp} + \boldsymbol{v} \times \boldsymbol{B}), \qquad \boldsymbol{B}'_{\perp} = \gamma(\boldsymbol{B}_{\perp} - \boldsymbol{v} \times \boldsymbol{E}/c^2). \tag{20.16}$$

The scalars $E^2 - c^2 B^2$ and $\boldsymbol{E} \cdot \boldsymbol{B}$ are invariant. A field that is purely electric for one observer can include magnetism for another.

For a field with $\boldsymbol{B} = 0$ and $\boldsymbol{E} \perp \boldsymbol{v}$, the transformed fields satisfy $E' = \gamma E$ and $B' = \gamma v E/c^2$. Then

$$E'^2 - c^2 B'^2 = \gamma^2 E^2 (1 - v^2/c^2) = E^2. \tag{20.17}$$

A magnetic component has appeared, but the invariant is unchanged. For a vacuum plane wave, both $E^2 - c^2 B^2$ and $\boldsymbol{E} \cdot \boldsymbol{B}$ vanish. No inertial boost can turn a nonzero such wave into a purely electric static field.

Exercises

Two observers disagree about elapsed time between events. Identify what each must know about the events and their relative motion before comparing measurements.

Answers are in Appendix A.

Part VIII

Geometry, Topology, and Continuous Symmetry

Chapter 21

Topology and global structure

Advanced foundation. Uses sets, functions, and vectors. It prepares manifolds, defects, gauge fields, and general relativity.

Geometry measures lengths and angles. Topology asks which features survive continuous deformation. A coffee-cup-shaped surface and a torus can be deformed into one another without cutting because each has one hole; a sphere cannot be changed into either without a singular step. The point is not the familiar slogan but the discipline behind it: state which transformations are allowed and identify a quantity they preserve.

21.1 Open sets, continuity, and homeomorphisms

A topology on a set X is a collection of subsets called open sets that contains X and the empty set, is closed under arbitrary unions, and is closed under finite intersections. These rules encode which points count as nearby without first assigning a distance. In ordinary Euclidean space, open balls generate the familiar topology.

A function $f : X \to Y$ is continuous when the preimage of every open set in Y is open in X. A homeomorphism is a continuous bijection with a continuous inverse. Homeomorphic spaces have the same topological structure, even when their shapes and distances look different.

21.2 Connectedness and compactness

A space is connected when it cannot be divided into two disjoint nonempty open subsets. A path-connected space allows a continuous path between every pair of points; path connectedness implies connectedness, although the converse needs qualifications.

Compactness is a global finiteness property: every open cover has a finite subcover. In $\mathbb{R}^n$, a set is compact exactly when it is closed and bounded. A continuous real-valued function on a compact

space reaches both a maximum and a minimum. This result quietly supports many variational arguments in physics.

21.3 Loops, winding, and the fundamental group

Two loops based at the same point are homotopic when one can be continuously deformed into the other while keeping the base point fixed. Their equivalence classes form the fundamental group $\pi_1(X)$. On a circle, loops are classified by an integer winding number, so $\pi_1(S^1) \simeq \mathbb{Z}$. On a sphere, every loop contracts to a point, so $\pi_1(S^2)$ is trivial.

For a loop on the unit circle written as $z(t) = \mathrm{e}^{\mathrm{i}\theta(t)}$, its winding number is

$$n = \frac{1}{2\pi} \int_0^1 \dot{\theta}(t)\, \mathrm{d}t = \frac{\theta(1) - \theta(0)}{2\pi} \in \mathbb{Z}. \tag{21.1}$$

The integer cannot change under a continuous deformation that keeps the loop on the circle. It can change only if the description becomes singular or leaves the space.

21.4 Surfaces and the Euler characteristic

For a polygonal decomposition of a closed surface, the Euler characteristic is

$$\chi = V - E + F. \tag{21.2}$$

Refining the decomposition changes V, E, F but preserves their combination. A sphere has $\chi = 2$, while an orientable surface with g handles has $\chi = 2 - 2g$. This global integer distinguishes a sphere from a torus even before a metric is chosen.

Worked example 21.1 (A loop around a missing point)**.** The punctured plane $\mathbb{R}^2 \setminus \{0\}$ has loops that wind around the missing origin. A loop with winding one cannot contract to a point without crossing the removed point. Radial projection onto the unit circle preserves this information, so its fundamental group is also $\mathbb{Z}$.

Exercises

Classify loops on a circle with winding numbers 2, -1, and 0. Calculate $V - E + F$ for the surface of a cube. Explain why cutting a torus can change a topological invariant even though stretching it cannot.

Answers are in Appendix A.

Chapter 22

Differential geometry

Advanced foundation. Uses multivariable calculus, linear algebra, and the preceding topology chapter.

A smooth manifold combines topology with calculus. Each sufficiently small region resembles an open subset of $\mathbb{R}^n$, while overlapping coordinate charts tell us how local descriptions fit together. This chapter develops the geometric tools before gravity assigns them physical meaning.

22.1 Charts, tangent vectors, and covectors

A chart assigns coordinates x^μ to points in an open region. Where two charts overlap, their transition functions must be smooth. A tangent vector at p can be defined as a directional derivative acting on smooth functions:

$$v[f] = v^\mu \partial_\mu f. \tag{22.1}$$

Under coordinates $x^\mu \to x'^\alpha$, its components obey

$$v'^\alpha = \frac{\partial x'^\alpha}{\partial x^\mu} v^\mu. \tag{22.2}$$

A covector $\omega = \omega_\mu \mathrm{d}x^\mu$ is a linear map from tangent vectors to numbers. Its components transform oppositely so the contraction $\omega_\mu v^\mu$ is coordinate independent.

22.2 Metrics, lengths, and volume

A metric g assigns an inner product to tangent vectors. In coordinates,

$$\mathrm{d}s^2 = g_{\mu\nu} \mathrm{d}x^\mu \mathrm{d}x^\nu. \tag{22.3}$$

The Euclidean plane in polar coordinates has $\mathrm{d}s^2 = \mathrm{d}r^2 + r^2 \mathrm{d}\phi^2$. The varying component $g_{\phi\phi} = r^2$ comes from the coordinate grid and does not imply curvature. The invariant volume element is $\sqrt{|g|}\, \mathrm{d}^n x$, where g is the determinant of the metric matrix.

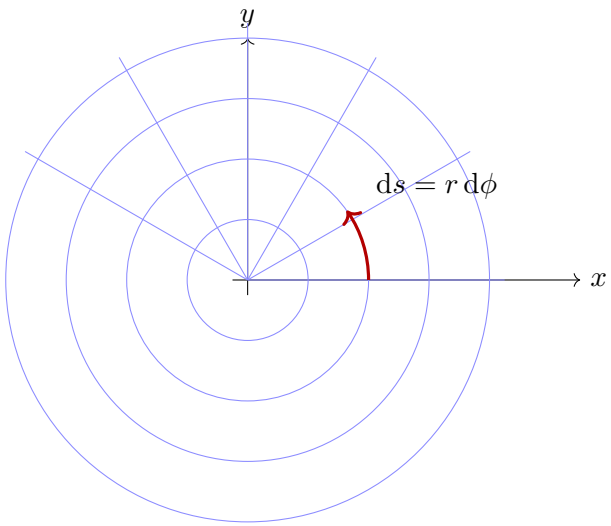


Figure 22.1: Polar coordinates curve the coordinate grid, not the Euclidean plane. An angular coordinate interval represents a longer arc at larger radius.

22.3 Connections and covariant differentiation

Vectors at different points belong to different tangent spaces, so subtracting their components requires a rule for comparison. A connection supplies that rule. For a vector field,

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu{}_{\mu\rho} V^\rho. \tag{22.4}$$

The connection coefficients are not tensor components; at one point they can be made zero by a suitable coordinate choice. The Levi–Civita connection is the unique torsion-free connection compatible with the metric:

$$\Gamma^\rho{}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}). \tag{22.5}$$

22.4 Geodesics and curvature

A geodesic parallel-transports its own tangent:

$$\frac{\mathrm{d}^2 x^\rho}{\mathrm{d}\lambda^2} + \Gamma^\rho{}_{\mu\nu} \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda} \frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} = 0. \tag{22.6}$$

It is locally straight in the connection's sense. For the Levi–Civita connection, sufficiently short geodesics extremize length or proper time, with causal qualifications in spacetime.

Curvature measures the failure of covariant derivatives to commute:

$$[\nabla_\mu, \nabla_\nu] V^\rho = R^\rho{}_{\sigma\mu\nu} V^\sigma. \tag{22.7}$$

It can also be detected by parallel transport around a small loop or by the relative acceleration of neighboring geodesics. Unlike a connection coefficient, curvature cannot be eliminated throughout a neighborhood by relabeling coordinates.

22.5 Differential forms

Differential forms are antisymmetric covariant tensors designed for integration. A one-form integrates along a curve, a two-form over a surface, and an n-form over an n-dimensional region. The exterior derivative satisfies $\mathrm{d}(\mathrm{d}\omega) = 0$, and the generalized Stokes theorem is

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega. \tag{22.8}$$

The fundamental theorem of calculus, the divergence theorem, and the ordinary Stokes theorem are instances of this statement. In electromagnetism, the field strength is naturally a two-form, and homogeneous Maxwell equations become a single relation $\mathrm{d}F = 0$.

Exercises

For the polar-plane metric, identify g_{rr} and $g_{\phi\phi}$ and calculate the area element. Explain why a nonzero $\Gamma^{\rho}{}_{\mu\nu}$ is insufficient evidence of curvature. State what kind of region can integrate a two-form.

Answers are in Appendix A.

Chapter 23

Lie groups and Lie algebras

Advanced foundation. Uses matrices, calculus, and basic group language. It prepares rotations, quantum angular momentum, gauge theory, and particle physics.

A Lie group is both a group and a smooth manifold, with smooth multiplication and inversion. It describes transformations that can be composed and varied continuously. Its Lie algebra captures the infinitesimal transformations near the identity. This local data often determines the finite transformations through exponentiation.

23.1 Groups and representations

A group has an associative product, an identity, and an inverse for every element. A representation assigns each abstract group element an invertible linear transformation while preserving multiplication. Physics uses representations because states and fields transform as vectors, spinors, tensors, or more general multiplets.

The planar rotations form $SO(2)$:

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad R(\theta_1)R(\theta_2) = R(\theta_1+\theta_2). \tag{23.1}$$

They commute. Three-dimensional rotations form $SO(3)$ and generally do not: rotations about different axes depend on order.

23.2 Generators and the exponential map

Near the identity, write a transformation as $I + \epsilon T + O(\epsilon^2)$. The matrix T is a generator. Finite one-parameter transformations are

$$g(\theta) = \exp(\theta T) = I + \theta T + \frac{\theta^2 T^2}{2!} + \cdots . \tag{23.2}$$

For $SO(2)$,

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I, \qquad \mathrm{e}^{\theta J} = I\cos\theta + J\sin\theta = R(\theta). \tag{23.3}$$

The power series turns the local generator into a finite rotation.

23.3 Commutators and structure constants

The Lie bracket measures the failure of infinitesimal transformations to commute. For matrix groups it is the commutator $[X, Y] = XY - YX$. In a basis T_a,

$$[T_a, T_b] = f_{ab}{}^{c} T_c. \tag{23.4}$$

where the structure constants encode the local algebra. Angular-momentum generators obey

$$[J_i, J_j] = \epsilon_{ijk} J_k. \tag{23.5}$$

in a real rotation convention, or acquire factors of i and $\hbar$ in the common quantum convention. The Jacobi identity ensures consistent composition of infinitesimal transformations.

23.4 $SO(3)$, $SU(2)$, and spin

The groups $SO(3)$ and $SU(2)$ have the same Lie algebra locally, but different global topology. The map from $SU(2)$ to $SO(3)$ is two-to-one: the matrices U and $-U$ describe the same ordinary spatial rotation. This is why spinors can change sign under a 2π rotation and return to themselves only after 4π, while observable probabilities remain unchanged.

Using Pauli matrices, an $SU(2)$ rotation can be written

$$U(\theta, \hat{\boldsymbol{n}}) = \exp\left(-\frac{\mathrm{i}\theta}{2}\hat{\boldsymbol{n}}\cdot\boldsymbol{\sigma}\right) = I\cos\frac{\theta}{2} - \mathrm{i}(\hat{\boldsymbol{n}}\cdot\boldsymbol{\sigma})\sin\frac{\theta}{2}. \tag{23.6}$$

The half-angle records the double-cover relationship, not an ordinary vector turning through half the physical angle.

23.5 Symmetry, conservation, and gauge groups

When a continuous transformation leaves an action invariant, Noether's theorem associates it with a conserved quantity. Spatial translations correspond to momentum, rotations to angular momentum, and time translations to energy. In quantum theory, Hermitian generators represent observables associated with continuous unitary transformations.

A global symmetry applies the same group element everywhere. A gauge symmetry permits the group element to depend on spacetime position. Ordinary derivatives then fail to transform covariantly, so a connection or gauge field is introduced. Electromagnetism uses $U(1)$; the Standard Model uses $SU(3)_C \times SU(2)_L \times U(1)_Y$. The group specifies transformation structure, while dynamics still require an action and measured parameters.

Worked example 23.1 (Order matters)**.** For small rotations ϵJ_x and δJ_y, reversing their order changes the result at order $\epsilon\delta$:

$$(I + \epsilon J_x)(I + \delta J_y) - (I + \delta J_y)(I + \epsilon J_x) = \epsilon\delta[J_x, J_y] = \epsilon\delta J_z. \tag{23.7}$$

The commutator says that the mismatch is an infinitesimal rotation about the third axis.

Exercises

Exponentiate the $SO(2)$ generator using $J^2 = -I$. Explain why $SO(2)$ is Abelian but $SO(3)$ is not. Evaluate $U(2\pi, \hat{\boldsymbol{n}})$ for a spinor and state why its minus sign does not alter a single measurement probability.

Answers are in Appendix A.

Part IX

Relativistic Gravitation

Chapter 24

General relativity

Advanced. Requires special relativity and the topology and differential-geometry chapters.

Special relativity gave us a way to compare measurements when gravity can be ignored. Now imagine dropping two nearby balls toward Earth. They both fall, but their paths can slowly approach one another because they point toward Earth's center. A freely falling observer can remove the common fall locally; the relative motion remains. General relativity describes that remaining effect with spacetime geometry. We will build the needed geometry from measurements, one idea at a time.

24.1 Equivalence principle and geometry

Inside a small freely falling laboratory, nearby loose objects appear almost weightless. Over a larger region their accelerations differ, revealing tidal effects. This suggests distinguishing the acceleration removable by choosing a freely falling frame from the relative acceleration that remains. The metric is the mathematical object that lets us make that distinction without privileging one coordinate system. $g_{\mu\nu}$ is the spacetime metric, $\Gamma^{\rho}{}_{\mu\nu}$ the Levi–Civita connection, $R^{\rho}{}_{\sigma\mu\nu}$ curvature, and $T_{\mu\nu}$ stress-energy. The indices are spacetime labels, not powers.

Locally, a freely falling laboratory removes the leading effect of a uniform gravitational field. Tidal effects remain and measure curvature. General relativity describes gravity with a spacetime metric,

$$\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^{\mu}\mathrm{d}x^{\nu}. \tag{24.1}$$

The equivalence principle is local: curvature cannot generally be removed throughout an extended region by changing coordinates.

Even flat space can have position-dependent metric components. In polar coordinates, a short displacement has squared length $\mathrm{d}\ell^2 = \mathrm{d}r^2 + r^2\mathrm{d}\phi^2$. The factor r^2 reflects the arc length associated with an angular change; it does not imply curvature. Likewise, nonzero connection coefficients

alone do not establish a gravitational tidal field. Curvature, built from derivatives and products of those coefficients, is the stronger test.

Worked example 24.1 (Objects inside a falling elevator)**.** Imagine an elevator and a loose ball falling together, far enough from obstacles for a short observation. Relative to the elevator, the ball initially appears to float because both accelerate downward under gravity. This is a local statement: if the elevator is very large, different parts can feel measurably different gravitational pulls. Those tidal differences are a clue to spacetime curvature, which cannot be removed by choosing a freely falling frame over an extended region.

24.2 Manifolds, tangent spaces, and coordinate changes

A smooth manifold is a space that looks like $\mathbb{R}^n$ in a sufficiently small neighborhood, even if no single coordinate chart covers it without distortion. A point p has a tangent space T_pM containing the possible infinitesimal directions through that point. A tangent vector can be defined by how it acts on smooth functions:

$$v[f] = \frac{\mathrm{d}}{\mathrm{d}t} f(\gamma(t))\bigg|_{t=0} \tag{24.2}$$

for any curve γ with $\gamma(0) = p$ and tangent v. In coordinates, $v = v^\mu \partial_\mu$, but the components change when coordinates change. M is a manifold, T_pM the tangent space at point p, $g_{\mu\nu}$ the metric components, and $\Gamma^\rho{}_{\mu\nu}$ connection coefficients. Greek indices label coordinates and are summed when repeated.

The dual space T_p^*M contains covectors, which act linearly on tangent vectors. The differential $\mathrm{d}f$ is the covector satisfying $\mathrm{d}f(v) = v[f]$. A metric is a smoothly varying inner product g_p on each tangent space:

$$\|v\|^2 = g_{\mu\nu} v^\mu v^\nu. \tag{24.3}$$

It determines lengths, angles, and causal character in relativity, but it need not be positive definite. The spacetime metric has one timelike and three spacelike directions.

Worked example 24.2 (A coordinate problem at the North Pole)**.** Latitude and longitude label most points on Earth, but every longitude meets at the North Pole. Walking through the pole does not tear or crease the surface; only that coordinate label fails there. We can describe a small neighborhood with another map. A manifold is built from such overlapping local coordinate maps, so a coordinate singularity alone is not evidence that the physical surface is singular.

24.3 Connections, parallel transport, and curvature

Vectors at different points belong to different tangent spaces, so there is no automatic way to subtract them. A connection defines such a comparison. Along a curve with tangent u^μ, parallel transport of V^μ means

$$\frac{\mathrm{d}V^\mu}{\mathrm{d}\lambda} + \Gamma^\mu{}_{\nu\rho} u^\nu V^\rho = 0. \tag{24.4}$$

The connection coefficients depend on coordinates, but the statement that a vector is parallel transported is geometric. For the unique torsion-free connection compatible with a metric, the coefficients are the Levi–Civita symbols used in the GR chapter. A coordinate component may change under relabeling even when the geometric vector does not; curvature is the coordinate-independent content measured by transport around loops.

Transport around a small closed loop need not return a vector to its original direction. The curvature tensor measures this failure:

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma. \tag{24.5}$$

Flat polar coordinates have nonzero connection coefficients but zero curvature. This is the clean distinction between coordinate effects and genuine gravitational tidal geometry.

24.4 Covariant derivatives and geodesics

A vector's components may change because the vector changes or because the coordinate basis changes. The ordinary derivative mixes those effects. A covariant derivative includes the basis change so that the result transforms as a tensor. A geodesic is a path whose tangent remains parallel to itself. In a small freely falling region, this describes motion without a force acting on the object. In the geodesic equation, τ is proper time for massive particles.

The Levi-Civita connection is

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}). \tag{24.6}$$

For a vector, $\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\rho}V^\rho$. A freely falling massive particle follows

$$\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}\frac{\mathrm{d}x^\beta}{\mathrm{d}\tau} = 0. \tag{24.7}$$

Light follows null geodesics with an affine parameter in place of proper time. Curvature can be defined by

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}. \tag{24.8}$$

We use $R_{\sigma\nu} = R^\rho{}_{\sigma\rho\nu}$ and $R = g^{\mu\nu}R_{\mu\nu}$. Sign conventions differ among texts, so tensor definitions and field equations must be used consistently.

To derive the geodesic equation, use an affine parameter λ and the path Lagrangian $L = \frac{1}{2}g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta$, where dots in this paragraph mean $\mathrm{d}/\mathrm{d}\lambda$. Its derivatives give

$$g_{\mu\nu}\ddot{x}^\nu + (\partial_\alpha g_{\mu\nu})\dot{x}^\alpha\dot{x}^\nu - \tfrac{1}{2}(\partial_\mu g_{\alpha\beta})\dot{x}^\alpha\dot{x}^\beta = 0. \tag{24.9}$$

The product $\dot{x}^\alpha\dot{x}^\beta$ is symmetric, so symmetrize the middle coefficient and multiply by the inverse metric. The combination of metric derivatives is precisely $\Gamma^\rho_{\alpha\beta}$, leaving $\ddot{x}^\rho + \Gamma^\rho_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$. For massive particles choose proper time; for light use a different affine parameter because proper time along a null path vanishes.

24.5 Einstein's field equations

The metric cannot be chosen arbitrarily once matter is specified. Einstein's equation is the gravitational field law relating a particular curvature tensor to stress-energy. It is not a consequence of the geodesic equation alone: one law determines geometry, while the other describes free motion within it. Its coefficient is fixed by requiring the weak-field limit to recover Newtonian gravity.

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \qquad G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R g_{\mu\nu}.} \tag{24.10}$$

Stress-energy includes energy density, momentum density, pressure, and stress. The geometric identity $\nabla_\mu G^{\mu\nu} = 0$ implies $\nabla_\mu T^{\mu\nu} = 0$. This is local covariant conservation; a globally conserved total energy requires additional structure and does not exist in every spacetime.

For weak, slowly varying gravity, $g_{00} \simeq -(1 + 2\Phi/c^2)$ and the geodesic equation gives $\ddot{\boldsymbol{r}} = -\boldsymbol{\nabla}\Phi$. Einstein's equations reduce to $\nabla^2\Phi = 4\pi G\rho$ when pressure and relativistic corrections are negligible and Λ is neglected.

We can check the force-law limit directly. Use $x^0 = ct$, a static weak field, $g_{00} \simeq -(1+2\Phi/c^2)$, and nearly Cartesian spatial coordinates. Then

$$\Gamma^i_{00} \simeq -\tfrac{1}{2}\partial_i g_{00} = \frac{\partial_i \Phi}{c^2}. \tag{24.11}$$

For slow motion, $\mathrm{d}t/\mathrm{d}\tau \simeq 1$ and the leading spatial geodesic equation is $\ddot{x}^i \simeq -c^2\Gamma^i_{00} = -\partial_i\Phi$. This recovers Newton's acceleration law. Deriving the source equation as well requires inserting the weak-field metric into the curvature tensors; the result is the Poisson equation stated above.

24.6 Schwarzschild geometry and tests

Spherical symmetry removes much of the complexity of the vacuum field equations. The Schwarzschild solution describes the exterior geometry under the stated assumptions. Rather than treating its metric as one intimidating formula, first read off what it says about a clock held at a fixed radius, then compare clocks at different radii.

Outside a spherical nonrotating mass in vacuum with $\Lambda = 0$,

$$\mathrm{d}s^2 = -\left(1 - \frac{r_s}{r}\right) c^2\mathrm{d}t^2 + \left(1 - \frac{r_s}{r}\right)^{-1} \mathrm{d}r^2 + r^2\mathrm{d}\Omega^2, \quad r_s = \frac{2GM}{c^2}. \tag{24.12}$$

Static clocks satisfy $\mathrm{d}\tau = \sqrt{1 - r_s/r}\,\mathrm{d}t$. For static emission and reception outside the horizon,

$$\frac{\nu_r}{\nu_e} = \sqrt{\frac{1 - r_s/r_e}{1 - r_s/r_r}}. \tag{24.13}$$

Light climbing outward redshifts. Leading weak-field predictions include deflection $\alpha \simeq 4GM/(bc^2)$ and periapsis advance per orbit $\Delta\phi \simeq 6\pi GM/[a(1-e^2)c^2]$.

The surface $r = r_s$ is an event horizon for a Schwarzschild black hole, not a material surface. The metric's singular form there is coordinate dependent; curvature diverges at $r = 0$ in the classical solution. Static observers cannot remain at the horizon.

For a static clock, $\mathrm{d}r = \mathrm{d}\Omega = 0$ and $\mathrm{d}s^2 = -c^2\mathrm{d}\tau^2$, giving $\mathrm{d}\tau = \sqrt{1 - r_s/r}\mathrm{d}t$. A fixed coordinate-time interval is therefore less proper time for a clock closer to the mass. In this stationary geometry, successive wave crests have the same coordinate-time separation at emission and reception, so their locally measured frequencies have the ratio shown above.

Far from the source, expand the clock factor:

$$\sqrt{1 - \frac{2GM}{rc^2}} \simeq 1 - \frac{GM}{rc^2} = 1 + \frac{\Phi}{c^2}. \tag{24.14}$$

This connects the relativistic clock effect to the Newtonian potential. Light deflection and periapsis advance require solving geodesics, not merely applying this clock factor; their formulas summarize that further calculation.

24.7 Gravitational waves and black-hole thermodynamics

A changing gravitational field can transport energy away from a source. For an orbiting binary, losing energy means the orbit tightens rather than simply slowing down at a fixed radius. Black-hole thermodynamics adds another layer: quantum fields in a black-hole background connect horizon area with entropy. These are distinct extensions of the geometry we have introduced.

Weak perturbations of flat spacetime have two transverse tensor polarizations propagating at c. Gravitational radiation is sourced at leading order by a changing mass quadrupole; isolated monopole and dipole radiation are excluded by conservation laws. For a circular weak-field binary of separation a,

$$P = \frac{32}{5}\frac{G^4\mu^2M^3}{c^5a^5}, \qquad M = m_1 + m_2. \tag{24.15}$$

Energy loss shrinks the orbit and increases its frequency. Strong-field merger physics generally requires numerical relativity.

Semiclassical theory assigns a Schwarzschild black hole temperature and entropy

$$T_H = \frac{\hbar c^3}{8\pi GMk_\mathrm{B}}, \qquad S_\mathrm{BH} = \frac{k_\mathrm{B}c^3A}{4G\hbar}. \tag{24.16}$$

These connect gravity, quantum theory, and thermodynamics; a complete microscopic account of general quantum spacetime remains beyond established theory.

The leading binary power law can be turned into an inspiral law using Newtonian orbital energy $E = -G\mu M/(2a)$:

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \frac{G\mu M}{2a^2}\dot{a} = -P \quad \Longrightarrow \quad \dot{a} = -\frac{64}{5}\frac{G^3\mu M^2}{c^5a^3}. \tag{24.17}$$

The negative sign shows shrinking separation. Since orbital frequency scales as $a^{-3/2}$, the signal grows faster as the system loses energy. This argument assumes a slowly evolving circular weak-field orbit. The Hawking temperature and entropy, by contrast, are semiclassical results quoted here; they cannot be obtained by Newtonian energy balance alone.

Exercises

Describe in words how a freely falling object moves in curved spacetime. Give one observation that distinguishes the theory from Newtonian gravity.

Answers are in Appendix A.

Part X

Quantum Physics

Chapter 25

Quantum principles and mathematical structure

Classical waves taught us to track amplitudes and phase, while statistical physics taught us to distinguish microscopic possibilities from measured averages. Quantum mechanics combines those lessons in a new state space: amplitudes evolve linearly, but measurements return probabilities. Quantum mechanics changes the object we predict. Instead of assigning a definite trajectory to every particle, we evolve a state and calculate probabilities for specified measurements. The algebra of vectors, operators, and complex amplitudes is the machinery for doing that. We will distinguish the theory's postulates from consequences derived from them, so the starting assumptions stay visible.

25.1 A frequency threshold for ejecting an electron

Shine light on a metal surface and measure whether electrons leave it. In a simple classical wave picture, increasing the light's intensity increases the energy delivered to the surface continuously. The experiment presents a sharper condition: below a material-dependent threshold frequency, electrons are not ejected by ordinary one-photon illumination no matter how many low-frequency photons arrive per second. Above threshold, the maximum electron kinetic energy grows with light frequency.

The quantum model treats light in this interaction as arriving in packets called photons. One photon of frequency ν carries energy $h\nu$, where h is Planck's constant. An electron needs at least energy W, the work function, to escape the material. Energy conservation for one successful event gives

$$K_{\max} = h\nu - W. \tag{25.1}$$

At threshold, $K_{\max} = 0$ and $\nu_0 = W/h$. Doubling the intensity at fixed frequency mainly increases how many photons reach the surface and thus how many electrons can be released; it does not double the energy of each photon. Doubling the frequency changes the energy per photon. This distinction between energy per event and number of events is the reason the experiment matters.

Worked example 25.1 (A threshold check)**.** Suppose the work function is $W = 2\,\text{eV}$. Light whose photons have energy $1.5\,\text{eV}$ cannot eject electrons in the one-photon model. Light with photon energy $3\,\text{eV}$ can give an emitted electron at most $1\,\text{eV}$ of kinetic energy. The difference is accounted for by the energy needed to leave the metal.

25.2 Why matter needs wave amplitudes

Atoms emit and absorb light at discrete frequencies, which means their internal energy changes in discrete amounts. A purely classical orbit around a nucleus does not by itself explain why only certain stationary energies occur. Quantum theory assigns wave-like amplitudes to matter as well as light. A particle with momentum magnitude p has a characteristic de Broglie wavelength $\lambda = h/p$; a larger momentum corresponds to a shorter wavelength.

The wavelength alone does not solve an atom. We still need a dynamical equation for the amplitude and boundary or normalization conditions for acceptable states. The next sections build that framework. The photon and de Broglie relations are empirical starting points; we will derive consequences from them and from the quantum postulates rather than claiming they follow from Newton's laws.

25.3 From two alternatives to a state

Imagine an experiment with two possible paths that later meet at a detector. If we can distinguish which path was taken, we can add the separate probabilities for reaching a detector position. When the alternatives remain indistinguishable, quantum experiments show an interference pattern instead. To predict it, we assign a complex amplitude to each alternative, add the amplitudes, and square the magnitude of the sum. The cross term is what can strengthen or cancel the observed pattern.

A state vector keeps track of such amplitudes in a chosen set of alternatives. For two orthogonal basis states $|0\rangle$ and $|1\rangle$, write

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle. \tag{25.2}$$

The symbols α and β are complex amplitudes. If we measure in this basis, the Born rule says $P(0) = |\alpha|^2$ and $P(1) = |\beta|^2$. Normalization, $|\alpha|^2 + |\beta|^2 = 1$, expresses certainty that one of the two complete outcomes occurs. Multiplying both amplitudes by the same phase changes no probability; changing their relative phase can change interference after the paths are recombined.

Worked example 25.2 (Equal probabilities, different future interference)**.** The states $(|0\rangle + |1\rangle)/\sqrt{2}$ and $(|0\rangle - |1\rangle)/\sqrt{2}$ each give 0 and 1 with probability $1/2$ if measured immediately in the 0/1 basis. If a later operation recombines the alternatives, one state's amplitudes can add where the other's cancel. A list of probabilities for one measurement therefore does not contain every prediction of the state.

25.4 Measurements and observables

A measurement needs a specified set of possible outcomes. An observable A is represented by a self-adjoint operator with eigenstates $|a_n\rangle$ and corresponding outcome values a_n: $A|a_n\rangle = a_n|a_n\rangle$. "Operator" means a mathematical action on a state vector. If these eigenstates form a complete orthonormal basis, we can expand any state as $|\psi\rangle = \sum_n c_n|a_n\rangle$. The coefficient $c_n = \langle a_n|\psi\rangle$ is the amplitude for outcome a_n.

The Born rule now gives $P(a_n) = |c_n|^2$. If we repeat the same preparation many times, the mean measured value is $\sum_n |c_n|^2 a_n$. We can write the same quantity compactly with the operator:

$$\langle A\rangle = \langle\psi|A|\psi\rangle = \sum_n |c_n|^2 a_n. \tag{25.3}$$

To see the equality, insert the state expansion on both sides of A and use orthonormality, $\langle a_m||a_n\rangle = \delta_{mn}$. The mixed terms vanish and each surviving term contributes its outcome times its probability. An average need not itself be an outcome; a two-outcome measurement with values 0 and 1 can have mean 1/2.

For position, alternatives vary continuously. A position wavefunction $\psi(\boldsymbol{r})$ is an amplitude density, and the probability of finding the particle in a region V is $\int_V |\psi(\boldsymbol{r})|^2 \mathrm{d}^3r$. Total probability one requires $\int |\psi|^2 \mathrm{d}^3r = 1$. The units of ψ reflect the spatial measure; this is the continuous version of adding the discrete probabilities.

For a nondegenerate outcome, an ideal projective measurement yielding a_n prepares the corresponding eigenstate for a subsequent immediate measurement of the same observable. Degenerate outcomes project onto an eigenspace instead. More general measurements require additional operators.

25.5 Operators and uncertainty

The order of two operator actions can matter. Apply the position and momentum operators to a smooth test wavefunction and compare their two orders: the product rule leaves a nonzero remainder. The resulting commutator controls a lower bound on the spreads of the corresponding measurement distributions.

In the position representation, $\hat{\boldsymbol{p}} = -\mathrm{i}\hbar\boldsymbol{\nabla}$, and $[x_i, p_j] = \mathrm{i}\hbar\delta_{ij}$. For suitable states and operator domains,

$$\Delta A\,\Delta B \geq \frac{1}{2}|\,\langle[A,B]\rangle\,|, \qquad \Delta x\,\Delta p_x \geq \hbar/2. \tag{25.4}$$

The spreads refer to repeated measurements on identically prepared states. This is not merely a statement about disturbing a particle with an instrument. Time usually enters nonrelativistic quantum mechanics as a parameter, so energy-time relations need interpretation different from the position-momentum commutator.

In one dimension,

$$(xp - px)\psi = -\mathrm{i}\hbar x\psi' + \mathrm{i}\hbar(x\psi)' = \mathrm{i}\hbar\psi. \tag{25.5}$$

For the uncertainty bound, define $\widetilde{A} = A - \langle A\rangle$ and $\widetilde{B} = B - \langle B\rangle$. Cauchy–Schwarz applied to $\widetilde{A}|\psi\rangle$ and $\widetilde{B}|\psi\rangle$ gives

$$(\Delta A)^2(\Delta B)^2 \geq |\left\langle \widetilde{A}\widetilde{B}\right\rangle|^2 \geq \left|\frac{\langle [A,B]\rangle}{2\mathrm{i}}\right|^2. \tag{25.6}$$

The second step retains the imaginary part of the complex expectation value. Taking square roots gives the stated inequality. A state with very small position spread must therefore have a sufficiently broad momentum distribution, even before specifying a measurement apparatus.

25.6 Schrodinger evolution and continuity

The time-dependent Schrodinger equation is the dynamical postulate for this nonrelativistic model. Once the Hamiltonian and initial state are given, it predicts the evolving amplitudes. For a time-independent potential, try separating spatial shape from time dependence; this reveals why energy eigenstates acquire a simple phase.

For a spinless nonrelativistic particle,

$$\mathrm{i}\hbar\partial_t\psi = \hat{H}\psi, \qquad \hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\boldsymbol{r},t). \tag{25.7}$$

A time-independent Hamiltonian has stationary states $\psi_n = \phi_n \mathrm{e}^{-\mathrm{i}E_n t/\hbar}$ with $\hat{H}\phi_n = E_n\phi_n$. Their probability densities are stationary, although the phase evolves. Superpositions can have time-dependent interference.

For real scalar V, subtracting the conjugate Schrodinger equation gives

$$\partial_t|\psi|^2 + \boldsymbol{\nabla}\cdot\boldsymbol{j} = 0, \qquad \boldsymbol{j} = \frac{\hbar}{m}\,\mathrm{Im}(\psi^*\boldsymbol{\nabla}\psi). \tag{25.8}$$

With electromagnetic coupling, $\hat{H} = [-\mathrm{i}\hbar\boldsymbol{\nabla} - q\boldsymbol{A}]^2/(2m) + q\phi$ and the current includes $-q\boldsymbol{A}|\psi|^2/m$. Gauge transformations must transform the wavefunction phase as well as the potentials.

Insert $\psi(\boldsymbol{r},t) = \phi(\boldsymbol{r})T(t)$ into the equation and separate variables:

$$\mathrm{i}\hbar\frac{\dot{T}}{T} = \frac{\hat{H}\phi}{\phi} = E, \qquad T(t) = T(0)\mathrm{e}^{-\mathrm{i}Et/\hbar}. \tag{25.9}$$

The spatial problem is the energy eigenvalue equation. To check probability conservation, differentiate $|\psi|^2$ and use both the equation and its conjugate:

$$\partial_t|\psi|^2 = \frac{\mathrm{i}\hbar}{2m}(\psi^*\nabla^2\psi - \psi\nabla^2\psi^*) = -\boldsymbol{\nabla}\cdot\left[\frac{\hbar}{2m\mathrm{i}}(\psi^*\boldsymbol{\nabla}\psi - \psi\boldsymbol{\nabla}\psi^*)\right]. \tag{25.10}$$

The real potential terms cancel. The bracket is the probability current, so the wave equation preserves total probability when no net probability escapes the boundary.

25.7 Ehrenfest's theorem and pictures of dynamics

A bridge to classical mechanics should compare measurable averages. Differentiate $\langle\psi|A|\psi\rangle$ using the product rule: one term comes from the bra, one from the operator if it depends on time, and one from the ket. Substituting Schrodinger evolution into the state terms produces a commutator.

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle A\rangle = \frac{\mathrm{i}}{\hbar}\langle[H,A]\rangle + \langle\partial_t A\rangle\,. \tag{25.11}$$

Thus $\mathrm{d}\langle x\rangle/\mathrm{d}t = \langle p\rangle/m$ and $\mathrm{d}\langle p\rangle/\mathrm{d}t = -\langle V'(x)\rangle$. These resemble Newton's equations, but generally $\langle V'(x)\rangle \neq V'(\langle x\rangle)$; a sufficiently narrow packet can make the classical approximation work.

In the Schrodinger picture states evolve; in the Heisenberg picture operators evolve. For time-independent H, $U(t) = \mathrm{e}^{-\mathrm{i}Ht/\hbar}$ is unitary, preserving probabilities and inner products.

For $H = p^2/(2m) + V(x)$, the commutators are $[H,x] = -\mathrm{i}\hbar p/m$ and $[H,p] = \mathrm{i}\hbar V'(x)$. Ehrenfest's formula then yields the stated mean position and momentum equations. Expand the force around $\bar{x} = \langle x\rangle$ to see the approximation involved:

$$\big\langle V'(x)\big\rangle = V'(\bar{x}) + \tfrac{1}{2}V^{(3)}(\bar{x})(\Delta x)^2 + \cdots\,. \tag{25.12}$$

The linear fluctuation term averages to zero. A narrow packet in a smoothly varying force can follow approximately classical mean motion; a quadratic potential makes the displayed correction vanish exactly.

Exercises

A state is a sum of two alternatives. Explain why amplitudes are added before probabilities are calculated. What does normalization guarantee?

Answers are in Appendix A.

Chapter 26

Exactly solvable quantum systems

The previous chapter introduced states and measurements. We now let the Hamiltonian evolve those states in concrete systems. Boundary conditions, just as in waves, select allowed patterns; the difference is that the patterns now describe probability amplitudes. Simple quantum models are valuable because each exposes a different mechanism: boundary conditions quantize a box, forbidden regions produce exponential tails, and operator algebra organizes an oscillator. In every case, begin with the potential, solve the appropriate differential or algebraic equation, and impose the conditions that make a state physically admissible.

26.1 Free particles and wave packets

A plane wave has one exact momentum and extends across all space. To describe a localized particle, superpose momenta. Their different propagation speeds then reshape the packet even without a force. Free spreading is therefore compatible with constant average momentum.

A momentum eigenfunction is proportional to $\mathrm{e}^{\mathrm{i}(kx-\omega t)}$ with $p = \hbar k$ and $E = \hbar\omega = \hbar^2 k^2/(2m)$. It is not square-integrable on an infinite line; physical localized states are superpositions. For an initially minimum-uncertainty Gaussian with position variance σ_0^2 and no initial position-momentum covariance,

$$\sigma_x^2(t) = \sigma_0^2 + \frac{\hbar^2 t^2}{4m^2\sigma_0^2}. \tag{26.1}$$

Localization produces momentum spread, which causes dispersive broadening.

In the Heisenberg description, $x(t) = x(0) + p(0)t/m$. The variance consequently satisfies

$$\sigma_x^2(t) = \sigma_x^2(0) + \frac{2t}{m}\,\mathrm{Cov}(x,p) + \frac{t^2}{m^2}\sigma_p^2(0), \tag{26.2}$$

where the covariance uses the symmetrized product. For the initially uncorrelated minimum-uncertainty Gaussian, covariance is zero and $\sigma_p = \hbar/(2\sigma_0)$. Substitution produces the spreading law above. A smaller initial width gives a larger momentum spread and hence faster eventual broadening.

26.2 Infinite and finite wells

For the infinite well, begin inside the region where the potential is zero. Solve there first, then demand that the wavefunction vanish at both walls. Those two boundary conditions eliminate most mathematical solutions and leave only a discrete set of energies. $V(x)$ is potential energy as a function of position, ϕ_n a stationary spatial amplitude, E_n its energy, and n an integer level label. The wavefunction must be normalized, so $|\phi|^2$ can be interpreted as a probability density. Nodes are points where the amplitude is zero, not places where the particle is classically stuck.

For $V = 0$ on $0 < x < L$ and infinite walls,

$$\phi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, \qquad E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, \ldots \tag{26.3}$$

Boundary conditions quantize the wave number. The ground state has nonzero kinetic energy because confinement forbids a zero-momentum constant wavefunction satisfying both walls.

For finite steps with equal mass and finite potential, ψ and ψ' are continuous. Bound states penetrate classically forbidden regions with decay constant $\kappa = \sqrt{2m(V - E)}/\hbar$. Singular delta potentials instead produce a derivative jump fixed by integrating the Schrodinger equation across the singularity.

Worked example 26.1 (A superposition in a box)**.** For $\psi(x, 0) = (\phi_1 + \phi_2)/\sqrt{2}$, the mean energy is $(E_1 + E_2)/2$, while an energy measurement yields E_1 or E_2 with equal probability. The density contains $\phi_1\phi_2 \cos[(E_2 - E_1)t/\hbar]$, so it oscillates despite the time-independent Hamiltonian.

The stationary equation inside the well is

$$-\frac{\hbar^2}{2m}\phi'' = E\phi \quad \Longrightarrow \quad \phi = A \sin kx + B \cos kx, \quad k^2 = \frac{2mE}{\hbar^2}. \tag{26.4}$$

At $x = 0$, $\phi = 0$ sets $B = 0$. At $x = L$, it requires $kL = n\pi$. The $n = 0$ case is the zero function, which cannot be normalized; negative n duplicates the same states up to a sign. Finally,

$$1 = |A|^2 \int_0^L \sin^2 \frac{n\pi x}{L} \mathrm{d}x = |A|^2 \frac{L}{2} \tag{26.5}$$

fixes $A = \sqrt{2/L}$ up to an irrelevant phase. For a finite barrier, the equation becomes $\phi'' = \kappa^2\phi$ in a forbidden region. This yields exponential tails instead of an abrupt zero at the interface.

26.3 Tunneling through a barrier

Divide the line into the left free region, the barrier, and the right free region. Solve the stationary equation in each, then match both the wavefunction and its derivative at each finite interface. Tunneling emerges from that matching; it does not require inserting a separate rule that permits a particle to cross.

For a rectangular barrier of height V_0 and width a, with zero potential on both sides and $0 < E < V_0$,

$$T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E(V_0 - E)}\right]^{-1}, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}. \tag{26.6}$$

For a thick barrier, the dominant suppression is $\mathrm{e}^{-2\kappa a}$. Transmission probability is a ratio of probability currents; when the asymptotic speeds differ, it is not just a squared amplitude ratio. Tunneling does not mean the particle temporarily violates energy conservation.

With unit incident amplitude, choose

$$\psi_I = \mathrm{e}^{\mathrm{i}kx} + r\mathrm{e}^{-\mathrm{i}kx}, \quad \psi_{II} = C\mathrm{e}^{\kappa x} + D\mathrm{e}^{-\kappa x}, \quad \psi_{III} = t\mathrm{e}^{\mathrm{i}kx}, \quad k = \frac{\sqrt{2mE}}{\hbar}. \tag{26.7}$$

There is no incoming wave from the right. The four matching conditions are

$$1 + r = C + D, \qquad \mathrm{i}k(1 - r) = \kappa(C - D), \tag{26.8}$$

$$t\mathrm{e}^{\mathrm{i}ka} = C\mathrm{e}^{\kappa a} + D\mathrm{e}^{-\kappa a}, \qquad \mathrm{i}kt\mathrm{e}^{\mathrm{i}ka} = \kappa(C\mathrm{e}^{\kappa a} - D\mathrm{e}^{-\kappa a}). \tag{26.9}$$

Eliminating C, D, r gives

$$\frac{1}{T} = 1 + \frac{(k^2 + \kappa^2)^2}{4k^2\kappa^2} \sinh^2(\kappa a). \tag{26.10}$$

Use $k^2 + \kappa^2 = 2mV_0/\hbar^2$ to recover the quoted formula. Equal asymptotic wave numbers make $T = |t|^2$ here. For $\kappa a \gg 1$, $\sinh^2 \kappa a \simeq \mathrm{e}^{2\kappa a}/4$, exposing the exponential suppression.

26.4 The harmonic oscillator

The oscillator potential is quadratic, so position and momentum enter symmetrically after suitable rescaling. Ladder operators exploit that symmetry. Instead of solving a new differential equation for every level, we will construct the spectrum by identifying an operator that raises or lowers energy by a fixed amount.

For $H = p^2/(2m) + m\omega^2 x^2/2$, define

$$a = \sqrt{\frac{m\omega}{2\hbar}}\, x + \frac{\mathrm{i}p}{\sqrt{2m\hbar\omega}}, \qquad [a, a^\dagger] = 1. \tag{26.11}$$

Then $H = \hbar\omega(a^\dagger a + 1/2)$, with

$$E_n = \hbar\omega(n + 1/2), \quad a|n\rangle = \sqrt{n}|n - 1\rangle, \quad a^\dagger|n\rangle = \sqrt{n + 1}|n + 1\rangle. \tag{26.12}$$

The ground state is Gaussian and has $\Delta x \Delta p = \hbar/2$. A coherent state satisfies $a|\alpha\rangle = \alpha|\alpha\rangle$; its expectation values follow the classical oscillator motion while retaining quantum fluctuations.

Multiply the definition of a by its adjoint and retain the order of x and p:

$$a^\dagger a = \frac{m\omega x^2}{2\hbar} + \frac{p^2}{2m\hbar\omega} + \frac{\mathrm{i}[x, p]}{2\hbar} = \frac{H}{\hbar\omega} - \frac{1}{2}. \tag{26.13}$$

The last term comes from the commutator and produces the zero-point energy. Positivity of $\|a|\psi\rangle\|^2$ prevents repeated lowering from continuing to negative occupation. The lowest state obeys $a|0\rangle = 0$. In position space,

$$\left(x + \frac{\hbar}{m\omega}\frac{\mathrm{d}}{\mathrm{d}x}\right)\phi_0 = 0 \quad \Longrightarrow \quad \phi_0(x) \propto \mathrm{e}^{-m\omega x^2/(2\hbar)}. \tag{26.14}$$

Repeated raising then generates excited states, with normalization fixing the square-root coefficients in the ladder formulas.

26.5 Angular momentum and spin

Quantum angular momentum describes how a state responds to rotations. Its components do not all commute, so we choose a compatible pair, conventionally J^2 and J_z. Raising and lowering the z component while keeping total angular momentum fixed produces a finite chain of allowed outcomes.

Angular momentum obeys $[J_i, J_j] = \mathrm{i}\hbar\epsilon_{ijk}J_k$. Simultaneous eigenstates satisfy

$$J^2|jm\rangle = \hbar^2 j(j+1)|jm\rangle, \qquad J_z|jm\rangle = \hbar m|jm\rangle, \quad m = -j, \ldots, j. \tag{26.15}$$

Orbital angular momentum has integer ℓ and angular wavefunctions $Y_{\ell m}$. Intrinsic spin may be integer or half-integer and is not literally a little sphere spinning in ordinary space.

For spin one-half, $\boldsymbol{S} = \hbar\boldsymbol{\sigma}/2$, where

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{26.16}$$

A spin state aligned along a unit vector $\hat{\boldsymbol{n}}$ has probability $\cos^2(\theta/2)$ of being measured spin-up along an axis making angle θ with $\hat{\boldsymbol{n}}$. Adding angular momenta gives $j = |j_1 - j_2|, \ldots, j_1 + j_2$.

Define $J_\pm = J_x \pm \mathrm{i}J_y$. The commutators imply $[J_z, J_\pm] = \pm\hbar J_\pm$ and

$$\|J_\pm|jm\rangle\|^2 = \hbar^2[j(j+1) - m(m \pm 1)]. \tag{26.17}$$

A norm cannot be negative, so the ladder stops at $m = j$ above and $m = -j$ below. For spin one-half, a state pointing at polar angle θ and azimuth ϕ can be written

$$|{+}\hat{\boldsymbol{n}}\rangle = \cos\frac{\theta}{2}|{+}z\rangle + \mathrm{e}^{\mathrm{i}\phi}\sin\frac{\theta}{2}|{-}z\rangle. \tag{26.18}$$

Projection onto $|{+}z\rangle$ and the Born rule give the $\cos^2(\theta/2)$ probability. The half-angle is a property of spinor rotations, not the classical projection of a rotating little ball.

26.6 Hydrogenic atoms

The Coulomb potential depends only on radius, so separate angular dependence using angular-momentum eigenfunctions. The remaining radial equation contains both attraction and an angular-momentum barrier. Bound-state energies are selected by regularity at the origin and decay at infinity, much as box energies were selected by the walls.

For a point nucleus of charge $+Ze$ and a single electron, use reduced mass μ and potential $-Ze^2/(4\pi\varepsilon_0 r)$. Separation gives $\psi_{n\ell m} = R_{n\ell}(r)Y_{\ell m}(\theta,\phi)$ and

$$E_n = -\frac{\mu Z^2 e^4}{2(4\pi\varepsilon_0)^2\hbar^2 n^2}, \qquad a_\mu = \frac{4\pi\varepsilon_0\hbar^2}{\mu e^2}. \tag{26.19}$$

For ordinary hydrogen, the ground-state binding energy is approximately 13.6 eV and the characteristic radius is about $a_0 = 5.29\times10^{-11}$ m. Allowed labels are $n \geq 1$, $0 \leq \ell \leq n-1$, and $-\ell \leq m \leq \ell$. Spin, relativistic effects, nuclear structure, and radiative corrections refine this Coulomb spectrum.

Write $u(r) = rR(r)$ and $\kappa = Ze^2/(4\pi\varepsilon_0)$. Separation gives

$$-\frac{\hbar^2}{2\mu}u'' + \left[\frac{\hbar^2\ell(\ell+1)}{2\mu r^2} - \frac{\kappa}{r}\right]u = Eu. \tag{26.20}$$

For the ground state, try $R(r) = A\mathrm{e}^{-r/a}$ with $\ell = 0$. Its radial Laplacian is $(a^{-2} - 2/(ar))R$, so the Schrodinger equation becomes

$$\left[-\frac{\hbar^2}{2\mu a^2} + \frac{1}{r}\left(\frac{\hbar^2}{\mu a} - \kappa\right)\right]R = ER. \tag{26.21}$$

The $1/r$ coefficient must vanish: $a = \hbar^2/(\mu\kappa) = a_\mu/Z$. The remaining constant gives $E_1 = -\mu\kappa^2/(2\hbar^2)$. Higher levels require solving the radial polynomial problem with the same boundary conditions; that further step yields the $1/n^2$ spectrum.

Exercises

For a particle in an infinite well, sketch the lowest stationary wavefunction. Identify where it must vanish and why its energy is not zero.

Answers are in Appendix A.

Chapter 27

Quantum approximation methods and scattering

Exact models are our training ground, not the usual destination. The same approximation habits learned from Taylor series, small oscillations, and geometric optics now return in quantum form: identify a small parameter, keep the leading correction, and test where the approximation breaks. Exact solutions are landmarks, not the typical case. When a problem differs slightly from a solvable one, we can organize corrections. When no nearby exact solution is available, we may seek a bound or a semiclassical approximation. The important question is not just how to apply a method, but which small parameter, gap, or length scale makes it trustworthy.

27.1 Time-independent perturbation theory

Start with a Hamiltonian whose eigenstates you know, and describe the remaining interaction as a perturbation. Expand both the energy and the state in the same bookkeeping parameter. Matching equal powers then turns one hard eigenproblem into a sequence of simpler equations involving the known states. H_0 is the solvable reference Hamiltonian, V the added perturbation, and λ a bookkeeping parameter that is set to one only after the expansion is organized.

Let $H = H_0 + \lambda V$. For a nondegenerate unperturbed level,

$$E_n = E_n^{(0)} + \lambda \langle n|V|n\rangle + \lambda^2 \sum_{m\neq n} \frac{|\langle m|V|n\rangle|^2}{E_n^{(0)} - E_m^{(0)}} + \cdots , \tag{27.1}$$

$$|n^{(1)}\rangle = \sum_{m\neq n} \frac{\langle m|V|n\rangle}{E_n^{(0)} - E_m^{(0)}} |m\rangle. \tag{27.2}$$

The states on the right are unperturbed. The expansion fails near degeneracy unless the perturbation is first diagonalized within the degenerate or nearly degenerate subspace.

Worked example 27.1 (Weak quartic anharmonicity)**.** For an added interaction $V = gx^4$, where g is a physical coupling distinct from the bookkeeping parameter above, the harmonic-oscillator

ground state has $\langle x^4 \rangle = 3[\hbar/(2m\omega)]^2$. The first-order shift is therefore $\Delta E_0 = 3g\hbar^2/(4m^2\omega^2)$. Its smallness relative to $\hbar\omega$ provides a practical perturbative criterion.

Use $E_n = E_n^{(0)} + \lambda E_n^{(1)} + \cdots$ and $|\psi_n\rangle = |n\rangle + \lambda|n^{(1)}\rangle + \cdots$. At first order,

$$(H_0 - E_n^{(0)})|n^{(1)}\rangle = -(V - E_n^{(1)})|n\rangle. \tag{27.3}$$

Project onto $\langle n|$ to obtain $E_n^{(1)} = \langle n|V|n\rangle$. Project onto $\langle m|$ for $m \neq n$ to obtain

$$\langle m|n^{(1)}\rangle = \frac{\langle m|V|n\rangle}{E_n^{(0)} - E_m^{(0)}}. \tag{27.4}$$

With $\langle n|n^{(1)}\rangle = 0$ as a normalization convention, these projections construct the first-order state. At second order, projection onto $\langle n|$ gives $E_n^{(2)} = \langle n|V|n^{(1)}\rangle$, producing the stated sum. Small denominators are a warning: even a modest matrix element can strongly mix nearly degenerate states.

27.2 Variational principle and WKB

The variational method asks how low an energy you can obtain within a chosen family of trial states. WKB asks how a wave evolves when its local wavelength changes slowly. These approaches use different information and answer different questions; neither requires an exact solution of the full eigenvalue problem.

For any normalized trial state in the Hamiltonian's domain,

$$E_0 \leq \langle \psi_{\text{trial}}|H|\psi_{\text{trial}}\rangle. \tag{27.5}$$

Expanding the trial state in energy eigenstates proves the bound: its expected energy is a weighted average no lower than the ground energy. Optimizing trial parameters improves an upper bound, not necessarily the exact wavefunction everywhere.

In one dimension where the local momentum varies slowly,

$$\psi(x) \approx \frac{1}{\sqrt{p(x)}}\left[A\mathrm{e}^{\frac{\mathrm{i}}{\hbar}\int^x p\mathrm{d}x'} + B\mathrm{e}^{-\frac{\mathrm{i}}{\hbar}\int^x p\mathrm{d}x'}\right], \quad p = \sqrt{2m(E-V)}. \tag{27.6}$$

WKB fails near turning points and requires matching. For two smooth simple turning points,

$$\int_{x_1}^{x_2} p\mathrm{d}x = (n + 1/2)\pi\hbar. \tag{27.7}$$

A smooth thick barrier gives leading transmission

$$T \sim \exp\left[-\frac{2}{\hbar}\int \sqrt{2m(V-E)}\mathrm{d}x\right], \tag{27.8}$$

where the integral extends over the forbidden interval.

For a normalized trial state $|\psi\rangle = \sum_n c_n|n\rangle$, the energy difference is

$$\langle\psi|H|\psi\rangle - E_0 = \sum_n |c_n|^2(E_n - E_0) \geq 0. \tag{27.9}$$

This is the entire upper-bound argument. A concrete trial width b for an oscillator, $\psi \propto \mathrm{e}^{-x^2/(2b^2)}$, gives

$$E(b) = \frac{\hbar^2}{4mb^2} + \frac{m\omega^2 b^2}{4}. \tag{27.10}$$

Narrowing lowers potential energy but raises kinetic energy. Setting $\mathrm{d}E/\mathrm{d}b = 0$ yields $b^2 = \hbar/(m\omega)$ and $E = \hbar\omega/2$; the result is exact because this trial family contains the true ground state.

For WKB, insert $\psi = A(x)\mathrm{e}^{\mathrm{i}S(x)/\hbar}$ into the stationary equation. At leading order, $S'^2 = 2m(E - V) = p^2$. At next order, $2A'S' + AS'' = 0$, or $(A^2S')' = 0$, so $A \propto p^{-1/2}$. This explains both the phase integral and the amplitude factor. At a turning point $p = 0$, that factor diverges and the approximation must be replaced locally and matched.

27.3 Time-dependent transitions

A time-dependent perturbation can transfer amplitude between energy eigenstates. Separate the phase each state would acquire on its own from the changing expansion coefficients. The resulting equation shows that transitions depend on both the coupling matrix element and the relative phase accumulated during the interaction.

To first order for a perturbation $V(t)$ switched on at $t = 0$,

$$c_f^{(1)}(t) = -\frac{\mathrm{i}}{\hbar}\int_0^t \langle f|V(t')|i\rangle\, \mathrm{e}^{\mathrm{i}(E_f - E_i)t'/\hbar}\mathrm{d}t'. \tag{27.11}$$

For weak coupling to a dense set of final states and a suitable long-time regime, Fermi's golden rule is

$$\Gamma_{i\to f} = \frac{2\pi}{\hbar}|V_{fi}|^2\rho(E_f), \tag{27.12}$$

with energy conservation selecting the resonant final energy, including any energy supplied by a periodic drive. It is a rate approximation, not a universal short-time law.

An isolated resonantly driven two-level system instead exhibits coherent Rabi oscillations, $P_e(t) = \sin^2(\Omega_R t/2)$ under the rotating-wave approximation and standard definition of Ω_R.

Write $|\psi\rangle = \sum_n c_n(t)\mathrm{e}^{-\mathrm{i}E_n t/\hbar}|n\rangle$ using the unperturbed eigenbasis. Substitution into Schrodinger's equation cancels the unperturbed evolution and leaves

$$\mathrm{i}\hbar\dot{c}_f = \sum_n V_{fn}(t)c_n(t)\mathrm{e}^{\mathrm{i}(E_f - E_n)t/\hbar}. \tag{27.13}$$

For a weak perturbation and initial state i, replace c_i by one and the other coefficients by zero on the right. Integration gives the first-order amplitude above. A rapidly oscillating phase tends to cancel contributions from different times; a resonant drive compensates that phase and lets amplitude accumulate. A dense continuum can support a rate description, while an isolated pair of levels can coherently exchange population back and forth.

27.4 Adiabatic evolution and geometric phase

Slow change is meaningful only relative to the system's internal time scales. If two levels approach each other, their small energy gap makes transitions easier and the needed change may have to be much slower. Even when the system remains in one instantaneous eigenstate, its phase records more than the integral of its energy.

A system follows an instantaneous nondegenerate eigenstate approximately when changes are slow compared with the relevant inverse energy gaps and transition matrix elements are small. Closing gaps can defeat this approximation. In a cyclic parameter change, the state can acquire both a dynamical phase and Berry phase

$$\gamma_n = \mathrm{i} \oint \langle n(\boldsymbol{R}) | \boldsymbol{\nabla}_R | n(\boldsymbol{R}) \rangle \cdot \mathrm{d}\boldsymbol{R}. \tag{27.14}$$

The accumulated geometric phase can have observable interference consequences.

For parameters changing in time, try $|\psi\rangle = \mathrm{e}^{-\frac{\mathrm{i}}{\hbar}\int E_n \mathrm{d}t} \mathrm{e}^{\mathrm{i}\gamma_n(t)} |n(t)\rangle$ and neglect transitions out of level n. Projecting Schrodinger's equation onto $\langle n(t)|$ gives

$$\dot{\gamma}_n = \mathrm{i}\langle n | \dot{n} \rangle. \tag{27.15}$$

Integrating along a closed parameter path produces the Berry phase. The eigenvector's phase convention can change the local integrand, but the closed-loop phase modulo 2π, with consistent endpoint identification, has observable interference consequences. This is another reason to distinguish relative phase from an arbitrary overall phase at one instant.

27.5 Scattering and cross sections

A scattering experiment prepares incoming flux and counts particles emerging into different directions. Solve for the outgoing amplitude far from the target, where its radial spreading is simple. The cross section translates that amplitude into a count rate normalized by the incident beam. In scattering, $f(\theta, \phi)$ is an amplitude while $\mathrm{d}\sigma/\mathrm{d}\Omega$ is a measurable cross section; amplitude is squared only after all alternatives have been added.

For an incoming plane wave and outgoing spherical wave in three dimensions,

$$\psi \sim \mathrm{e}^{\mathrm{i}kz} + f(\theta, \phi) \frac{\mathrm{e}^{\mathrm{i}kr}}{r}, \qquad \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = |f|^2 \tag{27.16}$$

for elastic scattering with equal incident and outgoing wave numbers. The differential cross section is scattered rate per solid angle divided by incident flux; it has units of area per solid angle.

For a short-range central potential,

$$f(\theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell + 1) \mathrm{e}^{\mathrm{i}\delta_\ell} \sin \delta_\ell P_\ell(\cos\theta), \quad \sigma_{\mathrm{tot}} = \frac{4\pi}{k^2} \sum_{\ell} (2\ell + 1) \sin^2 \delta_\ell. \tag{27.17}$$

At sufficiently low energy, the s wave often dominates. Long-range Coulomb scattering requires special treatment of the asymptotic form. The first Born approximation treats a weak scattering potential perturbatively; it is not justified merely because an integral is easy to evaluate.

For incident amplitude one, $j_{\mathrm{in}} = \hbar k/m$. Far away, the outgoing spherical term gives radial current $j_r \simeq (\hbar k/m)|f|^2/r^2$. A detector subtending solid angle $\mathrm{d}\Omega$ intercepts area $r^2\mathrm{d}\Omega$, so

$$\frac{j_r r^2 \mathrm{d}\Omega}{j_{\mathrm{in}}} = |f|^2 \mathrm{d}\Omega. \tag{27.18}$$

The distance cancels, explaining $\mathrm{d}\sigma/\mathrm{d}\Omega = |f|^2$. For a central potential, partial waves separate the angular-momentum channels; phase shifts measure how much each radial wave differs from free motion. The complicated interaction region is encoded in those shifts before the outgoing channels are recombined.

Exercises

Before using perturbation theory, what should you compare to judge whether the added interaction is small? Name one result that the approximation can predict.

Answers are in Appendix A.

Chapter 28

Identical particles, entanglement, and quantum information

A single quantum state was already a vector, but several quantum systems require a larger space built from tensor products. This is where the linear algebra introduced earlier becomes physically consequential: correlations can belong to the joint state even when no subsystem has a pure state of its own. With several quantum systems, the state space grows by tensor products. This introduces two ideas with no direct classical-particle counterpart: exchanging identical particles constrains the allowed states, and a joint state need not assign a pure state to each part. Density operators let us describe both incomplete preparation knowledge and subsystems of entangled states.

28.1 Exchange symmetry

Labels such as "particle 1" and "particle 2" are bookkeeping devices for identical particles. Swapping them must not produce a physically distinguishable labeling of the same configuration. Bosonic and fermionic exchange symmetry impose different amplitude relations, with important consequences even when the Hamiltonian contains no explicit exchange-force term.

For identical bosons the total state is symmetric under exchange; for identical fermions it is antisymmetric. Two fermions cannot occupy the same complete one-particle state. A Slater determinant implements antisymmetry for independent-particle orbitals.

For two electrons, a spin singlet is antisymmetric in spin and must be paired with a symmetric spatial state; spin triplets require antisymmetric spatial states. Exchange effects do not arise from an extra classical force, although they can strongly modify energies and correlations.

For orthonormal orbitals a, b, form

$$\psi_{\pm}(1,2) = \frac{\phi_a(1)\phi_b(2) \pm \phi_b(1)\phi_a(2)}{\sqrt{2}}. \tag{28.1}$$

Exchange preserves the plus state and reverses the sign of the minus state. If $a = b$, the antisymmetric numerator vanishes identically, so no normalized state exists: this is Pauli exclusion. For

electrons, apply antisymmetry to the complete spatial-and-spin state. A symmetric spatial factor is allowed when the spin factor is antisymmetric.

28.2 Density operators and subsystems

Suppose a preparation produces state $|\psi_i\rangle$ with probability p_i, and you do not know which occurred. Average the measurement predictions, not the amplitudes. The density operator collects that average into one object and also handles the state available to an observer who can access only part of a larger system. ρ is a density operator, ρ_{AB} the joint state, and $\rho_A = \mathrm{Tr}_B\, \rho_{AB}$ the state available when subsystem B is ignored. Purity $\mathrm{Tr}\,\rho^2$ distinguishes pure from mixed states.

A statistical mixture is represented by

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \qquad \mathrm{Tr}\,\rho = 1, \quad \rho \geq 0. \tag{28.2}$$

Expectations are $\langle A\rangle = \mathrm{Tr}(\rho A)$. A pure state has $\rho^2 = \rho$; a mixed state has $\mathrm{Tr}\,\rho^2 < 1$. Closed-system evolution obeys $\mathrm{i}\hbar\dot{\rho} = [H, \rho]$. The reduced state of subsystem A is $\rho_A = \mathrm{Tr}_B\, \rho_{AB}$.

For an observable A,

$$\langle A\rangle = \sum_i p_i \langle\psi_i|A|\psi_i\rangle = \mathrm{Tr}\left(\sum_i p_i |\psi_i\rangle\langle\psi_i| A\right). \tag{28.3}$$

This motivates ρ. A partial trace is defined by requiring all local predictions to agree: $\mathrm{Tr}_{AB}[\rho_{AB}(A\otimes I_B)] = \mathrm{Tr}_A(\rho_A A)$. In an orthonormal basis $|b\rangle$ of the inaccessible subsystem,

$$\rho_A = \sum_b \langle b|\rho_{AB}|b\rangle, \tag{28.4}$$

where the contraction is only over subsystem B. We are discarding access to its outcomes, not asserting that the global state ceased to exist.

Worked example 28.1 (A pure pair with an uncertain part)**.** Take $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. The pair has a definite pure state, but measuring only its first qubit gives 0 and 1 with equal probability. If we ignore the second qubit, the first is described by $\rho_A = (|0\rangle\langle 0| + |1\rangle\langle 1|)/2 = I/2$. Its purity is $\mathrm{Tr}(\rho_A^2) = 1/2$, while the joint state's purity is one. Losing access to a subsystem changes the description available to us.

28.3 Entanglement and correlations

A joint pure state is entangled when it cannot be written as one state for each subsystem multiplied together. The Bell state provides a small example whose entire algebra can be inspected. Its local uncertainty coexists with perfect correlations between selected joint outcomes. Entanglement is a property of factorization, not merely of a probability distribution.

The Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{28.5}$$

cannot be factored into a product of states of its two parts. Each subsystem has $\rho_A = I/2$, although the combined state is pure. Entanglement entropy for a pure bipartite state is $S_A = -k_{\mathrm{B}} \operatorname{Tr}(\rho_A \ln \rho_A)$.

Bell inequalities constrain local hidden-variable models under their stated assumptions; suitable quantum correlations violate them. Such correlations do not permit controllable faster-than-light signaling. Measurement outcomes are correlated, while each observer's unconditioned local statistics remain unaffected by the distant measurement choice.

Expand its density operator:

$$\rho_{AB} = \tfrac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|). \tag{28.6}$$

Tracing over B kills the cross terms because $\langle 0|1\rangle = 0$, leaving $\rho_A = (|0\rangle\langle 0| + |1\rangle\langle 1|)/2$. The eigenvalues are both $1/2$, so purity is $1/2$ and entropy is $k_{\mathrm{B}} \ln 2$. The global state's purity remains one. Local mixedness alone does not prove entanglement for an arbitrary mixed joint state, but for a pure joint state it signals entanglement.

28.4 Qubits, interference, and decoherence

A qubit's relative phase matters because future operations can convert it into an observable probability difference. Losing access to that phase information changes what interference can reveal. We can see the distinction with two Hadamard operations and with a simple model of environmental entanglement.

A qubit is $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. A Hadamard operation maps $|0\rangle$ to $(|0\rangle + |1\rangle)/\sqrt{2}$; a controlled-NOT flips a target when its control is $|1\rangle$. Together they can create the Bell state from $|00\rangle$.

A quantum computation manipulates amplitudes so useful outcomes interfere constructively. It does not make every amplitude individually readable. Unknown arbitrary quantum states cannot be perfectly cloned by a universal operation. Quantum error correction encodes information in a larger space and extracts error syndromes without directly measuring the encoded logical amplitudes.

Environmental entanglement suppresses interference in a subsystem's reduced state: this is decoherence. It explains the practical emergence of preferred stable observables, but should not be silently identified with every interpretive claim about definite measurement outcomes.

Since $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, applying H twice returns $|0\rangle$. The unwanted output amplitudes cancel. By contrast, an equal incoherent mixture remains $I/2$ under a Hadamard and gives equal output probabilities.

If an environment becomes correlated with the qubit, the joint state may be $\alpha|0\rangle|e_0\rangle + \beta|1\rangle|e_1\rangle$. After tracing out the environment, the qubit's off-diagonal element is

$$(\rho_{\mathrm{qubit}})_{01} = \alpha\beta^*\langle e_1|e_0\rangle. \tag{28.7}$$

Nearly orthogonal environmental records suppress this coherence. This calculation shows how interference can become inaccessible locally while the full system still undergoes quantum evolution.

Exercises

A pair of qubits is in an entangled pure state. Explain how one qubit can have a mixed reduced state even though the pair has a pure state.

Answers are in Appendix A.

Part XI

Quantum Computing

Chapter 29

Quantum computing from bits to algorithms

Quantum computing uses the state amplitudes and measurements we have just studied to process information. We will start with an ordinary bit, then add one qubit and one gate. Only after seeing how two gates make amplitudes cancel will we assemble a circuit and a small search algorithm.

29.1 Bits, states, and reversible operations

A classical bit is either 0 or 1. A register of n bits has one definite string at a time, although an algorithm may manipulate many possible inputs over many runs. A reversible classical operation is a permutation of those strings; it has an inverse and loses no information.

For two bits, a controlled-NOT maps (a, b) to $(a, a \operatorname{xor} b)$. Applying it twice returns the original pair. Reversibility matters because a quantum gate must be represented by a unitary transformation, and unitary transformations preserve inner products and total probability.

29.2 One qubit, one measurement at a time

A classical bit has a definite value, 0 or 1. For a qubit, choose two distinguishable basis states called $|0\rangle$ and $|1\rangle$. A general pure state combines them with complex amplitudes α and β:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle. \tag{29.1}$$

The vertical bars and angle brackets mark a state vector. They are labels in a mathematical state space, not positions of a particle. We will first ask only what happens if the qubit is measured in this 0/1 basis.

The Born rule says the probability of outcome 0 is $|\alpha|^2$ and the probability of outcome 1 is $|\beta|^2$. Because one of the two outcomes must occur, their probabilities add to one:

$$|\alpha|^2 + |\beta|^2 = 1. \tag{29.2}$$

This is normalization. An amplitude may be negative or complex; a probability cannot. For instance, $\alpha = 1/\sqrt{2}$ and $\beta = \mathrm{i}/\sqrt{2}$ give equal 1/2 probabilities in this basis.

Two states can have the same 0/1 probabilities yet differ physically. Compare $(|0\rangle + |1\rangle)/\sqrt{2}$ with $(|0\rangle - |1\rangle)/\sqrt{2}$. Both give half of each outcome when measured directly. A later gate can make their amplitudes add or cancel differently, revealing the relative sign. Multiplying the entire state by the same complex phase changes no measurement prediction; changing the phase of one term relative to the other can matter.

29.3 The Bloch sphere as a picture of one qubit

After normalization and removal of the irrelevant overall phase, a pure qubit needs two real angles. We can write it as

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + \mathrm{e}^{\mathrm{i}\phi}\sin\frac{\theta}{2}|1\rangle, \qquad 0 \le \theta \le \pi. \tag{29.3}$$

The angle θ sets the 0/1 probabilities: $P(0) = \cos^2(\theta/2)$ and $P(1) = \sin^2(\theta/2)$. The angle ϕ is a relative phase; it does not change these two probabilities directly, but it affects measurements in other bases. Put θ and ϕ on a unit sphere to obtain the Bloch-sphere picture. The north and south poles represent $|0\rangle$ and $|1\rangle$; points on the equator have equal 0/1 probabilities. The sphere describes possible pure states, not the qubit's location in physical space.

29.4 Gates as rotations and interference

Quantum gates are unitary matrices. The Pauli matrices generate rotations, while the Hadamard gate changes between computational and interference-friendly bases:

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \tag{29.4}$$

Applying H twice returns the input. The second application adds amplitudes with signs, so cancellation and reinforcement occur before measurement. This is the central visual idea of quantum algorithms. Gates are unitary matrices, so they preserve $|\alpha|^2 + |\beta|^2 = 1$.

For a general state, a phase gate changes β without changing computational-basis probabilities immediately. A later Hadamard converts that hidden phase into a measurable probability difference. Therefore an experiment must display both amplitudes and the final measurement histogram; either one alone hides part of the calculation.

Worked example 29.1 (Why two Hadamard gates undo each other)**.** Begin with $|0\rangle$. One Hadamard gate gives $(|0\rangle + |1\rangle)/\sqrt{2}$. Apply the gate again to each term: the $|0\rangle$ contribution becomes $(|0\rangle + |1\rangle)/2$ and the $|1\rangle$ contribution becomes $(|0\rangle - |1\rangle)/2$. Adding them cancels the $|1\rangle$ amplitude and leaves $|0\rangle$. This is interference in the arithmetic of amplitudes, before any measurement is made.

29.5 Registers, tensor products, and circuits

Two qubits occupy the tensor-product space with basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. A product state has amplitudes that factor, while an entangled state does not. A circuit is a time-ordered product of gates, with wires showing subsystems and measurement symbols showing where probabilities are extracted.

The controlled-NOT acts as

$$\mathrm{CNOT}|a, b\rangle = |a, a \text{ xor } b\rangle. \tag{29.5}$$

Starting from $|00\rangle$, apply H to the first qubit and then CNOT:

$$|00\rangle \xrightarrow{H\otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\mathrm{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{29.6}$$

The final state is entangled. Measuring one qubit predicts the corresponding result of the other, but neither result is individually determined before measurement.

29.6 A first algorithm: amplitude amplification

Suppose one item among N possibilities is marked. An oracle is an operation that recognizes the marked item and reverses the sign of its amplitude. The sign change alone does not make the marked result more likely; measurement probabilities are squares of amplitudes. We need a second operation that turns that sign difference into a larger amplitude. Grover's algorithm reflects all amplitudes about their average. Together, the two reflections increase the marked probability.

For a concrete case, start with four equally likely items. Their amplitudes are $(1/2, 1/2, 1/2, 1/2)$, so each has probability $1/4$. Let the last item be marked. The oracle changes the amplitudes to $(1/2, 1/2, 1/2, -1/2)$ without changing any probability. Their average is now $1/4$. Reflecting an amplitude a about this average sends it to $2(1/4) - a$. The first three amplitudes become 0, while the marked amplitude becomes 1. A measurement now returns the marked item with certainty in this ideal four-item example. This arithmetic shows where the amplification comes from before we write the general formula.

In the two-dimensional space spanned by the marked state $|w\rangle$ and the normalized unmarked state $|r\rangle$, write

$$|\psi\rangle = \sin\theta|w\rangle + \cos\theta|r\rangle, \qquad \sin\theta = \frac{1}{\sqrt{N}}. \tag{29.7}$$

The oracle reflection and diffusion reflection together rotate by approximately 2θ. After k iterations,

$$P_k = \sin^2[(2k+1)\theta]. \tag{29.8}$$

The quadratic speedup comes from rotation in amplitude space, not from reading all amplitudes individually. Measuring too late rotates past the marked direction, so the number of iterations is part of the algorithm.

29.7 Noise, error correction, and what the picture must show

Real qubits lose phase coherence, suffer gate errors, and can be measured imperfectly. A density matrix describes the resulting mixed state. Error correction encodes one logical qubit into several physical qubits so that error syndromes can be measured without directly revealing the logical state.

For a simple phase-damping model, the off-diagonal density-matrix element evolves as

$$\rho_{01}(t) = \rho_{01}(0)\mathrm{e}^{-t/T_2}, \tag{29.9}$$

while the diagonal probabilities remain fixed. A useful visual therefore shows the Bloch vector shrinking toward the measurement axis, alongside the circuit and the changing probability histogram. The picture must distinguish a changing amplitude from a changing classical probability and must report the noise model used.

Exercises

Apply a Hadamard gate twice to the state $|0\rangle$. What state results, and what does this show about interference?

Answers are in Appendix A.

Part XII

Atoms, Materials, Nuclei, and Particles

Chapter 30

Atomic and molecular physics

Quantum mechanics first gave us ideal particles and ideal potentials. Atoms ask what happens when those states interact with fields, nuclei, and other electrons; molecules then ask how the same quantum structure produces collective bonds and vibrations. The hydrogen atom gave us a starting model. Real atoms add electron-electron interactions, spin couplings, and corrections at smaller energy scales. Molecules add moving nuclei and bonding. We will build a hierarchy of approximations, using each simpler model to identify which additional terms are likely to matter next.

30.1 Atomic structure beyond hydrogen

Write the interactions before choosing orbitals. For a fixed nucleus, each electron is attracted to the nucleus and repelled by every other electron. The repulsion depends on pairs of coordinates, so it prevents the Hamiltonian from separating into independent one-electron hydrogen problems. Z is nuclear charge number, n principal quantum number, ℓ orbital angular-momentum label, and m its projection label.

In many-electron atoms, electron-electron repulsion prevents exact separation into independent hydrogenic problems. Useful approximations include a central mean field, Hartree–Fock orbitals with exchange, and correlation corrections. Shell structure, spin, and Pauli exclusion organize the periodic table.

Fine structure combines relativistic kinetic corrections, spin-orbit coupling, and the Darwin term. Hyperfine structure couples nuclear and electronic angular momentum. Radiative corrections and finite nuclear size produce additional shifts, including the Lamb shift. These corrections are smaller than gross Coulomb energies but essential for precision spectra.

Ignoring relativistic effects and nuclear motion, the starting Hamiltonian is

$$H = \sum_i \left[-\frac{\hbar^2}{2m_e}\nabla_i^2 - \frac{Ze^2}{4\pi\varepsilon_0 r_i} \right] + \sum_{i<j} \frac{e^2}{4\pi\varepsilon_0 |\boldsymbol{r}_i - \boldsymbol{r}_j|}. \tag{30.1}$$

The pair sum counts each electron pair once. A mean-field approximation replaces the detailed influence of other electrons by an effective one-electron potential. Because that potential depends on the occupied orbitals, solve for orbitals, rebuild the field, and repeat toward self-consistency. Antisymmetrization adds exchange; correlation describes physics beyond a single independent-particle determinant. This hierarchy explains why shell models are useful without making them exact.

30.2 Light-matter interaction and selection rules

A transition matrix element can vanish because of symmetry even when the photon has the right energy. Before integrating a complicated wavefunction, inspect how the initial state, final state, and interaction transform. Selection rules identify these zeros and save work while explaining which spectral lines are strong or absent in a given approximation. The symbols $\langle f|\boldsymbol{d}|i\rangle$ describe a transition matrix element between final and initial states. A selection rule says a matrix element vanishes by symmetry; it is not merely an energy mismatch.

When the optical wavelength is large compared with an atom, the electric dipole interaction is approximately $H' = -\boldsymbol{d}\cdot\boldsymbol{E}(t)$. Transition strength depends on $|\langle f|\boldsymbol{d}|i\rangle|^2$. For single-electron orbital states, electric dipole selection rules include

$$\Delta\ell = \pm 1, \qquad \Delta m = 0, \pm 1, \tag{30.2}$$

with parity changing. For total angular momentum, $\Delta J = 0, \pm 1$ except $J = 0 \to 0$; spin is conserved in the ideal spin-independent dipole approximation. Higher multipoles and state mixing can enable weaker transitions.

A finite excited-state lifetime τ produces a characteristic natural energy width of order $\hbar/\tau$. Doppler motion, collisions, and instrumentation broaden spectral lines further.

For a dipole transition, one component contains an integral of the form $\int \psi_f^* x \psi_i \mathrm{d}^3 r$. Under inversion, x changes sign. If the two states have the same parity, the integrand is odd and its integral vanishes. Opposite parity is therefore necessary. The dipole's vector character supplies the angular-momentum rules quoted above. Energy matching and a nonzero matrix element are separate requirements: satisfying the first does not guarantee the second.

Worked example 30.1 (Energy matching is only the first check)**.** Suppose two atomic states have energies E_i and $E_f > E_i$. A photon must supply $h\nu = E_f - E_i$ to drive an upward transition. Matching this energy is necessary, but the transition strength also depends on the dipole matrix element between the states. If symmetry makes that element zero, the electric-dipole transition is forbidden in that approximation even at the matching frequency. We therefore check both energy conservation and the interaction that couples the states.

30.3 Magnetic fields and resonance

A magnetic field distinguishes states with different projections of magnetic moment along it. First calculate their energy separation; then compare it with the energy of the applied oscillatory field.

Resonance occurs when the drive can efficiently connect those levels, subject to the coupling matrix element.

A magnetic moment contributes $H_Z = -\boldsymbol{\mu} \cdot \boldsymbol{B}$. In weak fields where angular momentum coupling remains valid, atomic shifts are often written $\Delta E = g_J \mu_B m_J B$, with $\mu_B = e\hbar/(2m_e)$. Resonant oscillatory fields drive transitions at the level splitting divided by $\hbar$.

Electron spin resonance and nuclear magnetic resonance probe different magnetic moments and environments. Nuclear chemical shifts and spin couplings reflect electronic structure; they are not described fully by the isolated-nucleus gyromagnetic ratio alone.

For adjacent allowed values of m_J in the stated linear Zeeman regime, the splitting magnitude is $|\Delta E| = |g_J|\mu_B B$. Matching to $\hbar\omega$ gives

$$\omega_{\text{res}} = \frac{|g_J|\mu_B B}{\hbar}. \tag{30.3}$$

The static field sets the energy separation, while a suitable transverse oscillating field induces transitions. As the magnetic interaction becomes comparable to internal angular-momentum couplings, the weak-field labels and effective g_J description must be reconsidered.

30.4 Molecules and the Born–Oppenheimer approximation

Start by holding the nuclei fixed and solving the electronic problem for that arrangement. Repeat for neighboring arrangements to obtain an electronic energy surface. The nuclei then move approximately on that surface. This approach turns the complicated joint problem into a fast electronic calculation followed by slower nuclear dynamics, provided nonadiabatic couplings are small.

Nuclei are much heavier than electrons. Solving electronic motion at approximately fixed nuclear positions produces potential energy surfaces for slower nuclear motion. This separation can fail near electronic degeneracies or during rapid nonadiabatic processes.

A diatomic molecule near equilibrium has approximate vibrational energies $E_v = \hbar\omega(v + 1/2)$ and rigid-rotor energies

$$E_J = \frac{\hbar^2 J(J+1)}{2I}, \qquad I = \mu R_e^2. \tag{30.4}$$

Electronic, vibrational, and rotational energy scales are typically successively smaller. Covalent bonding involves shared electronic states; ionic bonding involves charge transfer and electrostatic attraction; dispersion forces arise from correlated charge fluctuations.

For a diatomic molecule, let $U(R)$ include electronic energy and nuclear repulsion. Expand around its minimum R_e:

$$U(R) \simeq U(R_e) + \tfrac{1}{2}U''(R_e)(R - R_e)^2, \qquad \omega = \sqrt{\frac{U''(R_e)}{\mu}}. \tag{30.5}$$

This is the oscillator again, now for relative nuclear motion. Rotation at nearly fixed separation has classical energy $L^2/(2I)$; replacing L^2 by its eigenvalue $\hbar^2 J(J+1)$ yields the rotor spectrum. The approximations neglect bond stretching during rotation and anharmonicity at large vibration amplitude, giving clear directions for later corrections.

Exercises

Explain how a photon can cause a transition between two atomic energy levels. Why does matching the energy gap alone not guarantee that the transition occurs?

Answers are in Appendix A.

Chapter 31

Condensed matter and collective phenomena

Molecular vibrations showed how many-body motion can still be organized around modes. A crystal repeats that idea across an enormous lattice: translation symmetry creates phonons and bands, while interactions and collective order determine which simple pictures survive. A solid contains too many interacting particles to solve by listing all their coordinates. Its regularities suggest better variables: lattice waves, electron bands, collective order, and quasiparticles. The purpose of these descriptions is to retain the information needed for a particular scale and measurement. Their success comes from symmetry and controlled approximations, not from pretending the microscopic interactions disappear.

31.1 Crystals, reciprocal space, and phonons

A periodic arrangement makes waves emitted or scattered from different cells acquire related phases. Reciprocal space records the phase conditions for constructive addition. For vibrations, the same periodicity allows traveling patterns that diagonalize the coupling between lattice sites. a is lattice spacing, $\boldsymbol{R}$ a lattice translation, $\boldsymbol{k}$ crystal wave vector, $\boldsymbol{G}$ a reciprocal-lattice vector, and ω mode frequency.

A crystal is periodic under lattice translations. Reciprocal vectors satisfy $\boldsymbol{G} \cdot \boldsymbol{R} = 2\pi n$, organizing diffraction and wave states. Bragg's law is $2d \sin\theta = n\lambda$ when θ is measured from the lattice planes.

Small lattice vibrations become coupled harmonic modes. Quantization produces phonons with energies $\hbar\omega_{\boldsymbol{k},s}(n + 1/2)$. Acoustic branches have $\omega \to 0$ as $k \to 0$; multi-atom unit cells can also have optical branches. The Debye model predicts low-temperature lattice heat capacity $C_V \propto T^3$ in three dimensions, while the high-temperature classical limit is approximately $3Nk_{\mathrm{B}}$.

For a one-dimensional chain with spacing a, mass m, and nearest-neighbor spring constant K, the displacement satisfies

$$m\ddot{u}_j = K(u_{j+1} - 2u_j + u_{j-1}). \tag{31.1}$$

Try $u_j = u_0 \mathrm{e}^{\mathrm{i}(kja-\omega t)}$. Cancel the common phase and use $\mathrm{e}^{\mathrm{i}ka} + \mathrm{e}^{-\mathrm{i}ka} = 2\cos ka$:

$$\omega^2 = \frac{2K}{m}(1-\cos ka) = \frac{4K}{m}\sin^2\frac{ka}{2}. \tag{31.2}$$

For $|ka| \ll 1$, $\omega \simeq a\sqrt{K/m}|k|$, an acoustic branch. Quantizing each normal mode supplies phonons. In three dimensions, acoustic mode counting gives a low-frequency density proportional to ω^2; thermally populating modes up to scale $k_\mathrm{B}T/\hbar$ leads to energy proportional to T^4 and heat capacity proportional to T^3.

Worked example 31.1 (A repeating chain and an allowed wave)**.** Consider a chain with identical sites separated by distance a. A displacement pattern $u_j = A\cos(kja)$ assigns a displacement to site j. Increasing k by $2\pi/a$ leaves every site's phase changed by an integer multiple of 2π, so it describes the same sampled pattern on the lattice. This is why wave vectors in a crystal can be organized into one reciprocal-space interval. The physical motion remains a set of ordinary atomic displacements.

31.2 Bloch's theorem and electronic bands

A periodic Hamiltonian commutes with lattice translations. We can therefore choose states that acquire only a phase under each lattice translation. Extracting that phase as a plane-wave factor leaves a function with the periodicity of the lattice: this is the content of Bloch's theorem. The band label n distinguishes solutions with the same $\boldsymbol{k}$; it is not particle number. A gap is an energy interval with no allowed band state in the ideal model.

For a periodic single-particle potential,

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad u_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{R}) = u_{n\boldsymbol{k}}(\boldsymbol{r}). \tag{31.3}$$

Allowed energies form bands $E_n(\boldsymbol{k})$ separated in some cases by gaps. A partially filled band supports metallic behavior in the simple band picture; filled bands separated from empty bands by a gap give an insulator at zero temperature. Correlations and disorder can alter this classification.

A semiclassical wave packet in an isolated ordinary band has

$$\boldsymbol{v} = \frac{1}{\hbar}\boldsymbol{\nabla}_k E_n, \qquad \hbar\dot{\boldsymbol{k}} = -e(\boldsymbol{E} + \boldsymbol{v}\times\boldsymbol{B}). \tag{31.4}$$

Berry curvature can add an anomalous velocity term. Near an isotropic parabolic extremum, $E \simeq E_0 + \hbar^2k^2/(2m^*)$ defines effective mass. Crystal momentum is not generally the free-space mechanical momentum.

If $\psi(\boldsymbol{r}+\boldsymbol{R}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi(\boldsymbol{r})$, define $u(\boldsymbol{r}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\psi(\boldsymbol{r})$. Direct substitution gives $u(\boldsymbol{r}+\boldsymbol{R}) = u(\boldsymbol{r})$. Distinct solutions within a cell supply the band label n. Near a one-dimensional band minimum,

$$E(k) \simeq E_0 + \tfrac{1}{2}E''(k_0)(k-k_0)^2, \qquad \frac{1}{m^*} = \frac{E''(k_0)}{\hbar^2}. \tag{31.5}$$

Using $v = (1/\hbar)\mathrm{d}E/\mathrm{d}k$ and $\hbar\dot{k} = F$ gives $\dot{v} = F/m^*$ locally. The effective mass measures band curvature and hence response to force within this approximation.

31.3 Fermi gases and electron transport

At zero temperature, fermions fill available one-particle states from the bottom up. The Fermi wave number is determined by counting enough states to hold all electrons. Transport then asks how fields and scattering slightly change that occupation, rather than treating every electron as a freely accelerating isolated charge.

For a three-dimensional noninteracting spin-one-half electron gas,

$$k_F = (3\pi^2 n)^{1/3}, \qquad E_F = \frac{\hbar^2 k_F^2}{2m}. \tag{31.6}$$

At low temperature only states near the Fermi surface can change occupation, yielding electronic heat capacity proportional to T rather than the classical constant.

The Drude model with relaxation time τ gives $\sigma = ne^2\tau/m$; a band effective mass can replace m in a suitable simple band model. The single-carrier Hall coefficient is $R_H = -1/(ne)$ for electrons. Multiband transport, scattering anisotropy, and strong correlations require more complete models.

With two spin states per wave vector, counting inside a sphere of radius k_F gives

$$N = 2\frac{V}{(2\pi)^3}\frac{4\pi k_F^3}{3} \quad \Longrightarrow \quad n = \frac{k_F^3}{3\pi^2}. \tag{31.7}$$

For Drude transport, model momentum relaxation by $m\dot{\boldsymbol{v}}_d = -e\boldsymbol{E} - m\boldsymbol{v}_d/\tau$. In steady state $\boldsymbol{v}_d = -e\tau\boldsymbol{E}/m$. Since $\boldsymbol{J} = -ne\boldsymbol{v}_d$, we obtain $\sigma = ne^2\tau/m$. The model separates carrier density, inertia, and scattering time, while leaving the microscopic value of τ to additional physics.

31.4 Semiconductors and junctions

A filled valence band and an empty conduction band carry little ordinary direct current at zero temperature in the simple band model. Excitation or doping creates mobile carriers. Bringing differently doped regions together initially drives diffusion, which leaves fixed ionized dopants behind and builds an opposing electric field.

A semiconductor has a band gap small enough that thermal excitation and doping can control carrier density. Donors tend to supply electrons; acceptors tend to supply holes. A hole is a useful positive-charge quasiparticle describing an unfilled valence-band state.

A p–n junction develops a depletion region and built-in electric field. Under ideal diffusion-dominated conditions, its current-voltage relation is

$$I = I_s\left(\mathrm{e}^{eV/(n_{\mathrm{id}}k_{\mathrm{B}}T)} - 1\right), \tag{31.8}$$

with ideality factor $n_{\mathrm{id}} = 1$ in the simplest model. Recombination, series resistance, and breakdown change this relation. Photons above the gap can create electron-hole pairs, enabling photodetectors and solar cells.

At equilibrium, drift from the built-in field balances diffusion. A forward applied bias lowers the barrier for carrier injection. In the ideal diffusion model, the injected minority-carrier population grows by a Boltzmann factor $\mathrm{e}^{eV/(k_\mathrm{B}T)}$. Subtracting the equilibrium contribution explains the form $I_s(\mathrm{e}^{eV/(k_\mathrm{B}T)}-1)$. A full derivation of I_s requires solving minority-carrier diffusion and recombination with junction boundary conditions. Thus the exponential form and its prefactor carry different pieces of the physics.

31.5 Magnetism, superconductivity, and superfluidity

Collective order means the low-energy state is organized across many constituents. In a magnet, spin correlations organize orientation. In a superconductor or superfluid, a coherent phase becomes a central variable. These descriptions let us predict long-distance behavior without solving every microscopic degree of freedom at once.

Exchange interactions can favor aligned or alternating spins, producing ferromagnetic or antiferromagnetic order. A Heisenberg model uses $H = -\sum_{ij} J_{ij}\boldsymbol{S}_i \cdot \boldsymbol{S}_j$ with units of J_{ij} chosen for the spin convention. Collective spin excitations are magnons.

A superconductor exhibits dissipationless direct-current transport under suitable limits and the Meissner effect, expelling weak magnetic fields from its bulk. London theory gives magnetic penetration over a length λ_L; the coherence length ξ characterizes spatial variation of the order parameter. In conventional BCS superconductors, paired electrons form a coherent many-body state, with weak-coupling result

$$\Delta(0) \simeq 1.764 k_\mathrm{B} T_c, \qquad \Phi_0 = \frac{h}{2e}. \tag{31.9}$$

The first relation is not universal across all superconductors. The second is the usual flux quantum for charge-$2e$ condensates. Josephson junctions satisfy $I = I_c \sin\varphi$ and $\dot{\varphi} = 2eV/\hbar$ in the ideal relations.

Superfluidity involves coherent flow with restricted dissipation channels. A useful Landau criterion is $v_c = \min_p[\epsilon(p)/p]$ for elementary excitation production in the ideal setting; vortices and boundaries can lower observed critical velocities.

One useful derivation is the ideal Landau criterion. In the rest frame of an obstacle, creating an excitation with momentum $\boldsymbol{p}$ in a fluid moving at $\boldsymbol{v}$ can cost energy $\epsilon(p) - \boldsymbol{v}\cdot\boldsymbol{p}$. If this quantity becomes negative, creating the excitation can lower the energy. Requiring it to remain nonnegative for all momenta gives

$$v \le \min_p \frac{\epsilon(p)}{p}. \tag{31.10}$$

This identifies an energetic threshold for that excitation mechanism. It does not include every way real flow can dissipate, such as vortex nucleation at a boundary. Likewise, the quoted BCS gap relation needs a microscopic pairing calculation; it is not implied by phase coherence alone.

31.6 Topology when collective states need it

Topology ignores distances and angles and keeps only properties preserved by continuous deformation. A space is connected if it cannot be separated into two disjoint open pieces. A loop may or may not be continuously shrinkable to a point; the distinction detects holes. The circle has a noncontractible loop, while a disk does not. Orientability asks whether a consistent choice of local orientation exists across the whole space; a Mobius strip fails this test.

31.7 Topology and emergence

Some distinctions between phases survive smooth changes in material parameters. An integer invariant cannot vary continuously: changing it requires leaving the conditions under which it was defined, often through a gap closing. This offers a different kind of classification from a local order parameter such as magnetization.

Macroscopic phases can differ through global properties of their quantum states as well as local order parameters. In an ideal integer quantum Hall plateau,

$$\sigma_{xy} = \nu \frac{e^2}{h}, \qquad \nu \in \mathbb{Z}. \tag{31.11}$$

The integer relates to a topological invariant of occupied states. Edge transport and bulk topology are connected, subject to the system's symmetry and gap conditions. Fractional quantum Hall states require interactions and support excitations with fractionalized properties.

Quasiparticles, order parameters, and effective theories illustrate emergence: the most useful degrees of freedom at long distances need not resemble the microscopic ingredients directly.

Read the Hall formula as two separate statements. The combination e^2/h supplies the conductance scale; the integer ν contains the topological information. Dimensional analysis can identify a possible scale but cannot prove integer quantization or determine which integer a sample realizes. Those conclusions require the quantum states and their response. This is a useful place to remember the boundary between an instructive scaling argument and a full microscopic derivation.

Exercises

Describe how a repeating crystal structure changes the allowed electron states. What physical difference would you expect between a filled band and a partly filled band?

Answers are in Appendix A.

Chapter 32

Nuclear physics

Atomic physics taught us to separate energy scales and to distinguish an allowed transition from a likely one. Nuclear physics applies the same discipline at a smaller length scale, where binding, tunneling, statistics, and relativistic reaction kinematics meet. Nuclear physics brings together binding energy, quantum statistics, tunneling, and reaction kinematics at much shorter distances. Start by accounting for mass and charge consistently, then distinguish whether a process is energetically allowed from whether it happens rapidly. Favorable energy release alone does not determine a decay or reaction rate.

32.1 Nuclear scales and binding

A bound nucleus has less mass than its separated constituents because energy was released when the bound state formed. The binding energy measures the energy required to separate it again. Before inserting tabulated masses, decide whether they refer to bare nuclei or neutral atoms so that electron contributions are handled consistently. $A = Z + N$ is mass number, Z proton number, and N neutron number. B is binding energy and Q reaction energy release; positive Q does not guarantee a rapid reaction.

A nucleus contains Z protons and N neutrons, with mass number $A = Z + N$. A useful empirical radius scale is $R \simeq r_0 A^{1/3}$ with $r_0 \simeq 1.2\,\mathrm{fm}$. Nuclear binding energy is

$$B = [Z m_p + N m_n - M_{\mathrm{nucleus}}]c^2. \tag{32.1}$$

If atomic masses are used, electron masses and binding must be accounted for consistently. Binding reflects residual strong interactions, opposed in part by proton Coulomb repulsion.

A schematic semi-empirical mass formula is

$$B = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} + \delta(A,Z). \tag{32.2}$$

Its terms represent volume attraction, surface correction, Coulomb energy, neutron-proton asymmetry, and pairing. Shell effects cause additional structure and explain enhanced stability at magic nucleon numbers.

The empirical relation $R \propto A^{1/3}$ implies volume proportional to nucleon number, suggesting roughly constant interior density. If each nucleon interacts attractively with a limited number of nearby neighbors, the bulk binding scales as A. Surface nucleons have fewer neighbors, producing a correction proportional to area, $A^{2/3}$. Electrostatic repulsion scales roughly as Z^2/R, hence $Z^2/A^{1/3}$. These arguments motivate the first three mass-formula terms; their coefficients and the remaining refinements require nuclear data and more detailed theory.

32.2 Radioactive decay

A constant decay rate means that each surviving nucleus has the same small probability $\lambda \mathrm{d}t$ of decaying in the next interval $\mathrm{d}t$. The expected number lost is proportional to the number still present. Exponential decay follows from this local probability statement, without requiring every nucleus to carry a predetermined decay time. In decay, λ is a probability rate and $t_{1/2} = \ln 2/\lambda$ the half-life.

For independent nuclei with constant decay rate λ,

$$\dot{N} = -\lambda N, \quad N = N_0 \mathrm{e}^{-\lambda t}, \quad t_{1/2} = \frac{\ln 2}{\lambda}, \quad \mathcal{A} = \lambda N. \tag{32.3}$$

The exponential law is an excellent intermediate-time description; exact quantum survival amplitudes need not be exponential at arbitrarily short or long times. Competing decay modes have total rate $\lambda = \sum_i \lambda_i$ and branching fractions λ_i/λ.

Alpha decay emits a helium nucleus and is enabled by tunneling through a Coulomb barrier. Beta-minus decay changes a neutron into a proton, electron, and electron antineutrino; beta-plus decay changes a proton into a neutron, positron, and electron neutrino when energetically allowed. Gamma decay changes nuclear excitation without changing A or Z.

Separate $\mathrm{d}N/\mathrm{d}t = -\lambda N$ and integrate:

$$\int_{N_0}^{N(t)} \frac{\mathrm{d}N'}{N'} = -\lambda \int_0^t \mathrm{d}t' \quad \Longrightarrow \quad N = N_0 \mathrm{e}^{-\lambda t}. \tag{32.4}$$

Set $N(t_{1/2}) = N_0/2$ to obtain $t_{1/2} = \ln 2/\lambda$. For several independent channels, their probabilities add over a short interval, giving total rate $\sum_i \lambda_i$. The integrated probability for channel i is $\int_0^\infty \lambda_i \mathrm{e}^{-\lambda t} \mathrm{d}t = \lambda_i/\lambda$, the branching fraction.

Worked example 32.1 (What a half-life predicts)**.** If a large sample has 800 undecayed nuclei and a half-life of one hour, the decay model predicts about 400 after one hour and 200 after two. It does not predict which particular nucleus will decay next. The expected sample count follows a smooth exponential because many independent random decay events are being combined; a small sample would show noticeable statistical fluctuations around that expectation.

32.3 Reaction energetics and cross sections

List the initial and final particles and compare their total rest energies. This determines the energy release, but momentum must also be conserved. For an endothermic reaction against a stationary target, some incident energy necessarily remains as collective motion, so the beam energy threshold is generally larger than the rest-energy deficit alone.

A reaction's energy release is

$$Q = (M_{\text{initial}} - M_{\text{final}})c^2. \tag{32.5}$$

Positive Q is exothermic. Negative Q requires sufficient incident energy, with recoil making the laboratory threshold exceed $|Q|$ in general. A beam with flux Φ incident on N_t thin-target nuclei gives event rate approximately $R = \Phi N_t \sigma$ when attenuation and multiple interactions are negligible.

Fusion of light nuclei and fission of sufficiently heavy nuclei can release binding energy because products may be more tightly bound per nucleon. Their rates depend on barriers, quantum states, and reaction channels, not simply on a favorable Q value. Stellar fusion occurs through tunneling at energies below the classical Coulomb barrier.

The cross section is an effective interaction area, not necessarily the geometric area of a nucleus. A beam flux Φ has units of particles per area per time, so one target produces rate $\Phi\sigma$. For N_t independently exposed targets, rates add to $R = \Phi N_t \sigma$. If a target is thick enough to attenuate the beam, its flux changes with depth and this simple multiplication must be replaced by a transport calculation. This connects a microscopic interaction probability to an experimentally measured event count.

32.4 Nuclear models and dense matter

Choose the model according to the feature you want to explain. Smooth binding trends suggest a liquid-drop picture; unusually stable nucleon numbers suggest shells; low-lying rotations suggest collective deformation. Different descriptions can be useful for the same nucleus because they organize different observables.

The liquid-drop model explains bulk trends; the shell model treats nucleons in an effective mean field with strong spin-orbit effects. Collective rotational and vibrational models describe other spectra. No single simple model captures all nuclear structure.

At extreme density, neutron-rich matter is governed by degeneracy, nuclear interactions, and relativistic gravity. Its equation of state relates pressure to energy density and composition. Neutron-star structure therefore links microscopic many-body physics to macroscopic gravitational observations.

For dense matter, the key output needed by a stellar model is an equation of state. Microscopic calculations determine how pressure changes with energy density and composition; gravitational balance then determines the density profile. This is a chain of models with distinct jobs. Agreement with one nuclear binding trend does not by itself establish the high-density equation of state needed inside a neutron star.

Exercises

Two nuclei have different binding energies. Explain what additional information you need to decide whether a proposed reaction releases energy.

Answers are in Appendix A.

Chapter 33

Relativistic quantum theory and particle physics

The quantum chapters allowed a fixed number of particles, while relativity allows energy to become new particles. Quantum field theory is the next synthesis: fields are fundamental, particles are excitations, and symmetries organize the interactions. Quantum mechanics and special relativity meet most naturally in quantum field theory, where particle number can change and fields supply the underlying degrees of freedom. A full development requires a substantial course of its own. Here we build the entry points carefully: relativistic dispersion, field equations, quantized modes, and the role of interactions and symmetries. Treat the more advanced results as a map for further study, with their assumptions visible.

33.1 Why quantum fields are needed

At low enough energies, a fixed number of particles can be an excellent approximation. Once enough energy is available to produce additional particles, that approximation cannot describe all possible outcomes. A field description accommodates sectors with different occupation numbers while preserving a common dynamical framework. ϕ is a scalar field, ψ a spinor field, m mass, and ∂_μ a spacetime derivative.

Relativity permits energy to create particle-antiparticle pairs. A fixed-particle-number wavefunction is therefore insufficient for general high-energy processes. Quantum field theory assigns operators to fields; particles are excitations of those fields, and interactions can change particle number.

For the remainder of this chapter use $\hbar = c = 1$ unless restored explicitly. Keep the metric signature $(-,+,+,+)$ and define $\Box = \partial_\mu \partial^\mu = -\partial_t^2 + \nabla^2$. A free scalar field obeys

$$(\Box - m^2)\phi = 0, \tag{33.1}$$

which gives $E^2 = \boldsymbol{p}^2 + m^2$. The Lagrangian density $\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2$ yields this equation with our signature.

Check the scalar equation by inserting a plane wave $\phi \propto \mathrm{e}^{-\mathrm{i}Et+\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{x}}$. With this chapter's signature,

$$\Box\phi = (E^2 - \boldsymbol{p}^2)\phi, \qquad (\Box - m^2)\phi = 0 \quad \Longrightarrow \quad E^2 = \boldsymbol{p}^2 + m^2. \tag{33.2}$$

The equation can also be obtained from the displayed Lagrangian density: $\partial\mathcal{L}/\partial(\partial_\mu\phi) = -\partial^\mu\phi$ and $\partial\mathcal{L}/\partial\phi = -m^2\phi$. The field Euler–Lagrange equation then gives $-\Box\phi + m^2\phi = 0$. This checks the signs against a physical dispersion relation before more complicated calculations are attempted.

Worked example 33.1 (Why a fixed particle count fails)**.** In a sufficiently energetic collision, some incoming kinetic energy can become the rest energy of new particles while total energy and momentum remain conserved. A theory that insists on exactly the original particles in every possible final state cannot describe that outcome. Quantum field theory assigns states to fields and allows their excitations to be created or removed. The collision must still satisfy conservation laws; "particle creation" is not energy appearing from nowhere.

33.2 Spinors, antiparticles, and quantization

A relativistic equation first order in time and space requires coefficients that cannot all be ordinary numbers. Matrices can cancel the unwanted cross terms when the operator is squared. The Dirac algebra is precisely the condition that makes this construction recover the relativistic energy relation while introducing spinor components. The metric convention fixes signs in $\Box$ and the gamma-matrix algebra; changing signature requires changing those definitions consistently.

A Dirac equation can be written

$$(\mathrm{i}\gamma^\mu\partial_\mu - m)\psi = 0, \qquad \{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}I, \tag{33.3}$$

using this explicitly chosen gamma-matrix convention with the mostly-plus metric. Squaring the operator reproduces the stated Klein–Gordon dispersion. Other texts often use the opposite metric and corresponding Clifford-algebra convention.

A free field decomposes into harmonic modes, promoted to creation and annihilation operators. Bosonic operators obey commutators and fermionic ones anticommutators. In local relativistic quantum theories satisfying the usual assumptions, the spin-statistics connection links integer spin with bosons and half-integer spin with fermions. Antiparticles have the same mass and spin as their partners and opposite additive internal charges.

Multiply the Dirac operator by its opposite-mass partner:

$$(\mathrm{i}\gamma^\mu\partial_\mu + m)(\mathrm{i}\gamma^\nu\partial_\nu - m) = -\tfrac{1}{2}\{\gamma^\mu, \gamma^\nu\}\partial_\mu\partial_\nu - m^2 = \Box - m^2. \tag{33.4}$$

Only the symmetric matrix combination survives because the derivatives commute. With the explicitly chosen anticommutator, each component satisfies the scalar mass-shell equation. Quantization then promotes the field's mode amplitudes to operators; the oscillator algebra familiar from quantum mechanics becomes the starting point for particle creation and annihilation. The interpretation of antiparticles requires that field framework, rather than treating negative-energy solutions as ordinary classical trajectories.

33.3 Interactions, diagrams, and effective theories

Begin with a free theory you can solve and add interaction terms allowed by the symmetries. Perturbation theory organizes their repeated action in powers of coupling. Diagrams are a way to label those algebraic contributions and their integrals, so their meaning should be traced back to amplitudes rather than imagined particle paths. A Feynman diagram organizes an amplitude, not a literal classical trajectory.

An interaction term couples fields. Perturbation theory expands amplitudes in coupling constants; Feynman diagrams organize integrals in that expansion. Internal lines are propagators, not directly observed little particles following classical paths. Observable rates come from amplitudes, interference, phase space, and the appropriate normalization.

Loop integrals require regularization and renormalization. Measured masses and couplings depend on how parameters are defined and, for running couplings, on the energy scale. Effective field theory organizes allowed interactions by symmetry and powers of a small ratio E/Λ, where Λ is a higher scale. It can make precise low-energy predictions without claiming validity at arbitrarily high energy.

If two alternatives contribute amplitudes $\mathcal{M}_1$ and $\mathcal{M}_2$, their probability factor is

$$|\mathcal{M}_1 + \mathcal{M}_2|^2 = |\mathcal{M}_1|^2 + |\mathcal{M}_2|^2 + 2\,\mathrm{Re}(\mathcal{M}_1\mathcal{M}_2^*). \tag{33.5}$$

The interference term means separate diagrams are generally not independent events with separately addable probabilities. In an effective theory, omitted contributions can often be ordered as powers of E/Λ. Identifying that ratio and the relevant symmetries tells you what accuracy to expect and when new degrees of freedom may need to be included.

33.4 The Standard Model's structure

Read the particle table as a classification of fields and their transformation properties. The gauge groups indicate different internal symmetries and constrain which interactions can occur. Mass, electric charge, color, and flavor are distinct labels; belonging to one generation does not mean the particles have similar masses or identical forces.

The Standard Model is a gauge theory with group $SU(3)_C \times SU(2)_L \times U(1)_Y$. Its matter fields form three generations of quarks and leptons:

Generation	Quarks	Leptons
1	up, down	electron, electron neutrino
2	charm, strange	muon, muon neutrino
3	top, bottom	tau, tau neutrino

Quarks carry color and fractional electric charges; leptons do not carry color. Gluons mediate the strong interaction, the photon electromagnetism, and $W^{\pm}, Z$ the weak interaction. The Higgs

field's nonzero vacuum value is associated with electroweak symmetry breaking and masses for the weak bosons; Yukawa couplings generate charged-fermion masses. Most of the proton's mass is associated with QCD dynamics, rather than simply the sum of its valence-quark masses.

The minimal original Standard Model has massless neutrinos. Observed flavor oscillations require neutrino mass differences and therefore an extension of that minimal formulation; the detailed mass mechanism is not fixed by oscillation data alone.

A helpful learning order is to start with the familiar electromagnetic coupling, then ask which additional charges and fields are needed for weak and strong processes. The table supplies the matter content; the gauge structure supplies interactions; the Higgs sector and its couplings supply part of the mass mechanism. The classification alone does not calculate all particle masses. Many parameters must be measured, and composite masses also require interaction energy, just as binding changed the mass of a nucleus.

33.5 Strong and weak interactions

The strength and character of an interaction can depend on the scale at which it is probed. Strong interactions organize quarks into hadrons, while weak interactions connect different flavors. Neutrino oscillations give a particularly transparent example of flavor change arising from coherent phase evolution rather than a sequence of classical conversions.

Quantum chromodynamics is the $SU(3)$ gauge theory of color. Its coupling weakens at short distances (asymptotic freedom) and becomes strong at large distances. Isolated colored quarks and gluons are not observed as asymptotic free particles; they form color-neutral hadrons. Baryons include three-valence-quark states such as protons and neutrons; mesons include quark-antiquark states, with sea constituents and gluons also contributing.

Weak interactions can change flavor and violate parity. Quark flavor mixing is described by the CKM matrix; lepton mixing by the PMNS matrix. Charge conjugation and parity together are also violated in some processes. In the two-flavor vacuum approximation, restoring $\hbar, c$,

$$P(\nu_\alpha \to \nu_\beta) = \sin^2(2\theta)\sin^2\left(\frac{\Delta m^2 c^3 L}{4\hbar E}\right), \quad \alpha \neq \beta. \tag{33.6}$$

This assumes relativistic coherent propagation. Three-flavor effects and matter interactions modify the expression.

For two flavors, use mass eigenstates $|\nu_1\rangle, |\nu_2\rangle$ and write

$$|\nu_\alpha\rangle = \cos\theta|\nu_1\rangle + \sin\theta|\nu_2\rangle, \tag{33.7}$$

$$|\nu_\beta\rangle = -\sin\theta|\nu_1\rangle + \cos\theta|\nu_2\rangle. \tag{33.8}$$

Their propagation phases differ by $\Delta\varphi \simeq \Delta m^2 c^3 L/(2\hbar E)$ in the relativistic coherent limit. Projection onto the second flavor gives an amplitude proportional to $\sin\theta\cos\theta(\mathrm{e}^{-\mathrm{i}\varphi_2} - \mathrm{e}^{-\mathrm{i}\varphi_1})$. Squaring yields

$$P_{\alpha\to\beta} = \sin^2(2\theta)\sin^2(\Delta\varphi/2). \tag{33.9}$$

This recovers the stated formula. Both mixing and a nonzero relative phase are necessary; equal masses would eliminate this oscillation phase difference.

33.6 Symmetries, conservation, and open questions

Before calculating a proposed reaction, apply the conservation laws relevant to the interaction and regime. They can rule out a channel before any difficult integral is done. After that, distinguish what the established model predicts from what it leaves as measured input or an unresolved question.

Energy-momentum and electric charge conservation strongly constrain reactions. Additional quantum numbers may be exact, approximate, or violated by particular interactions; a conservation rule from one regime should not be assumed universally.

Established quantum field theories do not by themselves provide a complete quantum theory of gravity, identify the microscopic nature of dark matter, or explain every cosmological initial condition. These are open problems. Proposed answers should be distinguished from the tested framework used to calculate spectra and scattering.

Conservation is necessary but not sufficient for an appreciable rate. A channel can conserve energy, momentum, and charge yet be suppressed by a small coupling, a symmetry of the matrix element, or limited final-state phase space. This repeats a lesson from atomic selection rules and nuclear barriers: first ask whether the process is allowed, then ask what amplitude and accessible states determine how often it occurs.

Exercises

Why can a relativistic collision require a description in which particle number changes? State what a Feynman diagram represents in a calculation.

Answers are in Appendix A.

Part XIII

Astrophysics, Cosmology, and Synthesis

Chapter 34

Stars, compact objects, and the expanding universe

The final physics unit brings earlier scales together. Gravity supplies structure, quantum and nuclear physics supply matter, thermodynamics supplies pressure and entropy, and relativity supplies the geometry of compact objects and the expanding universe. Astrophysics is therefore a synthesis, not a separate beginning. Astrophysics combines the previous modules on a grand scale. Stellar pressure is supplied by matter and radiation; gravity sets the required support; nuclear reactions and transport determine energy flow. Cosmology replaces the internal profile of one object with the evolution of a nearly homogeneous universe. In both settings, state the symmetry assumptions before reducing field equations to a manageable model.

34.1 Stellar structure and energy transport

Build a star out of thin spherical shells. Each shell adds mass, contributes or transports energy, and must be supported against the gravitational pull of the enclosed mass. This construction explains why several coupled differential equations are needed: no single pressure balance determines the temperature, luminosity, and density by itself. In stellar equations, M_r is mass enclosed inside radius r and L_r luminosity crossing that shell. A subscript r means "at radius r," not a new physical kind of mass or luminosity.

A spherical nonrotating star in Newtonian hydrostatic equilibrium satisfies

$$\frac{\mathrm{d}M_r}{\mathrm{d}r} = 4\pi r^2 \rho, \qquad \frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{GM_r\rho}{r^2}, \qquad \frac{\mathrm{d}L_r}{\mathrm{d}r} = 4\pi r^2 \rho \epsilon, \tag{34.1}$$

where M_r is enclosed mass, L_r is outward luminosity, and ϵ is net local energy generation per mass per time. An equation of state and an energy-transport law complete the model. In an optically thick radiative region,

$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3\kappa_R \rho L_r}{16\pi a c\, r^2 T^3}, \tag{34.2}$$

where κ_R is the Rosseland mean opacity and a the radiation constant. Convection becomes important when the required temperature gradient is sufficiently steep; convection modeling is an additional physical problem.

For a self-gravitating stationary system with negligible surface terms, the virial theorem gives $2\langle K\rangle + \langle U\rangle = 0$ for an inverse-distance potential. A star can heat as it loses total energy through gravitational contraction: its thermal kinetic energy increases while gravitational energy becomes more negative.

A shell of thickness $\mathrm{d}r$ has mass $\mathrm{d}M_r = 4\pi r^2 \rho \mathrm{d}r$, giving the first structure equation. Hydrostatic balance is the fluid relation $\mathrm{d}p/\mathrm{d}r = -\rho g$ with $g = GM_r/r^2$. Net energy production in the shell is $\epsilon \mathrm{d}M_r$, giving $\mathrm{d}L_r/\mathrm{d}r = 4\pi r^2 \rho \epsilon$.

For optically thick radiative diffusion, energy density is $u = aT^4$ and flux is approximately $F = -[c/(3\kappa_R \rho)]\mathrm{d}u/\mathrm{d}r$. Use $\mathrm{d}u/\mathrm{d}r = 4aT^3 \mathrm{d}T/\mathrm{d}r$ and $L_r = 4\pi r^2 F$ to obtain the temperature gradient above. Central regularity, surface conditions, an equation of state, and material rates complete the boundary-value problem.

34.2 Degeneracy pressure and compact remnants

Fermions cannot all collapse into the same low-momentum state. Increasing their density fills states to larger momenta, producing pressure even at zero temperature. The relation between momentum and energy changes as particles become relativistic, and that change alters how strongly pressure grows during compression.

Electron degeneracy supports white dwarfs. For nonrelativistic degenerate electrons, $p \propto n_e^{5/3}$; in the ultrarelativistic limit, $p \propto n_e^{4/3}$. The change in scaling leads to a limiting white-dwarf mass of order a solar mass, conventionally about $1.4 M_\odot$ for common idealized composition assumptions.

Neutron stars require relativistic hydrostatic balance. Writing ρ for total energy density divided by c^2, the Tolman–Oppenheimer–Volkoff equation is

$$\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{G(\rho + p/c^2)(M_r + 4\pi r^3 p/c^2)}{r^2[1 - 2GM_r/(rc^2)]}, \qquad \frac{\mathrm{d}M_r}{\mathrm{d}r} = 4\pi r^2 \rho. \tag{34.3}$$

Pressure contributes to gravitation, and the equation of state controls the possible masses and radii. Sufficiently compact collapse can form a black hole. Rotation, magnetic fields, and accretion complicate these idealized pictures.

State counting gives $p_F \propto n^{1/3}$. For nonrelativistic particles, a characteristic energy is $p_F^2/(2m)$, so energy density and degeneracy pressure scale as $n^{5/3}$. For ultrarelativistic particles, energy is proportional to $p_F c$, giving $n^{4/3}$. To see why the latter produces a mass scale, estimate required gravitational pressure as GM^2/R^4 and relativistic electron pressure as proportional to $(M/R^3)^{4/3}$ at fixed composition. Both scale as R^{-4}, so reducing radius no longer changes their relative strength for a fixed mass. Detailed stellar integration supplies the coefficient and composition dependence of the limiting mass.

34.3 Homogeneity, isotropy, and expansion

Homogeneity means no preferred position in the idealized spatial geometry; isotropy means no preferred direction. These assumptions leave a scale factor to describe how separations between comoving locations change in time. A comoving coordinate stays fixed while its physical separation from another comoving location can grow. ρ is mass-equivalent energy density, p pressure, $a(t)$ the cosmological scale factor, and $H = \dot{a}/a$ the expansion rate.

On sufficiently large scales, the cosmological principle models space as statistically homogeneous and isotropic. The FLRW metric is

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2(t)\left[\frac{\mathrm{d}r^2}{1-kr^2} + r^2\mathrm{d}\Omega^2\right]. \tag{34.4}$$

Here take r dimensionless, a to have dimensions of length, and $k = 0, \pm 1$. The expansion rate is $H = \dot{a}/a$. Comoving wavelength grows with a, so

$$1 + z = \frac{a(t_0)}{a(t_{\mathrm{emission}})}. \tag{34.5}$$

Nearby galaxies approximately satisfy $v \simeq H_0 d$ when peculiar motion is negligible. At large distances, cosmological recession is not simply a special-relativistic velocity in one global inertial frame.

In a spatially flat slice, two nearby comoving points separated by fixed Δr have distance $d = a(t)\Delta r$. Differentiation gives $\dot{d} = (\dot{a}/a)d = Hd$. This is the geometric origin of the local Hubble relation. Photon wavelength stretches with the same scale factor, so

$$\frac{\lambda_0}{\lambda_e} = \frac{a_0}{a_e} = 1 + z, \qquad \frac{\nu_0}{\nu_e} = \frac{a_e}{a_0}. \tag{34.6}$$

For gravitationally bound systems, fixed comoving separation is generally not the appropriate description; local dynamics must also be considered.

34.4 Friedmann equations and cosmic contents

The expansion rate depends on both geometry and the matter content. To solve for $a(t)$, first specify an equation of state, then determine how its density changes with expansion. The continuity equation supplies that density law, and the Friedmann equation turns it into a time evolution.

For a perfect fluid with mass-equivalent density ρ,

$$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \tag{34.7}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}, \tag{34.8}$$

$$\dot{\rho} + 3H\left(\rho + \frac{p}{c^2}\right) = 0. \tag{34.9}$$

Here ρ, p exclude the explicitly displayed cosmological constant. For a separately conserved component with constant $p = w\rho c^2$,

$$\rho \propto a^{-3(1+w)}. \tag{34.10}$$

Nonrelativistic matter has $w \simeq 0$ and dilutes as a^{-3}; radiation has $w = 1/3$ and scales as a^{-4} because its photons also redshift. Vacuum energy has $w = -1$ and constant density.

In flat universes dominated by a single component and with no separate Λ, $a \propto t^{2/3}$ for matter and $a \propto t^{1/2}$ for radiation. Positive constant vacuum energy instead produces exponential expansion when it dominates. The critical density is $\rho_c = 3H^2/(8\pi G)$. Numerical estimates of cosmological parameters depend on observations and models; no particular current fit is required for these relations.

Worked example 34.1 (Temperature of expanding radiation)**.** For equilibrium freely redshifting radiation, $\rho_{\mathrm{rad}} \propto a^{-4}$ and $u = a_{\mathrm{rad}}T^4$ imply $T \propto a^{-1}$. Thus radiation observed at redshift z had temperature $T(z) = T_0(1 + z)$ if no additional entropy was transferred into it during the interval being modeled.

For a comoving volume $V \propto a^3$, local energy balance has the form $\mathrm{d}(\rho c^2 V) = -p\mathrm{d}V$. Expanding the differential gives

$$\mathrm{d}\rho = -\left(\rho + \frac{p}{c^2}\right)\frac{\mathrm{d}V}{V} = -3\left(\rho + \frac{p}{c^2}\right)\frac{\mathrm{d}a}{a}. \tag{34.11}$$

For constant w, integrate to obtain $\rho \propto a^{-3(1+w)}$. In a flat expanding universe containing only this component, $\dot{a}/a \propto a^{-3(1+w)/2}$. For $w > -1$, integration from a vanishing scale factor gives

$$a^{3(1+w)/2} \propto t, \qquad a(t) \propto t^{2/[3(1+w)]}. \tag{34.12}$$

Matter and radiation give the quoted powers. The case $w = -1$ must be treated separately: constant density gives constant H and therefore $a \propto \mathrm{e}^{Ht}$. The pressure contribution explains why radiation dilutes faster than conserved particle number alone.

34.5 Thermal history and structure formation

Compare two clocks: the time $1/\Gamma$ for a microscopic reaction and the expansion time $1/H$. When reactions are fast, populations can adjust as the universe cools. When expansion is faster, they may retain a nonequilibrium abundance. Structure growth asks a related competition question: can gravitational attraction amplify a density excess despite the expanding background?

The hot expanding-universe model organizes early particle reactions, light-element nucleosynthesis, formation of neutral atoms, and the release of radiation now observed as the cosmic microwave background. Recombination and photon decoupling are related but distinct processes: atomic populations evolve while the photon scattering rate becomes too small to maintain tight coupling.

A reaction remains near equilibrium when its rate Γ greatly exceeds H. Freeze-out occurs roughly when $\Gamma \lesssim H$, although accurate abundances require kinetic equations. For cold nonrelativistic matter on subhorizon scales in the linear pressureless approximation, density contrast obeys

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G\rho_m\delta = 0. \tag{34.13}$$

Expansion competes with gravitational growth. Pressure, radiation, free streaming, and nonlinear collapse require extensions.

For a flat matter-only background, $H = 2/(3t)$ and $\rho_m = 1/(6\pi G t^2)$. Try a power law $\delta = t^p$ in the linear growth equation:

$$p(p-1) + \frac{4}{3}p - \frac{2}{3} = 0 \quad \Longrightarrow \quad p = \frac{2}{3} \text{ or } -1. \tag{34.14}$$

The growing mode has $\delta \propto t^{2/3} \propto a$, and the other mode decays. This provides a concrete prediction within the pressureless linear model. Once $|\delta|$ becomes order one, neglecting nonlinear terms is no longer justified.

34.6 Dark sectors and inflation

Keep the observational role of a component separate from a proposed microscopic explanation. Extra gravitational clustering and accelerated expansion point to different features of cosmological models. A scalar field is one way to produce negative pressure, but deriving its possible behavior does not establish that a particular field drove the early universe.

Dark matter denotes gravitating matter inferred from multiple gravitational effects whose dominant microscopic identity is not established. Dark energy is a description of the component associated with accelerated expansion; a cosmological constant is one model. These terms summarize different phenomena and should not be treated as interchangeable.

Inflation proposes an early accelerated-expansion phase that can address horizon and flatness problems and generate primordial perturbations. Its specific field content and dynamics are model dependent. A homogeneous canonical scalar field in units $c = \hbar = 1$ has $\rho = \dot{\phi}^2/2 + V(\phi)$ and $p = \dot{\phi}^2/2 - V(\phi)$; potential domination yields negative pressure. This mechanism is a theoretical framework, not a complete experimentally selected account of the earliest universe.

In the scalar-field units specified above,

$$\rho + 3p = 2\dot{\phi}^2 - 2V(\phi). \tag{34.15}$$

The acceleration equation with no separate cosmological constant therefore permits $\ddot{a} > 0$ when $V > \dot{\phi}^2$. Potential domination gives $p \simeq -\rho$, analogous to a nearly constant vacuum component. Establishing a complete inflationary model additionally requires the field's dynamics, an end to the accelerated era, and predictions for perturbations. The inequality identifies the mechanism, not all of those ingredients.

Exercises

What balances gravity inside an ordinary star? Explain how a measurement of redshift can give evidence about cosmic expansion.

Answers are in Appendix A.

Chapter 35

Computational physics and model validation

A calculation is not finished when an equation has been written, and it is not validated merely because an animation looks plausible. This chapter gathers the book's recurring habits—units, limits, conservation, approximation, and comparison—and turns them into a computational workflow. A numerical solution is another approximation, layered on top of the physical model. It can make an unsolvable analytic problem accessible, but smooth output alone does not establish accuracy. The aim of this chapter is to make computational choices explainable: derive the discrete rule, estimate its error, and test it against something whose answer you understand.

35.1 Discretization and numerical error

A computer represents a function at finitely many points. To approximate a derivative from those values, expand nearby samples about the point of interest and combine them so unwanted Taylor terms cancel. The first surviving neglected term tells you how error should change when the grid is refined. h is a spatial or temporal step, Δx and Δt grid spacings, $Q(h)$ a computed quantity at resolution h, and Q_* its limiting value. A numerical index such as j labels a grid cell and n a time level.

Numerical physics approximates a well-posed mathematical model, whose own physical approximations remain separate from numerical error. A forward difference has error $O(h)$, while

$$f'(x) \simeq \frac{f(x+h) - f(x-h)}{2h} \tag{35.1}$$

has truncation error $O(h^2)$ for a sufficiently smooth function. Taking h extremely small can amplify floating-point cancellation. Numerical integration likewise balances resolution, smoothness, and cost.

Convergence means the computed result approaches a stable limit as resolution improves. Agreement at one step size is insufficient evidence. Compare conserved quantities, analytic limiting cases, and independent formulations whenever feasible.

Subtract the expansions of $f(x+h)$ and $f(x-h)$:

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{h^3}{3} f^{(3)}(x) + O(h^5). \tag{35.2}$$

Division by $2h$ gives the centered derivative with leading error $h^2 f^{(3)}/6$. In the truncation-dominated regime, halving h should reduce that error by about four. For a numerical quantity $Q(h) = Q_* + Ch^2 + \cdots$, the extrapolation

$$Q_* \simeq \frac{4Q(h/2) - Q(h)}{3} \tag{35.3}$$

cancels the leading error. Check that successive refinements actually show the expected trend before relying on the extrapolation; rounding error or unresolved features can break it.

Worked example 35.1 (A timestep that changes the answer)**.** For $\dot{x} = -x$ with $x(0) = 1$, forward Euler gives $x_{n+1} = x_n(1 - \Delta t)$. One step of size 1 predicts $x(1) = 0$. Two steps of size 0.5 predict $x(1) = 0.25$. Four steps of size 0.25 predict about 0.316, moving toward the exact $\mathrm{e}^{-1} \approx 0.368$. A plotted curve can look smooth at every resolution, so comparing resolutions is part of checking the calculation.

35.2 Time integration and Hamiltonian systems

An integrator advances initial data by a finite time step. Its local update may look accurate while gradually distorting the geometry of a long trajectory. Oscillators are excellent test cases because their exact phase-space motion is bounded, making artificial growth or damping easy to detect.

Forward Euler uses $y_{n+1} = y_n + hf(t_n, y_n)$ and is first-order globally. Fourth-order Runge–Kutta combines several slope estimates for smooth ordinary differential equations. Stiff systems may require implicit or specially designed solvers rather than impractically tiny explicit steps.

For $\ddot{\boldsymbol{r}} = \boldsymbol{a}(\boldsymbol{r})$, velocity Verlet is

$$\boldsymbol{r}_{n+1} = \boldsymbol{r}_n + h\boldsymbol{v}_n + \tfrac{1}{2}h^2\boldsymbol{a}_n, \tag{35.4}$$

$$\boldsymbol{v}_{n+1} = \boldsymbol{v}_n + \tfrac{h}{2}(\boldsymbol{a}_n + \boldsymbol{a}_{n+1}). \tag{35.5}$$

For separable conservative Hamiltonians it is a second-order symplectic scheme, often maintaining bounded long-term energy error for suitable steps. Symplectic does not mean exact energy conservation at every step.

For $\dot{x} = v$, $\dot{v} = -\omega^2 x$, forward Euler gives $x' = x + hv$ and $v' = v - h\omega^2 x$. Its next-step energy is

$$E' = \tfrac{1}{2}m(v - h\omega^2 x)^2 + \tfrac{1}{2}m\omega^2(x + hv)^2 = (1 + h^2\omega^2)E. \tag{35.6}$$

Energy grows on every nonzero step even though the exact system conserves it. Velocity Verlet can instead be organized as a half-step velocity update, full-step position update, and final half-step velocity update:

$$\boldsymbol{v}_{n+1/2} = \boldsymbol{v}_n + \tfrac{h}{2}\boldsymbol{a}_n, \quad \boldsymbol{r}_{n+1} = \boldsymbol{r}_n + h\boldsymbol{v}_{n+1/2}, \quad \boldsymbol{v}_{n+1} = \boldsymbol{v}_{n+1/2} + \tfrac{h}{2}\boldsymbol{a}_{n+1}. \tag{35.7}$$

Combining these expressions recovers the displayed Verlet formulas. Its structure improves conservative long-time behavior for suitable steps, while convergence and stability still need checking.

35.3 PDE stability and eigenvalue problems

A grid error can be analyzed as a sum of Fourier modes, just like a physical disturbance. Insert one mode into the update rule and calculate how much its amplitude changes in one step. Stability requires that numerical evolution not amplify modes that the underlying diffusive physics should damp. Stability is a property of an update rule; accuracy is how closely its result approaches the continuum solution.

For explicit centered-space diffusion in one dimension,

$$u_j^{n+1} = u_j^n + r(u_{j+1}^n - 2u_j^n + u_{j-1}^n), \qquad r = D\Delta t/(\Delta x)^2, \tag{35.8}$$

linear stability requires $r \leq 1/2$. For a common explicit centered wave scheme, the one-dimensional Courant condition is $v\Delta t/\Delta x \leq 1$. Conditions depend on the actual algorithm, geometry, and dimension; these bounds are not universal for every discretization.

Discretizing a Schrodinger or normal-mode operator produces a matrix eigenproblem. Boundary conditions, grid spacing, and domain size affect the spectrum. Verify normalization, orthogonality, and convergence of the lowest states before trusting highly excited numerical modes.

Write $u_j^n = g^n \mathrm{e}^{\mathrm{i}kj\Delta x}$ and substitute into the diffusion update. Cancel the common factor to obtain

$$g = 1 + r(\mathrm{e}^{\mathrm{i}k\Delta x} - 2 + \mathrm{e}^{-\mathrm{i}k\Delta x}) \tag{35.9}$$

$$= 1 - 4r\sin^2(k\Delta x/2). \tag{35.10}$$

For $r \geq 0$, the largest amplification is one and the smallest is $1 - 4r$. Requiring $g \geq -1$ gives $r \leq 1/2$. Halving Δx therefore requires reducing the allowed Δt by four for this method. Spatial refinement without the accompanying time-step change can make a previously stable simulation fail.

35.4 Monte Carlo and statistical sampling

Sampling replaces a difficult sum or integral by an average over representative configurations. Independent random samples make the error analysis simple; a Markov chain trades independence for the ability to reach complicated distributions. Its acceptance rule must preserve the target distribution, and its exploration must be broad enough to represent that distribution in practice.

Monte Carlo integration estimates an expectation using samples, with typical independent-sample error scaling as $N^{-1/2}$ for finite variance. In Markov-chain Monte Carlo, samples are correlated; the effective sample size is smaller. Metropolis sampling accepts a proposed configuration change with probability

$$P_{\text{accept}} = \min(1, \mathrm{e}^{-\beta\Delta E}) \tag{35.11}$$

for a symmetric proposal in a canonical ensemble. Asymmetric proposals require an additional proposal-probability ratio. Check equilibration, autocorrelation, and sampling across distinct metastable regions.

For canonical weights $\pi_i \propto \mathrm{e}^{-\beta E_i}$ and symmetric proposal probabilities, detailed balance requires

$$\pi_i P_{i\to j} = \pi_j P_{j\to i}, \qquad \frac{P_{i\to j}}{P_{j\to i}} = \mathrm{e}^{-\beta(E_j - E_i)}. \tag{35.12}$$

Metropolis acceptance satisfies the required ratio by accepting energy-lowering moves and accepting energy-raising moves with the Boltzmann factor. This ensures the desired stationary weights when combined with the proposal, but does not guarantee rapid exploration. A chain trapped in one region can look quiet while badly underestimating uncertainty.

35.5 Uncertainty and data

Suppose you repeat a timing measurement and obtain slightly different answers. Averaging helps with independent fluctuations, but it will not repair a clock that runs consistently slow. We therefore separate random variation from systematic error before using statistics. When measured quantities enter a formula, the next question is how sensitively the answer depends on each input. In uncertainty formulas, σ is a standard deviation and C_{ij} is a covariance; covariance can be positive or negative because two measured quantities may vary together or in opposition.

For N independent observations with finite variance, the sample mean is $\bar{x} = N^{-1}\sum_i x_i$ and its estimated standard error is $s/\sqrt{N}$, where

$$s^2 = \frac{1}{N-1}\sum_i (x_i - \bar{x})^2. \tag{35.13}$$

This does not remove calibration errors or other systematic biases. For a differentiable function $f(\boldsymbol{x})$ with small errors and covariance matrix C,

$$\sigma_f^2 \simeq \sum_{ij} \frac{\partial f}{\partial x_i} C_{ij} \frac{\partial f}{\partial x_j}. \tag{35.14}$$

For independent inputs, the off-diagonal terms vanish. A least-squares fit with independent Gaussian errors minimizes $\chi^2 = \sum_i [y_i - f(x_i;\boldsymbol{a})]^2/\sigma_i^2$. Good agreement is judged relative to uncertainties and degrees of freedom, not simply by how smooth a curve looks.

Common pitfall. Measurement uncertainty describes limited knowledge of a result; quantum uncertainty can remain even for an ideally prepared state and perfect apparatus. Neither means that physical predictions are arbitrary.

Write each input as its estimated value plus a small fluctuation, $x_i = \bar{x}_i + \delta x_i$. A first-order Taylor expansion gives

$$\delta f \simeq \sum_i f_i \delta x_i, \qquad f_i = \left.\frac{\partial f}{\partial x_i}\right|_{\bar{\boldsymbol{x}}}. \tag{35.15}$$

Square this expression and average. With zero-mean fluctuations, $\langle \delta x_i \delta x_j \rangle = C_{ij}$, which produces the covariance formula above. Its cross terms describe inputs that move together. For example, if $v = d/t$ and the errors in d and t are independent,

$$\left(\frac{\sigma_v}{v}\right)^2 \simeq \left(\frac{\sigma_d}{d}\right)^2 + \left(\frac{\sigma_t}{t}\right)^2. \tag{35.16}$$

The minus sign in $\partial v/\partial t = -d/t^2$ disappears upon squaring. This is why independent fractional errors add in quadrature even when one quantity is in a denominator.

Exercises

A length of 2.0 m is measured with an uncertainty of 0.1 m. What is the fractional uncertainty? A timing device always reads 0.2 s too high. Explain why repeating the measurement does not remove that error.

Answers are in Appendix A.

35.6 A practical verification checklist

Treat verification as part of solving the problem, not a final cosmetic step. For example, a numerical orbit should approach a known circular solution in the appropriate initial state, show predictable refinement behavior, and conserve the relevant quantities to the expected accuracy. Use the following sequence to make those questions routine.

1. Check the continuum equations, units, parameter regime, and boundary conditions.
2. Reproduce an analytic special case or a trusted limiting solution.
3. Refine time step, spatial grid, domain size, or sample count separately.
4. Monitor relevant invariants and calculate uncertainty rather than relying only on plots.
5. Distinguish numerical artifacts from model limitations and measurement disagreement.

When a check fails, change one thing at a time. If halving the time step changes the result but enlarging the spatial domain does not, the evidence points toward time discretization rather than the outer boundary. If the numerical result converges but still disagrees with experiment, revisit model assumptions and parameter uncertainty. Separating these questions prevents you from spending computation to refine the wrong physical model.

Exercises

A calculation gives $Q(0.1) = 1.42$ and $Q(0.05) = 1.36$ when only the timestep is changed. What does the difference suggest, and what further value would you calculate to test convergence? Name one conserved quantity or limiting case you could check.

Answers are in Appendix A.

Chapter 36

Connections across the subject

We can now look backward without flattening the differences between subjects. The same mathematical form can describe a spring, a circuit, a field mode, or a quantum oscillator, but the symbols acquire different physical meanings. These connections are the book's invitation to continue learning beyond its last page. By now the same mathematical patterns have appeared in very different settings. Recognizing them lets you reuse an argument without confusing the physical meanings of its symbols. We close the main text by tracing four recurring ideas: local balance, oscillation, symmetry, and scale. Each is a guide to choosing where a new calculation should begin.

36.1 Conservation from local balance

Imagine drawing a boundary around whatever quantity you want to track: mass, charge, energy, or probability. Its amount inside can change because it crosses the boundary or because a local source creates or removes it. Write that statement before selecting a coordinate system or a constitutive law. When the same symbol appears in different chapters, its meaning must be re-established from context. The recurring forms are the important objects: a local balance, an oscillator, a symmetry, or a small parameter. Always translate the symbols back into the physical quantities before claiming that two equations describe the same physics.

Many laws share the structure

$$\partial_t \rho + \boldsymbol{\nabla} \cdot \boldsymbol{j} = s. \tag{36.1}$$

Here ρ is a stored density, $\boldsymbol{j}$ its flux, and s a source. Integrating over a fixed volume gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_V \rho \mathrm{d}V = - \oint_{\partial V} \boldsymbol{j} \cdot \mathrm{d}\boldsymbol{S} + \int_V s \mathrm{d}V. \tag{36.2}$$

Mass continuity, charge conservation, heat flow, and quantum probability conservation are instances. Their different flux laws create different dynamics.

For an insulated boundary with zero normal flux and no sources, the integrated amount remains constant. If the flux is instead $\boldsymbol{j} = -D\boldsymbol{\nabla}\rho$, the same conservation statement becomes diffusion. If the flux is a velocity times density, it becomes transport. The shared balance law does not make these systems physically identical; the flux relation determines how each one evolves.

36.2 The oscillator as a recurring model

Whenever you find a stable equilibrium, expand the energy around it. The vanishing linear term and positive quadratic term are the invitation to use oscillator mathematics. In several coordinates, diagonalize the quadratic form to discover the independent normal modes.

A spring, an LC circuit, a molecular vibration, an electromagnetic cavity mode, and a free quantum-field mode all reduce to oscillator mathematics in an appropriate regime. Frequencies come from restoring forces and inertia or their generalized analogues. Damping represents coupling to other degrees of freedom; quantization turns normal modes into discrete excitations.

A useful translation table is $m \leftrightarrow L$, $k \leftrightarrow 1/C$, and $x \leftrightarrow Q$ for a spring and an LC circuit. Their energies become $\frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$ and $\frac{1}{2}L\dot{Q}^2 + Q^2/(2C)$. The variables have different units, but the equations have the same structure. Quantizing a normal mode then adds the common spectrum $\hbar\omega(n + 1/2)$. This is how one solved model reaches from mechanics to circuits, vibrations, and fields.

Worked example 36.1 (A spring and an electric circuit)**.** A spring cart obeys $m\ddot{x} + kx = 0$. In an ideal inductor–capacitor circuit, charge q on the capacitor obeys $L\ddot{q} + q/C = 0$. Displacement x and charge q are different physical quantities, but both equations describe exchange between two energy stores. Their angular frequencies are $\sqrt{k/m}$ and $1/\sqrt{LC}$. The shared mathematics helps us solve the second system after learning the first, while the units keep their meanings distinct.

36.3 Symmetry, gauge freedom, and observables

Ask what a proposed transformation changes physically. Rotating an apparatus can relate distinct physical configurations through a symmetry of the laws. Changing gauge can alter the mathematical representation of the very same fields. Keeping these roles separate helps you identify which quantities should be compared with measurements.

A physical symmetry relates states while preserving the laws. Gauge freedom is a redundancy of description; gauge-related potentials describe the same physical situation when matter fields are transformed consistently. Continuous symmetries generate conservation laws, while symmetry breaking and boundary conditions determine which states occur.

Only observable predictions can be compared directly with experiments. Coordinate components, wavefunction phases, and gauge potentials may be essential calculation tools without being individually observable in every form.

An additive constant in a Newtonian potential changes the displayed energy value but not the force $-\boldsymbol{\nabla}V$ or energy differences. A global wavefunction phase changes amplitudes but not measurement probabilities. These examples teach a practical habit: before assigning physical meaning to a symbol, examine which changes of description leave all predictions invariant. More elaborate gauge theories extend this habit with corresponding transformations of matter fields.

36.4 Scale and effective descriptions

The first decision in a new problem is often the description itself. Estimate the relevant lengths, times, energies, and speeds before choosing equations. A small ratio can justify removing a detail; a ratio approaching one warns that the omitted physics may become important.

The same matter can be modeled as particles, a continuous fluid, a thermodynamic system, or a quantum many-body state. The useful description depends on resolution and the question asked. A hierarchy of small parameters can organize the choice:

Parameter or comparison	Physical significance
v/c	Relativistic kinematic corrections
$GM/(rc^2)$	Strength of relativistic gravitational effects
$\hbar/S_{\text{characteristic}}$	Semiclassical action scale; not a complete criterion for every state
$n\lambda_T^3$	Quantum degeneracy of a dilute particle gas
Mean free path divided by system size	Validity of a local continuum description
Wavelength divided by apparatus size	Wave optics versus a possible ray approximation
Interaction energy divided by a level gap	Suitability of nondegenerate perturbation theory

No one approximation wins everywhere. A good solution identifies the relevant regime and estimates its own neglected effects.

For example, a gas can be dilute enough for classical particle statistics while still requiring a fluid description over distances much larger than its mean free path. A slowly moving object can need quantum mechanics, while a rapidly moving dilute beam can be treated with relativistic particle kinematics. These comparisons answer different questions, so no single small parameter decides every approximation. State what you want to predict, identify the controlling ratios, and let those estimates guide the calculation.

Exercises

Choose two systems from different parts of the book that obey an oscillator equation. Identify the changing quantity and restoring effect in each.

Answers are in Appendix A.

Appendix A

Answers to chapter exercises

The answers follow the order of the chapter exercises. A sketch answer describes the features that must appear; equivalent correct drawings are acceptable.

1. **Functions, trigonometry, and calculus.** $v(t) = 6t - 2$. Its integral is $3T^2 - 2T$, equal to $x(T) - x(0)$. Since sine accepts an angle, $[\omega t] = 1$ and $[\omega] = T^{-1}$.

2. **Complex numbers, vectors, and linear algebra.** $|3 - 4\mathrm{i}| = 5$ and its angle is $-\arctan(4/3)$, so $z = 5\mathrm{e}^{-\mathrm{i}\arctan(4/3)}$. A quarter-turn sends $(1, 0)$ to $(0, 1)$. Direct multiplication gives $A(1, 1) = 3(1, 1)$ and $A(1, -1) = 1(1, -1)$.

3. **Fields and vector calculus.** $\boldsymbol{\nabla} f = (2x, 2y, 2z)$ and $\nabla^2 f = 6$. For $\boldsymbol{A} = (-y, x, 0)$, $\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0$. The field is nonzero and circulates; zero divergence only says it has no local net source.

4. **Physical models.** The average speed is $1.2\,\mathrm{m/s}$. Stops or changes of pace make the instantaneous speed differ. Steady motion is a straight distance–time line; speeding up is a curve whose slope increases.

5. **Change and accumulation.** The average velocity is $(14 - 2)/4 = 3\,\mathrm{m/s}$. A suitable curve begins with slope below 3 and ends above it while keeping the same endpoints. Constant $-2\,\mathrm{m/s}$ for $3\,\mathrm{s}$ gives displacement $-6\,\mathrm{m}$ and distance $6\,\mathrm{m}$.

6. **Vectors and motion.** After $2\,\mathrm{s}$, $\boldsymbol{v} = (3, -20)\,\mathrm{m/s}$. Velocity points right and down; acceleration points straight down. The horizontal component stays constant because $a_x = 0$.

7. **First forces.** $a = 6/2 = 3\,\mathrm{m/s^2}$ in the force direction. The table force is $5\,\mathrm{N}$ up and weight is $5\,\mathrm{N}$ down. If $x < 0$, $F = -kx > 0$, so the spring force points right toward equilibrium.

8. **Newtonian conservation.** $v(1) = 10 - 10 = 0$ and $\Delta z = 10(1) - 10(1)^2/2 = 5\,\mathrm{m}$. The impulse gives $\Delta v = -6/2 = -3\,\mathrm{m/s}$. With drag, acceleration changes with velocity, so a constant-acceleration formula does not apply.

9. **Gravitation and rigid bodies.** Increasing r reduces GMm/r^2 and raises $V = -GMm/r$ toward zero. Conserved angular momentum removes the angular rate from the radial problem; energy then controls the remaining motion.

10. **Oscillations.** In phase means equal displacements; out of phase means opposite displacements. The coupling spring is unchanged in the first mode and stretched in the second, producing different effective restoring strengths and frequencies.

11. **Lagrangian mechanics.** The coordinate is θ. $T = m\ell^2\dot{\theta}^2/2$ and $V = mg\ell(1-\cos\theta)$. Fixed endpoints remove the integration-by-parts boundary term so the interior variation yields the Euler–Lagrange equation.

12. **Fluids and transport.** Inflow and outflow are $\rho\boldsymbol{v}\cdot\mathrm{d}\boldsymbol{S}$ with outward signs. Mass conservation needs a momentum equation and constitutive information such as pressure, viscosity, and an equation of state, plus initial and boundary data.

13. **Waves.** $f_1 = v/(2L)$ is halved when L doubles. The fundamental has nodes only at the two fixed ends and one antinode at the center.

14. **Optics.** Entering a larger refractive index, the ray bends toward the normal. Wave optics is required when phase, interference, diffraction, polarization, or features comparable with wavelength matter.

15. **Thermodynamics.** With work by the gas positive, $\Delta U = Q - W_{\text{by}}$. Heating gives $Q > 0$; expansion usually gives $W_{\text{by}} > 0$. Their difference fixes the internal-energy change.

16. **Statistical mechanics.** At low temperature the lower level dominates and $U \to E_{\text{low}}$. At high temperature, nondegenerate levels approach equal occupation and $U \to (E_{\text{low}} + E_{\text{high}})/2$.

17. **Electrostatics.** A positive point charge has radial outward field. $\boldsymbol{E}$ gives force per charge and direction; ϕ gives scalar potential energy per charge, with $\boldsymbol{E} = -\boldsymbol{\nabla}\phi$.

18. **Circuits.** Capacitor voltage and current magnitude decay exponentially as $\mathrm{e}^{-t/(RC)}$; the current direction is opposite the charging direction. The time constant is RC.

19. **Electromagnetic waves.** Changing electric and magnetic fields sustain one another through Maxwell's curl equations in vacuum. Measurable properties include frequency, wavelength, intensity, polarization, or propagation direction.

20. **Special relativity.** Each observer must identify the same pair of events, their spacetime separation, the relative frame velocity, and which clock measures proper time. For spatially separated events, simultaneity conventions matter.

21. **Topology.** The loop classes are $2, -1, 0$; the sign records orientation. For a cube, $8-12+6 = 2$. Cutting permits paths and identifications forbidden to a continuous deformation and can therefore change genus or connectedness.

22. **Differential geometry.** $g_{rr} = 1$, $g_{\phi\phi} = r^2$, and $\sqrt{g}\,\mathrm{d}r\mathrm{d}\phi = r\,\mathrm{d}r\mathrm{d}\phi$. Connection coefficients can be nonzero in flat curvilinear coordinates; curvature is the invariant test. A two-form integrates over an oriented two-dimensional surface.

23. **Lie theory.** Since $J^2 = -I$, even and odd powers give $\mathrm{e}^{\theta J} = I\cos\theta + J\sin\theta$. Planar rotations add angles and commute; rotations about distinct 3D axes generally do not. $U(2\pi) = -I$ for a spinor; the common minus phase cancels from probabilities.

24. **General relativity.** A freely falling object follows a timelike geodesic, locally inertial but with neighboring geodesics showing tidal curvature. Gravitational time dilation, twice-Newtonian light bending, or gravitational waves distinguish the theory from Newtonian gravity.

25. **Quantum principles.** Indistinguishable alternatives contribute amplitudes, whose cross terms produce interference. Normalization guarantees total probability one for a complete measurement.

26. **Solvable quantum systems.** The ground state is a single positive half-sine on $0 < x < L$, vanishing at both walls. A constant zero-momentum state cannot satisfy both boundaries, so the lowest allowed wave number and energy are nonzero.

27. **Quantum approximations.** Compare perturbation matrix elements with relevant unperturbed energy gaps. The method predicts energy shifts and state mixing; small denominators warn that nondegenerate theory fails.

28. **Entanglement.** A pure joint state can contain correlations that prevent factorization. Tracing over the inaccessible qubit removes joint coherences from the local description, producing a mixed reduced density operator.

29. **Quantum computing.** $H^2|0\rangle = |0\rangle$. The second Hadamard recombines paths so the $|1\rangle$ amplitudes cancel and the $|0\rangle$ amplitudes reinforce.

30. **Atomic physics.** A photon supplies the energy gap through a nonzero interaction matrix element. Energy matching is necessary, while symmetry or polarization can make the relevant dipole matrix element vanish.

31. **Condensed matter.** Periodicity produces Bloch states and energy bands. A completely filled isolated band has no nearby empty states for ordinary acceleration, while a partly filled band can support current in the simple band picture.

32. **Nuclear physics.** List all initial and final particles and compare consistent total masses: $Q = (M_i - M_f)c^2$. Momentum, charge, and other relevant quantum numbers must also be conserved; a positive Q does not determine the rate.

33. **Particle physics.** Relativistic energy can create particle–antiparticle excitations, so fixed particle number is insufficient. A Feynman diagram labels a term in an amplitude expansion, including vertices and propagators; it is not a literal trajectory.

34. **Astrophysics and cosmology.** A pressure gradient balances gravity inside a stable star, with the pressure supplied by thermal, radiation, or degeneracy physics. A systematic relation between cosmological redshift and distance supports an expanding scale factor after local motions are accounted for.

35. **Uncertainty and data.** The fractional uncertainty is $0.1/2.0 = 0.05$, or 5%. Repetition reduces independent random error but preserves a fixed 0.2 s calibration bias.

36. **Numerical verification.** The change from 1.42 to 1.36 shows unresolved timestep error. Calculate $Q(0.025)$ and compare successive differences with the method's expected order. Energy, momentum, probability norm, or a known limiting solution can provide a check.

37. **Connections.** Examples include a spring, with displacement restored by $-kx$, and an LC circuit, with charge restored by capacitor voltage $-Q/C$. Both reduce to a second-order oscillator equation with different physical variables.

Appendix B

Constants, units, and useful identities

B.1 Common constants

Use this appendix after deciding the symbolic structure of a calculation. Carrying symbols until the final substitution makes cancellations and dimensional checks visible. Numerical constants then set the scale rather than hide the reasoning. Distinguish definitions that fix a constant exactly from measured quantities whose quoted digits are only approximate.

Exact SI-defining values are distinguished below from rounded reference values. Rounded values suffice for the calculations in this book and are not a precision metrology table.

Quantity	Symbol	Value in SI units
Vacuum light speed	c	$299\,792\,458\ \mathrm{m\,s^{-1}}$ (exact)
Planck constant	h	$6.62607015 \times 10^{-34}\ \mathrm{J\,s}$ (exact)
Elementary charge	e	$1.602176634 \times 10^{-19}\ \mathrm{C}$ (exact)
Boltzmann constant	k_B	$1.380649 \times 10^{-23}\ \mathrm{J\,K^{-1}}$ (exact)
Avogadro constant	N_A	$6.02214076 \times 10^{23}\ \mathrm{mol^{-1}}$ (exact)
Reduced Planck constant	$\hbar = h/(2\pi)$	$\simeq 1.05457 \times 10^{-34}\ \mathrm{J\,s}$
Gravitational constant	G	$\simeq 6.6743 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$
Electron mass	m_e	$\simeq 9.1094 \times 10^{-31}\ \mathrm{kg}$
Proton mass	m_p	$\simeq 1.6726 \times 10^{-27}\ \mathrm{kg}$
Vacuum permittivity	ε_0	$\simeq 8.8542 \times 10^{-12}\ \mathrm{F\,m^{-1}}$

Useful conversions include $1\,\mathrm{eV} = 1.602176634 \times 10^{-19}\,\mathrm{J}$, $1\,\mathrm{fm} = 10^{-15}\,\mathrm{m}$, $1\,\mathrm{angstrom} = 10^{-10}\,\mathrm{m}$, and $\hbar c \simeq 197.33\,\mathrm{MeV\,fm}$. The fine-structure constant is $\alpha = e^2/(4\pi\varepsilon_0\hbar c) \simeq 1/137$. In SI, $\mu_0\varepsilon_0 c^2 = 1$; μ_0 is not an independently exact fixed decimal constant under the modern SI definition.

B.2 Integrals and expansions

These formulas are tools to reconstruct, not merely entries to memorize. A Gaussian integral appears whenever energy is quadratic, and differentiating with respect to its parameter generates useful moments. A Taylor expansion replaces a complicated function locally by the lowest terms that matter at the accuracy you need.

For $a > 0$,

$$\int_{-\infty}^{\infty} \mathrm{e}^{-ax^2}\mathrm{d}x = \sqrt{\pi/a}, \qquad \int_{-\infty}^{\infty} x^2\mathrm{e}^{-ax^2}\mathrm{d}x = \frac{\sqrt{\pi}}{2a^{3/2}}, \tag{B.1}$$

$$\int_0^{\infty} x^n\mathrm{e}^{-ax}\mathrm{d}x = \frac{n!}{a^{n+1}} \quad (n = 0, 1, \ldots). \tag{B.2}$$

Common small-x expansions are

$$\mathrm{e}^x = 1+x+x^2/2+\cdots, \quad \ln(1+x) = x-x^2/2+\cdots, \quad (1+x)^a = 1+ax+\tfrac{1}{2}a(a-1)x^2+\cdots. \tag{B.3}$$

An expansion is useful only within its convergence or asymptotic regime and with a sufficiently small neglected contribution.

For example, differentiate $I(a) = \int \mathrm{e}^{-ax^2}\mathrm{d}x = \sqrt{\pi/a}$ with respect to $a > 0$:

$$-I'(a) = \int x^2\mathrm{e}^{-ax^2}\mathrm{d}x = \frac{\sqrt{\pi}}{2a^{3/2}}. \tag{B.4}$$

Further differentiation supplies higher even moments. To justify the starting Gaussian integral, square it, regard the product as an integral over the plane, and use polar coordinates: $I(a)^2 = 2\pi\int_0^{\infty} r\mathrm{e}^{-ar^2}\mathrm{d}r = \pi/a$. A change of geometry turns a difficult one-dimensional antiderivative into an easy definite integral.

B.3 Natural units and restoring dimensions

Natural units reduce clutter by using constants as conversion factors between units. Setting $c = 1$ does not make a physical speed disappear; it means that length and time are measured in compatible units. Restore constants by identifying the dimensions of the physical quantity, not by placing c or $\hbar$ wherever an expression happens to look familiar.

Setting $c = \hbar = 1$ makes time and length inverse-energy quantities, while mass and momentum have energy units. Temperature can also be measured in energy if $k_\mathrm{B} = 1$. Restore constants using physical dimensions: $E^2 = p^2 + m^2$ becomes $E^2 = p^2c^2 + m^2c^4$, while a Compton length $1/m$ becomes $\hbar/(mc)$. Dimensional restoration may not determine numerical factors or distinguish quantities with identical dimensions.

Appendix C

Guided diagrams and thought exercises

These exercises require only pencil and paper. A diagram is useful when it makes a relation visible: slope as a rate, area as accumulation, arrows as vectors, contours as constant values, and transformations as changes of description. Draw labels before adding detail, and use equations to check the picture.

C.1 Graphs, forces, and conservation

1. Draw a position–time curve that begins with zero slope and becomes steadily steeper. Beneath it, draw the corresponding qualitative velocity–time graph. State the sign of acceleration.

2. Draw a box on a horizontal surface with a rightward applied force larger than leftward friction. Add weight and normal force. Write the two component equations and mark which arrows cancel.

3. Draw two carts before and after a sticking collision. Choose arrow lengths proportional to momentum and use the drawing to explain why kinetic energy can decrease while momentum is conserved.

4. Sketch $V_{\text{eff}}(r) = -\kappa/r + \ell^2/(2\mu r^2)$. Mark a stable circular orbit, a bound radial motion, and an unbound energy.

C.2 Fields, waves, and thermal processes

1. Draw a small box in a vector field. Label flux entering and leaving each pair of faces, then explain how their difference becomes a divergence.

2. Draw the first three fixed-end string modes. Mark nodes and wavelengths and infer their frequency ratios.

3. Draw rays from an object through a converging thin lens using the central ray and the initially parallel ray. Use similar triangles to recover the sign of magnification.

4. Sketch an isothermal expansion and an isochoric heating path on a p–V diagram. Mark the work as an area and explain which path has zero boundary work.

5. Draw field lines and equipotential curves for a positive point charge. Explain geometrically why they meet at right angles.

C.3 Geometry, relativity, and quantum theory

1. Draw two loops on a punctured plane, one winding once around the missing point and one winding zero times. Explain why only the second can contract without crossing the puncture.

2. Draw a polar coordinate grid. Mark equal angular intervals at two radii and explain the metric factor $r^2 \mathrm{d}\phi^2$.

3. Draw two small rotations about different axes in opposite orders. Use the commutator to label the residual rotation.

4. Draw a spacetime diagram with light cones, a timelike worldline, and a spacelike-separated event. Add two different simultaneity lines.

5. Draw the lowest three infinite-well wavefunctions, including their nodes. Beside each, sketch the probability density and label its energy scaling with n^2.

6. On a Bloch sphere, mark $|0\rangle$, $|1\rangle$, $(|0\rangle + |1\rangle)/\sqrt{2}$, and $(|0\rangle + \mathrm{i}|1\rangle)/\sqrt{2}$. State which pairs have equal computational-basis probabilities but different relative phases.

Answers

The velocity graph in the first problem is positive and increasing, so acceleration is positive. In the force diagram, $N = mg$ while $F_{\text{app}} - f = ma_x$. A sticking collision preserves the vector sum of momentum; deformation and thermal energy account for lost kinetic energy. The minimum of V_{eff} marks a stable circular orbit. Fixed-string frequencies have ratio $1 : 2 : 3$. Isochoric work is zero because $\mathrm{d}V = 0$. Electric field is $-\boldsymbol{\nabla}\phi$, so it is normal to constant-ϕ curves. Winding distinguishes the punctured-plane loops. Polar arc length is $r\mathrm{d}\phi$. Noncommuting rotations leave a commutator-order residual. Timelike events lie within a light cone, while spacelike simultaneity depends on frame. Infinite-well nodes increase with n and $E_n \propto n^2$. Both equatorial qubit states give equal 0/1 probabilities, but their azimuths record different phases.

Appendix D

Review problems

Try each problem before reading its solution. State assumptions and check dimensions even when the algebra is short.

1. A particle starts from rest under linear drag and a constant force F_0 in one dimension. Find its velocity and the time at which it reaches half its terminal velocity.

2. A uniform solid cylinder rolls without slipping down a height h. Find its final center-of-mass speed and compare it with a frictionless sliding point mass.

3. For $V(r) = -\kappa/r$, find the circular-orbit radius and energy in terms of reduced mass μ and angular momentum ℓ.

4. Two equal masses have the coupled-spring system in the oscillations chapter. If $x_1(0) = A$, $x_2(0) = 0$, and both velocities vanish, find $x_1(t), x_2(t)$.

5. A string of length L is fixed at both ends. What happens to all mode frequencies if tension is multiplied by four while linear density stays fixed?

6. Two ideal polarizers are oriented at 45° to one another. Unpolarized light of intensity I_0 enters the first. Find the final intensity.

7. One mole of a monatomic ideal gas is heated reversibly at constant pressure from T to $2T$. Find Q, W_{by}, ΔU, and ΔS.

8. For N independent nondegenerate two-level systems with gap Δ, find the high- and low-temperature limits of internal energy and entropy.

9. A uniformly charged solid sphere has radius R and total charge Q. Find the electric field inside and outside, and the potential at the center with zero at infinity.

10. A capacitor initially charged to voltage V_0 discharges through resistance R. Find $V(t)$ and total heat dissipated.

11. A vacuum plane wave has peak electric field E_0. Find average intensity and mean pressure on a perfectly reflecting normal surface.

12. A particle has speed $0.8c$. Find γ, its kinetic energy in units of mc^2, and its momentum in units of mc.

13. An emitter is static at Schwarzschild radius-coordinate $r_e = 4GM/c^2$. What frequency is measured by a static observer at infinity relative to the emitted frequency?

14. In the infinite well $0 < x < L$, find the ground-state $\langle x \rangle$, $\langle p \rangle$, and $\langle p^2 \rangle$, interpreting the latter through kinetic energy.

15. A harmonic oscillator in its ground state is perturbed by $V' = fx$, with constant force parameter f. Find the exact energy shift by completing the square and compare with first-order perturbation theory.

16. A spin initially in $|+z\rangle$ is measured along $+x$, then along $+z$, with the first outcome recorded but not selected. What is the final probability of $+z$?

17. Trace out either qubit of $|\Phi^+\rangle$. Find the reduced purity and entropy in units of k_{B}.

18. A radioactive sample has two independent decay channels with rates λ_1, λ_2. Find its half-life and the fraction of decays in channel 1.

19. Show that a spatially flat matter-only expanding universe has $a(t) \propto t^{2/3}$, taking $a = 0$ at $t = 0$.

20. For the explicit diffusion scheme, insert an error mode $u_j^n = A^n \mathrm{e}^{\mathrm{i}kj\Delta x}$. Derive its amplification factor and stability bound.

21. A polyhedral surface has $V = 12$, $E = 30$, and $F = 20$. Find its Euler characteristic and, if it is a closed connected orientable surface, its genus.

22. For the unit sphere with metric $\mathrm{d}s^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2$, find the area element and integrate it over the sphere.

23. Let J_x, J_y, J_z satisfy $[J_x, J_y] = J_z$ and cyclic permutations. Expand $\mathrm{e}^{\epsilon J_x}\mathrm{e}^{\delta J_y}$ and the reversed product through order $\epsilon\delta$ and find their difference.

24. For $U(\theta) = \exp(-\mathrm{i}\theta\sigma_z/2)$, calculate $U(2\pi)$ and $U(4\pi)$. Explain the distinction between the spinor state vector and its measurement probabilities.

Appendix E

Solutions and reasoning

1. Solve $m\dot{v} = F_0 - bv$: $v = (F_0/b)(1 - \mathrm{e}^{-bt/m})$. Thus $v_\infty = F_0/b$ and $t_{1/2} = (m/b)\ln 2$. The velocity approaches the terminal value monotonically for positive F_0, b.

2. With $I = mR^2/2$, energy conservation gives $mgh = 3mv^2/4$, hence $v = \sqrt{4gh/3}$. Sliding gives $\sqrt{2gh}$; rolling stores some energy in rotation. Enough static friction is assumed to enforce rolling.

3. Set $V'_{\text{eff}} = \kappa/r^2 - \ell^2/(\mu r^3) = 0$, yielding $r_c = \ell^2/(\mu\kappa)$. Then $E_c = -\kappa/(2r_c) = -\mu\kappa^2/(2\ell^2)$. The second derivative at r_c is positive, so the orbit is radially stable.

4. Both normal coordinates initially equal A. Therefore $x_1 = A[\cos(\omega_+ t) + \cos(\omega_- t)]/2$ and $x_2 = A[\cos(\omega_+ t) - \cos(\omega_- t)]/2$. A mass initially at rest can later move through exchange of mode energy contributions.

5. Since $v = \sqrt{\mathcal{T}/\mu}$ doubles, every $f_n = nv/(2L)$ doubles. The spatial node locations for each mode do not change.

6. The first transmits $I_0/2$. The second transmits another factor $\cos^2 45^\circ = 1/2$, leaving $I_0/4$.

7. For one mole, $C_V = 3R/2$ and $C_p = 5R/2$. Thus $Q = 5RT/2$, $W_{\text{by}} = R\Delta T = RT$, $\Delta U = 3RT/2$, and $\Delta S = C_p \ln 2 = (5R/2)\ln 2$. Here the initial temperature is denoted by T.

8. At low T, all systems approach the unique ground state: $U \to 0$ and $S \to 0$. At high T, each has equal probabilities: $U \to N\Delta/2$, $S \to Nk_{\text{B}} \ln 2$. The bounded spectrum explains why the high-temperature energy saturates.

9. Gauss's law gives $\boldsymbol{E} = Qr\hat{\boldsymbol{r}}/(4\pi\varepsilon_0 R^3)$ inside and $Q\hat{\boldsymbol{r}}/(4\pi\varepsilon_0 r^2)$ outside. Integrating inward from infinity gives $\phi(0) = 3Q/(8\pi\varepsilon_0 R)$. Inside, $\phi(r) = Q(3 - r^2/R^2)/(8\pi\varepsilon_0 R)$.

10. $V = V_0 \mathrm{e}^{-t/(RC)}$. Integrating V^2/R from zero to infinity gives heat $CV_0^2/2$, exactly the initial stored energy.

11. Since $\langle E^2 \rangle = E_0^2/2$, $I = \varepsilon_0 c E_0^2/2$. Reflection reverses normal photon momentum, so $\bar{p} = 2I/c = \varepsilon_0 E_0^2$. Here p is pressure, not momentum.

12. $\gamma = 1/\sqrt{1-0.64} = 5/3$. Thus $K/(mc^2) = 2/3$ and $p/(mc) = \gamma(v/c) = 4/3$. The invariant check is $(5/3)^2 - (4/3)^2 = 1$.

13. Since $r_e = 2r_s$, $\nu_\infty/\nu_e = \sqrt{1 - r_s/r_e} = 1/\sqrt{2}$. This assumes a static emitter, rather than a freely falling one whose motion adds a Doppler effect.

14. Symmetry gives $\langle x \rangle = L/2$ and a real stationary wavefunction gives zero mean momentum. Since $E_1 = \pi^2\hbar^2/(2mL^2)$, $\langle p^2 \rangle = 2mE_1 = \pi^2\hbar^2/L^2$. Strict operator-domain issues for momentum in a hard-wall interval do not change this kinetic-energy result.

15. Completing the square gives

$$\tfrac{1}{2}m\omega^2x^2 + fx = \tfrac{1}{2}m\omega^2\left(x + \frac{f}{m\omega^2}\right)^2 - \frac{f^2}{2m\omega^2}.$$

    Every level shifts by $-f^2/(2m\omega^2)$. First order vanishes because $\langle x \rangle = 0$ in an unperturbed number state; the leading shift is second order.

16. The x measurement gives either outcome with probability $1/2$. Each x eigenstate gives $+z$ with probability $1/2$, so the final probability is $1/2$. Without the intervening measurement, it would have been 1 for no intervening dynamics.

17. $\rho_A = (|0\rangle\langle 0| + |1\rangle\langle 1|)/2 = I/2$. Then $\operatorname{Tr}\rho_A^2 = 1/2$ and $S_A/k_\mathrm{B} = \ln 2$. The global purity is still 1.

18. Survival depends on the sum of rates: $N = N_0\mathrm{e}^{-(\lambda_1+\lambda_2)t}$. Therefore $t_{1/2} = \ln 2/(\lambda_1 + \lambda_2)$ and the channel-1 branching fraction is $\lambda_1/(\lambda_1 + \lambda_2)$.

19. Conservation gives $\rho = C/a^3$. The Friedmann equation implies $\dot{a}^2 = (8\pi GC/3)/a$, so $a^{1/2}\mathrm{d}a = \sqrt{8\pi GC/3}\mathrm{d}t$. Integration yields $a^{3/2} \propto t$ and $a \propto t^{2/3}$.

20. Substitution yields $A = 1 + r(\mathrm{e}^{\mathrm{i}k\Delta x} - 2 + \mathrm{e}^{-\mathrm{i}k\Delta x}) = 1 - 4r\sin^2(k\Delta x/2)$. Requiring $|A| \le 1$ for all resolved modes and $r \ge 0$ gives $r \le 1/2$.

21. $\chi = V - E + F = 12 - 30 + 20 = 2$. Since $\chi = 2 - 2g$, the genus is $g = 0$; the surface is topologically a sphere under the stated assumptions.

22. The metric determinant is $g = \sin^2\theta$, so $\mathrm{d}A = \sin\theta\,\mathrm{d}\theta\mathrm{d}\phi$. Therefore $A = \int_0^{2\pi}\mathrm{d}\phi\int_0^{\pi}\sin\theta\mathrm{d}\theta = 4\pi$.

23. To the requested order, the first product contains $\epsilon\delta J_xJ_y$ and the reversed product contains $\epsilon\delta J_yJ_x$. Their difference is $\epsilon\delta[J_x, J_y] = \epsilon\delta J_z$.

24. Since $\sigma_z^2 = I$, $U(2\pi) = \cos\pi I - \mathrm{i}\sin\pi\sigma_z = -I$, while $U(4\pi) = I$. The first rotation changes a spinor's overall sign; every probability formed from an absolute square is unchanged.

Appendix F

Further study and a route through the material

These established textbook references provide deeper derivations and problems. They are reading pointers, not claims that this book reproduces any particular edition. No specific edition is required for the topic mapping.

Subject	References
Broad foundations	Halliday, Resnick, and Walker, *Fundamentals of Physics*; Feynman, Leighton, and Sands, *The Feynman Lectures on Physics*.
Mathematical methods	Boas, *Mathematical Methods in the Physical Sciences*; Arfken, Weber, and Harris, *Mathematical Methods for Physicists*.
Geometry and topology	Nakahara, *Geometry, Topology and Physics*; Lee, *Introduction to Smooth Manifolds*.
Lie groups and algebras	Hall, *Lie Groups, Lie Algebras, and Representations*; Georgi, *Lie Algebras in Particle Physics*.
Classical mechanics	Taylor, *Classical Mechanics*; Goldstein, Poole, and Safko, *Classical Mechanics*.
Fluids	Acheson, *Elementary Fluid Dynamics*.
Optics	Hecht, *Optics*.
Thermal physics	Schroeder, *An Introduction to Thermal Physics*; Reif, *Fundamentals of Statistical and Thermal Physics*.
Electromagnetism	Griffiths, *Introduction to Electrodynamics*; Jackson, *Classical Electrodynamics*.
Relativity	Taylor and Wheeler, *Spacetime Physics*; Carroll, *Spacetime and Geometry*.
Quantum mechanics	Griffiths and Schroeter, *Introduction to Quantum Mechanics*; Shankar, *Principles of Quantum Mechanics*; Sakurai and Napolitano, *Modern Quantum Mechanics*.
Condensed matter	Kittel, *Introduction to Solid State Physics*; Ashcroft and Mermin, *Solid State Physics*.

Subject	References
Nuclear physics	Krane, *Introductory Nuclear Physics*.
Particles and fields	Griffiths, *Introduction to Elementary Particles*; Peskin and Schroeder, *An Introduction to Quantum Field Theory*.
Cosmology	Ryden, *Introduction to Cosmology*.
Computational physics	Newman, *Computational Physics*.

Suggested sequence

First become fluent with dimensions, calculus, Newtonian conservation, oscillators, and the mathematical meaning of boundary conditions. Next connect fields and fluxes through fluid mechanics and electromagnetism, and connect microscopic probabilities to thermodynamics. Then study special relativity and basic quantum mechanics before proceeding to atoms, materials, nuclei, and fields. General relativity and cosmology benefit from all of these foundations.

For each chapter, be able to explain the physical picture, derive at least one central equation, solve a representative problem, and identify a regime in which the model fails. That combination is more useful than memorizing an isolated list of formulas.

Building this document

The source is a single self-contained file with no external figures, bibliography database, or shell-escape requirement. It uses common LaTeX packages from a standard TeX Live or MiKTeX installation. From the source directory, compile twice so the table of contents and internal references settle:

```
pdflatex -interaction=nonstopmode -halt-on-error \
  "Physics Calc.tex"
pdflatex -interaction=nonstopmode -halt-on-error \
  "Physics Calc.tex"
```

Alternatively, compile with:

```
latexmk -pdf "Physics Calc.tex"
```

You can also upload the source as the main file in an online LaTeX editor with the listed packages available.